Shape transitions and rare reversals in the elephant random walk

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Abstract

At every superdiffusive memory, the initial biases for which the elephant random walk's limit density is bimodal form one open interval centred at $1/2$. Outside it, the density is unimodal with a unique local maximum. The proof controls first differences, their boundary histories and the dominant half of each finite count law; every finite law has at most two modal plateaus. For a fair first step, the density has two inflection points when unimodal and four when bimodal. A certified computation proves a unique modal transition at $a_* = 0.6688128600146\ldots$, enclosed in an interval shorter than $3.3 \cdot 10^{-15}$, with the critical density unimodal. Bimodality for $a \geq 4/5$ also has a proof without computation. As $a \to 1$, a two-component Landau approximation holds uniformly over the initial bias, in total variation and scaled uniform density distance. It determines the locations and heights of the fair-start peaks. For rare reversals, an explicit total variation approximation by a truncated Luria–Delbrück law gives a Landau local limit theorem with a rate. When $\log N \leq x = N\varepsilon \leq 2\log N$, every mode lies at distance $x\log x + \lambda_{\mathrm{L}} x + O(\sqrt{x\log x})$ from an end. For even N, the origin and each end are equally likely at a unique reversal probability, with $N\varepsilon = 2\log N - \log\log N - \log 16 + o(1)$. At this crossing, every even $N \geq 8$ has two symmetric modal intervals and a unique interior minimum, by finite certified computations and analytic bounds.

Keywords: elephant random walk; random recursive tree; compound Poisson approximation; Landau distribution; bimodality; computer-assisted proof.
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1 Introduction

An elephant and an ant are playing hide and seek. The elephant counts while the ant hides among many identical anthills, then opens its eyes and knows at once which one the ant is in: the ant's shoes are outside the door.

(a riddle my Indian mother told me)

In the elephant random walk of Schütz and Trimper [1], each step copies a uniformly chosen earlier step and reverses its sign with probability ε. The first step is fair unless we say otherwise. We ask how memory and initial bias determine the modes. A finite-step second-difference bound connects rare reversals, where the modes lie near the ends, to a unimodal-or-bimodal superdiffusive limit at fixed memory. In our rare-reversal regime $\varepsilon \to 0$ and $x = N\varepsilon$ is of order $\log N$, so the expected number of reversals $x - \varepsilon$ still grows. Most limit theorems instead fix ε [3, 7, 8, 9].

For the number M_N of steps that disagree with the first step, we give a total variation approximation of order $x^2 \log^2 N/N$ by a truncated Luria–Delbrück law: the compound Poisson law with jumps $d \le N-1$ at rates $x/(d(d+1))$. Its explicit Landau local limit rate (Theorem 1.1) locates every mode about $x \log x - 0.2228\, x$ from an end. The second difference changes sign at most four times (Theorem 1.4).

The count M_N is the resistant-cell count in Bokes, Hlubinová and Singh's reversible fluctuation assay [27]: one sensitive founder, uniform parent choice and equal switching probabilities at birth. Their formal asymptotics give the first-order center coefficient and Landau shape [27, Eqs. (6)–(10), (50)–(52), (58)–(60)]; we give error bounds and exact crossing and shape results.

For even N the crossing equates the origin's mass with that of each end. In the broadcasting representation it compares a color tie with a unanimous outcome, rather than the full majority-vote reconstruction error. It is unique, at $x = 2\log N - \log\log N - \log 16 + o(1)$. For every even $N \ge 8$ its law has two symmetric modal intervals and a unique interior minimum. The persistent walk, which copies the last step, crosses at about $\log N$ switches [16]; the early-family calculation below explains why.

At fixed memory $a = 1 - 2\varepsilon \in (\frac{1}{2}, 1)$, the bimodal biases form a centred open interval, proving the conjecture in [10, Remark 2.1(b)]. The fair law has a unique modal transition at $a_* = 0.6688128600146\ldots$, with critical unimodality (Theorems 4.4 and 4.6). Theorem 4.7 strengthens the weak bridge of Corollary 4.1 to uniform and total variation approximations, uniformly over the bias, and locates and measures the peaks.

The original memory parameter is $1-\varepsilon$, the parameter a of [10] is $1-2\varepsilon$, and the broadcasting flip probability is ε (Lemma 3.1) [5, 6].

1.1 Setting and notation

The walk takes N steps $X_1, \ldots, X_N \in \{+1, -1\}$, with $+1$ denoting right. Put $A_N = \#\{i \le N : X_i = +1\}$ and $P_N(k) = \mathbb{P}(A_N = k)$. The position $2A_N - N$ is at the origin for $A_N = N/2$ and at an end for $A_N \in \{0, N\}$. Unless stated otherwise, $\mathbb{P}(X_1 = +1) = \frac{1}{2}$, giving $P_N(k) = P_N(N-k)$. We write $\mathbb{P}_+$ and $\mathbb{E}_+$ for conditioning on $X_1 = +1$. For even N put $h = N/2$, $C_N = P_N(h)$ and $E_N = P_N(N) = P_N(0)$, the center and end masses. A reference to C_N presupposes even N; the crossing solves $C_N = E_N$. All logarithms are natural, $L = \log N$, $H_k = \sum_{j \le k} 1/j$ is the k-th harmonic number, and γ is Euler's constant. Limits, $O(\cdot)$ and $o(\cdot)$ refer to $N \to \infty$, over even N when C_N appears. The constants implied by $O(\cdot)$ are absolute unless we name what they depend on.

The reversal probability is $\varepsilon \in (0,1)$, $x = N\varepsilon$, and the crossing is ε_N, with $x_N = N\varepsilon_N$.

1.2 Main results

The law $f_N(m) = \mathbb{P}(M_N = m)$ does not depend on X_1, and $P_N(k) = \frac{1}{2}[f_N(N-k) + f_N(k)]$ (Lemma 2.2). Let CP_N be the law of $\sum_{d=1}^{N-1} d\,\mathcal{N}_d$, where the $\mathcal{N}_d$ are independent Poisson variables with means $x/(d(d+1))$. Put $\lambda_m = (m - x\log x)/x$, and let

$$f_L(\lambda) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-i\lambda t} e^{B(t)}\,dt, \qquad B(t) = -it\log|t| - \frac{\pi}{2}|t|,$$

be the Landau density.

Theorem 1.1 (the approximation)**.** *For $N \geq 1$ and $0 < \varepsilon < 1$ the total variation distance between f_N and CP_N is at most an explicit $\delta_\Sigma = O(x^2 \log^2 N/N)$ (Theorem 3.8). Suppose that $x \to \infty$ and $x^3 \log^2 N/N \to 0$. Then, uniformly over integers $m \geq 0$,*

$$\bigl|x f_N(m) - f_L(\lambda_m)\bigr| = O\Bigl(\frac{1+\log x}{x} + \frac{x^3 \log^2 N}{N}\Bigr).$$

The weak convergence $(M_N - x\log x)/x \Rightarrow f_L$ needs only $x \to \infty$ and $x^2 \log^2 N/N \to 0$. Theorem 3.9 gives explicit finite-N bounds.

Theorem 1.2 (the modes)**.** *The Landau density has a unique mode $\lambda_{\mathrm{L}} \approx -0.222783$ (Lemma 3.10, computer-assisted). For $\log N \leq x \leq 2\log N$ and all sufficiently large N, every mode $a^* \geq N/2$ of P_N satisfies*

$$\bigl|N - a^* - x\log x - \lambda_{\mathrm{L}} x\bigr| \leq K\sqrt{x\log x},$$

and the same bound holds for every mode of f_N in place of $N - a^*$ (Corollary 3.11). We do not compute K.

Theorem 1.3 (the center and the crossing)**.** *Let $N \geq 2$ be even.*

(a) The ratio C_N/E_N is strictly increasing in $\varepsilon \in (0,1)$, and the crossing ε_N exists, is unique and lies in $(0, \frac{1}{2})$ (Proposition 2.3).

(b) For $0 < \varepsilon < \frac{1}{2}$ we have $C_N \geq \frac{4\varepsilon}{N+2}(1-\varepsilon)^{2-12/(N+4)} \geq \frac{4\varepsilon}{N+2}(1-2\varepsilon)$. If $\varepsilon(1+\log N) \leq \frac{1}{100}$, then $C_N \leq \frac{4\varepsilon}{N+2} N^{2\varepsilon} e^{5200\varepsilon}$ (Theorem 2.6, whose constant 5200 uses an enclosure in ball arithmetic).

(c) As $N \to \infty$ over even N, the crossing satisfies $x_N + \log x_N = 2\log N - \log 8 + O(\log^2 N/N)$, and so

$$x_N = 2\log N - \log\log N - \log 16 + O\Bigl(\frac{\log\log N}{\log N}\Bigr)$$

(Theorem 2.7).

So $C_N = \frac{4\varepsilon}{N+2}(1 + O(\varepsilon \log N))$ when $\varepsilon \log N$ is small, and the error factor is $1 + O(\log^2 N/N)$ at the crossing. For the persistent walk with switch probability q, the crossing has $T = Nq$ close to $\log N$. We compare the two crossings to the next order in Theorem S1.7.

For the shape between the humps put $D_N(a) = P_N(a+1) - 2P_N(a) + P_N(a-1)$ and $m^* = N - a^*$ for a mode $a^* \geq N/2$.

Theorem 1.4 (the shape)**.** *(a) For $N \geq 3$ and $0 < \varepsilon < \frac{1}{2}$, delete zeros from the signs of D_N and merge equal neighbors. The resulting pattern is a subsequence of $(+,-,+,-,+)$ (Theorem 3.16).*

(b) *For every even $N \geq 8$, the crossing law has exactly two symmetric intervals of maximizing points. It strictly rises to each interval and strictly falls from it toward the center, its unique interior minimum (Corollary 3.19). This uses the finite endpoint-dip certificates of Proposition 3.7. The whole statement, certificates included, is kernel-checked in Lean (Section 5).*

Between the humps we expect more, namely strict convexity on $[2m^*, N - 2m^*]$. Section S5 proves this window for every large N, and at the crossing for $5 \leq N \leq 2000$ by a certified computation.

Theorem 1.4(b) does not require this window.

Theorem 1.5 (the fixed-memory limit law)**.** *Fix $0 < \varepsilon < \frac{1}{4}$ and put $a = 1 - 2\varepsilon$. With a fair first step, $(2A_N - N)/N^a$ converges almost surely to a limit with an even density g_a. Either g_a is nonincreasing on $[0, \infty)$, or there is $c > 0$ such that g_a is nondecreasing on $[0, c]$ and nonincreasing on $[c, \infty)$ with $g_a(0) < g_a(c)$. These monotonicities are strict, so g_a has one or two point modes (Theorem 4.2). The second case holds for $0 < \varepsilon \leq \frac{1}{10}$ (Theorem 4.3). The density has exactly two inflection points in the first case and four in the second. With explicit rationals $z_- < z_+$ such that $z_- = 0.66881286001459775\ldots$ and $z_+ - z_- < 3.3 \cdot 10^{-15}$, there is a unique $a_* \in (z_-, z_+)$ such that the first case holds for $a \leq a_*$ and the second for $a > a_*$ (Theorem 4.4, computer-assisted). For arbitrary initial bias the limit is bimodal exactly when $b_a < q < 1 - b_a$, for some $b_a \in [0, 1/2]$, and otherwise has one local maximum (Theorem 4.6). As $a \to 1$, its density admits a two-component Landau approximation in total variation and scaled uniform distance, uniformly over the bias (Theorem 4.7).*

Why twice as many reversals? An endpoint has mass about $\frac{1}{2}e^{-x}$. In the persistent walk, let q be the switch probability and $T = Nq$. Its runs have typical length N/T, and the right-step count spreads over about $N/\sqrt{T}$ values. Its central mass is of order $\sqrt{T}/N$, which balances e^{-T} at $T \approx \log N$. An elephant reversal instead starts a family of descendants, governed by Pólya's urn. Reaching the middle requires an early reversal whose family occupies about half the walk. This costs order ε, and the exact middle costs another $1/N$. Thus the center has mass about $4\varepsilon/N = 4x/N^2$, balancing the endpoint at $x \approx 2\log N$. Most of the roughly $2\log N$ reversals there occur too late to produce a large family. As in the riddle, the evidence is left at the door: a walk that ends in the middle usually shows a reversal among its first few steps.

1.3 Prior work

The counting chain predates the elephant walk. Drezner and Farnum's generalized binomial model [2], in the formulation of James, James and Qi [4, p. 2339], has next-success probability $(1-\theta)p+\theta b/t$. Taking $p = 1/2$ and $\theta = a = 1-2\varepsilon$ gives our $r_t(b) = \varepsilon+(1-2\varepsilon)b/t$. For $0 < \varepsilon \leq 1/2$ this is the same transition rule; the fair-start chain also has their initialization. Other initial biases change only that initialization. Heyde [3] established the fixed-parameter limit regimes, including almost-sure convergence for $\theta > 1/2$. James, James and Qi [4, Theorem 2.2] treat time-varying dependence parameters. The results below concern the shapes of these laws and the rare-reversal regime.

Further fixed-memory results include [7, 8, 9]. Murahara and Tokumitsu [13] give optimal Gaussian local and central limit rates for fixed copying probabilities at most 3/4; Vishwakarma [14] studies moment recurrences in the three regimes. The superdiffusive limit has a smooth positive density with known moments [11, Theorems 1.3 and 1.4], unimodal conditional on the first step [10,

Theorem 1.1]. Trinas and Valle's uniform local limit theorem [12, Theorem 3.1] transfers our first-difference bound to every initial bias. Their exact moments and certified recursion disprove the log-concavity conjecture in [10, Remark 2.3] [12, Theorem 4.4]; conditional unimodality alone does not control the mixture. Theorem 4.2 treats its fair-start shape and curvature.

Engländer and Volkov's coin-turning walk switches at time n with probability p_n; for $p_n = a/n$ eventually, its occupation fraction tends to $\mathrm{Beta}(a, a)$ [21, Section 4, Theorem 1]. Fan and Qi [17] vary the elephant's memory within a walk. Our reversal probability instead varies with its final length, giving a triangular array, as in Roy, Takei and Tanemura's varying memory windows [18].

Kürsten's coupling [19] copies a parent with probability $1 - 2\varepsilon$ and otherwise assigns a fair sign; Baur and Bertoin [22] give the urn connection. Our regime is strongly supercritical in Baur's classification [20], following Bertoin [23]. Baur's Corollary 1 gives a Poisson limit for the largest nonroot clusters. Miyazaki and Takei derive root-cluster limits via the minimal elephant walk [24, Theorems 2.3, 2.4 and Section 2.3]. These results do not estimate our exact central mass.

The untruncated law is the Lea–Coulson Luria–Delbrück law. Pakes proved its weak Landau limit [28, p. 993]; Möhle gave density formulas and a uniform lattice local limit for integer parameters, without a rate [29, Theorem 4.1 and Corollaries 4.1–4.2]. Kessler and Levine give the small-parameter profile, Landau limit and dip threshold, with mode near $\mu N(\log \mu N - 0.22)$ [25, Section 2 and Eqs. (14), (18), (19)]. Bertoin proves the same weak limit for vertices cut off from a recursive-tree root at cut probability of order $k^{-1/4}$ on a tree of size k [31, Proposition 1]. In the Iksanov–Möhle coupling [32, Section 2 and Proposition 1], the Meir–Moon cutting procedure [33] removes subtrees whose sizes have law $1/(j(j+1))$ independently up to first passage. These weights also give the expected $N/(d(d+1))$ families of size d in Section 3. Schweinsberg [30] and Bertoin [31] obtain this stable law for root-cluster fluctuations at retention $1 - c/\log n$. Our additions are the total variation bound, explicit local rate with truncation at $N - 1$, transfer to the elephant walk and shape theorems.

Broadcasting papers study root-color reconstruction [5, 6], color-count limits and martingale error bounds [34, 35]. We estimate the tie probability as the flip probability tends to zero. Gut and Stadtmüller's count [36, (1.5)] has next-step probability pk/n, whereas ours has $\varepsilon + (1 - 2\varepsilon)k/n$, so their chain cannot leave zero. The irreversible mutation expansion of Kessler and Levine [26, Eqs. (10), (13), (14)] has the same first derivative at $\varepsilon = 0$ as our law, but not our uniform relative bound at $N\varepsilon \asymp \log N$.

1.4 Organization

Sections 2–4 give the crossing and shape results, proved in Appendices A–G. Section 5 describes the numerical and formal checks. Sections S1–S5 add second-order center and crossing formulas, the tail, odd lengths, the assay likelihood bound and the convexity window. They are not used above. Section S6 completes the proofs of Appendix G, making this version self-contained.

2 The elephant walk and its crossing

Definition 2.1 (elephant walk)**.** Fix $0 < \varepsilon < 1$. Let X_1 be uniform on $\{\pm 1\}$. For $t \geq 1$ let U_t be uniform on $\{1, \dots, t\}$, and put $X_{t+1} = X_{U_t}$ with probability $1 - \varepsilon$ and $X_{t+1} = -X_{U_t}$ with probability ε. All these choices are independent. We call the event $X_{t+1} = -X_{U_t}$ a flip. As in Section 1.1, $A_N = \#\{i \leq N : X_i = +1\}$ and $P_N(k) = \mathbb{P}(A_N = k)$. The crossing ε_N is the value of ε with $C_N = E_N$ (it exists and is unique by Proposition 2.3), and $x_N = N\varepsilon_N$.

A flip is the reversal of the introduction. Let M_t count the $i \le t$ with $X_i \neq X_1$, and put

$$r_t(b) = \varepsilon + (1-2\varepsilon)\frac{b}{t}, \qquad f_N(m) = \mathbb{P}_+(M_N = m).$$

Lemma 2.2. *The process $(M_t)_{t\ge 1}$ is a Markov chain with $M_1 = 0$ and $\mathbb{P}(M_{t+1} = b+1 \mid M_t = b) = r_t(b) = 1 - \mathbb{P}(M_{t+1} = b \mid M_t = b)$, and it is independent of X_1. For $0 \le k \le N$, $P_N(k) = \frac{1}{2}[f_N(N-k) + f_N(k)]$. In particular $C_N = f_N(h)$, $E_N = \frac{1}{2}(1-\varepsilon)^{N-1}$, and $P_N(k) = P_N(N-k)$.*

Proof. Given $X_1, \dots, X_t$, the copied step X_{U_t} differs from X_1 with probability M_t/t. So $X_{t+1} \neq X_1$ with conditional probability $(1-\varepsilon)M_t/t + \varepsilon(1 - M_t/t) = r_t(M_t)$. This depends only on M_t and not on X_1, which proves the first claim. On $\{X_1 = +1\}$ we have $A_N = N - M_N$, and on $\{X_1 = -1\}$ we have $A_N = M_N$. Averaging over X_1 gives the formula for $P_N(k)$, which does not change under $k \mapsto N-k$. Finally $M_N \le N-1$, so $f_N(N) = 0$, and $f_N(0) = \prod_{t=1}^{N-1}(1 - r_t(0)) = (1-\varepsilon)^{N-1}$. □

So the end is that of the persistent walk with switch probability ε. More flips always raise the center relative to the end.

Proposition 2.3. *Let $N \ge 2$ and $1 \le k \le N-1$. Then $f_N(k)/f_N(0)$ and $P_N(k)/E_N$ are strictly increasing in $\varepsilon \in (0,1)$. In particular C_N/E_N is strictly increasing, the crossing ε_N exists and is unique, and $\varepsilon_N < \frac{1}{2}$. Also $P_N(k) \ge 2E_N$ for $\varepsilon \ge \frac{1}{2}$, and in particular $C_N \ge 2E_N$ there.*

Proof. We compare each path of M with the path that never flips, one step at a time. The probability of a path of M is a product of $N-1$ factors, namely $r_s(b_s)$ for an up step at time s and $1 - r_s(b_s)$ otherwise, where b_s is the state at time s. The path that never flips has probability $f_N(0) = (1-\varepsilon)^{N-1}$, which is one factor $1-\varepsilon$ for each step. So we divide each factor by $1-\varepsilon$. With $y = b_s/s \in [0,1)$ the two kinds of factors become $\frac{\varepsilon + (1-2\varepsilon)y}{1-\varepsilon}$ and $1 - y\frac{1-2\varepsilon}{1-\varepsilon}$, whose derivatives in ε are $\frac{1-y}{(1-\varepsilon)^2} > 0$ and $\frac{y}{(1-\varepsilon)^2} \ge 0$. Both factors are positive, since $r_s(b_s) = \varepsilon(1-y) + (1-\varepsilon)y$ lies strictly between 0 and 1. So the ratio for each path is nondecreasing in ε, and it is strictly increasing if the path has an up step. Summing over the paths with $k \ge 1$ up steps proves the claim for $f_N(k)/f_N(0)$. By Lemma 2.2, $P_N(k)/E_N = [f_N(k) + f_N(N-k)]/f_N(0)$, and both k and $N-k$ are at least 1. So the claim for $P_N(k)/E_N$ follows.

It remains to locate the crossing. The ratio C_N/E_N is continuous and strictly increasing in ε. As $\varepsilon \to 0$ it tends to 0, since $C_N \to 0$ (at $\varepsilon = 0$ the chain never leaves 0) and $E_N \to \frac{1}{2}$. For $\varepsilon \ge \frac{1}{2}$ every $r_s(b)$ and every $1 - r_s(b)$ lies in $[1-\varepsilon, \varepsilon]$, so each path of M has probability at least $(1-\varepsilon)^{N-1} = 2E_N$. Every m with $0 \le m \le N-1$ is the end of some path, so $f_N(m) \ge 2E_N$. Both k and $N-k$ lie in $\{1, \dots, N-1\}$, so $P_N(k) = \frac{1}{2}[f_N(N-k) + f_N(k)] \ge 2E_N$, and in particular $C_N \ge 2E_N$ there. So the ratio equals 1 at exactly one ε, and this ε lies in $(0, \frac{1}{2})$. □

2.1 First order, one early flip

When $\varepsilon = 0$ the chain M never leaves 0. After a first flip, and to first order in ε, it is Pólya's urn.

Lemma 2.4 (Pólya's urn)**.** *Let $2 \le v \le N$. At time v an urn holds one ball of one color and $v-1$ of the other. At each time $k = v, \dots, N-1$ we draw a ball uniformly and return it with one more ball of its color. Let $\beta_v(d)$ be the probability that at time N there are d balls of the first color. Then $\beta_v(d) = \binom{N-d-1}{v-2}/\binom{N-1}{v-1}$ for $1 \le d \le N-v+1$, and $\beta_v(d) = 0$ for other d. Also β_v is nonincreasing on $\{1, \dots, N-v+1\}$ and $\sum_d d\,\beta_v(d) = N/v$. For $1 \le d \le N-1$,*

$$\sum_{v=2}^{N} \beta_v(d) = \frac{N}{d(d+1)}, \qquad \sum_{v=2}^{N} (v-2)\,\beta_v(d) = \frac{2N(N-d-1)}{d(d+1)(d+2)}.$$

Proof. We compute the probability of one order of the draws and count the orders. At time k the urn holds k balls. So if s of them are of the first color, the next draw is of that color with probability s/k. Take one order of the $N-v$ draws in which i draws are of the first color. The draws of the first color contribute the numerators $1, 2, \dots, i$, the other draws contribute $v-1, v, \dots, N-i-2$, and the denominators are $v, v+1, \dots, N-1$. So this order has probability $\frac{i!\,(v-1)v\cdots(N-i-2)}{v(v+1)\cdots(N-1)} = \frac{(v-1)\,i!\,(N-i-2)!}{(N-1)!}$, whatever the order is. There are $\binom{N-v}{i}$ orders. With $d = i+1$ this gives $\beta_v(d) = a_d(v-1)\binom{N-v}{d-1}$ with $a_d = \frac{(d-1)!(N-d-1)!}{(N-1)!}$, and writing out the factorials shows that this is $\binom{N-d-1}{v-2}/\binom{N-1}{v-1}$. From the last form, $\beta_v(d)/\beta_v(d-1) = 1 - \frac{v-2}{N-d} \le 1$ for $2 \le d \le N-v+1$, so β_v is nonincreasing on its support. The fraction of balls of the first color is a martingale that starts at $1/v$. So the mean number of such balls at time N is N/v.

For the two sums we use the first form of $\beta_v(d)$ and put $i = N-v$, so that $v-1 = \binom{N-1-i}{1}$ and $(v-1)(v-2) = 2\binom{N-1-i}{2}$. The Chu–Vandermonde identity says that $\sum_i \binom{i}{a}\binom{N-1-i}{b} = \binom{N}{a+b+1}$ (sort the subsets of $\{1, \dots, N\}$ of size $a+b+1$ by their element number $a+1$, which we call $i+1$). With $a = d-1$ and $b = 1, 2$ it shows that the two sums are $a_d\binom{N}{d+1}$ and $2a_d\binom{N}{d+2}$, and these are the values in the statement. □

Define the hitting probability $V_s(b)$ from state b at time s by $V_N(b) = \mathbf{1}\{b = h\}$ and $V_s(b) = r_s(b)V_{s+1}(b+1) + (1 - r_s(b))V_{s+1}(b)$ for $s < N$. So $V_s(b) = \mathbb{P}_+(M_N = h \mid M_s = b)$ whenever $\mathbb{P}_+(M_s = b) > 0$, and $V_s(b)$ is a polynomial in ε. In particular it is defined at $\varepsilon = 0$, without conditioning on an event of probability zero. For $1 \le t \le h$ let

$$\pi_t = \beta_{t+1}(h) = \frac{t}{N-t}\prod_{i=1}^{t-1}\frac{h-i}{N-i}, \qquad c_1 = \sum_{t=1}^{h}\pi_t, \qquad w_t = \frac{\pi_t}{c_1},$$

so that $(w_t)_{1 \le t \le h}$ is a probability vector.

The first-order coefficient in (b) is known. With equal switching probabilities ε and one initial sensitive cell, the resistant-cell count of Bokes, Hlubinová and Singh [27, Section 2 and Eq. (1)] is M. Their first-switch and Pólya-urn argument gives $\varepsilon N/(m(m+1))$ as the first-order part of $f_N(m)$ [27, Eqs. (6)–(10)], hence $\frac{4\varepsilon}{N+2}$ at $m = h$. We retain the proof because (a) and (c) underpin the uniform bounds of Theorem 2.6 and Appendix A.

Proposition 2.5 (first order, as in [27, Eqs. (6)–(10)])**.** *Let $N \ge 2$ be even.*

(a) The center is $C_N = \sum_{t=1}^{h} \varepsilon(1-\varepsilon)^{t-1}V_{t+1}(1)$.

(b) The center C_N is a polynomial in ε with $C_N(0) = 0$ (extended as a polynomial to $\varepsilon = 0$). At $\varepsilon = 0$ we have $V_{t+1}(1) = \pi_t$, and $C_N'(0) = c_1 = \frac{4}{N+2}$ exactly.

(c) The weights satisfy $\sum_t w_t t = 3 - \frac{6}{h+2}$, and $w_t \le 2t2^{-t}$ for $1 \le t \le h$.

Proof. (a) We condition on the time of the first flip. Under $\mathbb{P}_+$ the first flip happens at step $t+1$ exactly when $M_s = 0$ for $s \le t$ and $M_{t+1} = 1$. Since $r_s(0) = \varepsilon$, this has probability $(1-\varepsilon)^{t-1}\varepsilon$, and by the Markov property the chain then ends at h with probability $V_{t+1}(1)$. A path with no flip ends at $0 \ne h$. So C_N is the sum of $\varepsilon(1-\varepsilon)^{t-1}V_{t+1}(1)$ over $1 \le t \le N-1$. Only $t \le h$ contribute. Indeed at time $t+1$ there are t steps equal to X_1, this number never decreases, and at the center it is h. So $V_{t+1}(1) = 0$ for $t > h$.

(b) Every transition probability is affine in ε, so C_N and $V_{t+1}(1)$ are polynomials in ε. By (a), $C_N(0) = 0$ and $C_N'(0) = \sum_t V_{t+1}(1)|_{\varepsilon=0}$. At $\varepsilon = 0$ we have $r_s(b) = b/s$. So from time $t+1$ on, M

counts the balls of the first color in the urn of Lemma 2.4 with $v = t+1$, and $V_{t+1}(1) = \beta_{t+1}(h) = \pi_t$. Since $\beta_v(h) = 0$ for $v > h+1$, Lemma 2.4 with $d = h$ gives $c_1 = \sum_{v=2}^{N} \beta_v(h) = \frac{N}{h(h+1)} = \frac{4}{N+2}$.

(c) In the same way the second sum of Lemma 2.4, with $d = h$ and $v = t+1$, gives $\sum_t (t-1)\pi_t = \frac{2N(h-1)}{h(h+1)(h+2)}$. Dividing by $c_1 = \frac{N}{h(h+1)}$ gives $\sum_t w_t(t-1) = \frac{2(h-1)}{h+2} = 2 - \frac{6}{h+2}$, and adding $\sum_t w_t = 1$ gives the first claim. For the second claim, each factor $\frac{h-i}{N-i}$ in π_t is at most $\frac{1}{2}$, so $\pi_t \le \frac{t}{N-t} 2^{1-t}$. Hence $w_t = \frac{N+2}{4}\pi_t \le \frac{N+2}{2(N-t)}\, t 2^{-t}$. For $t \le h$ we have $N - t \ge h$ and $\frac{N+2}{2h} = 1 + \frac{2}{N} \le 2$, so $w_t \le 2t2^{-t}$. □

To first order, one flip at step $t+1$ starts Pólya's urn with t balls of one color and one of the other. As $N \to \infty$, the latter fraction tends to Beta$(1, t)$, with density $t2^{1-t}$ at $\frac{1}{2}$. Thus $\sum_t t2^{1-t} = 4$ explains $4/(N+2)$, and $w_t \le 2t2^{-t}$ restricts this term to early flips.

2.2 Two-sided bounds on the center

Theorem 2.6. *Let $N \ge 2$ be even.*

(a) For $0 < \varepsilon < \frac{1}{2}$, $C_N \ge \frac{4\varepsilon}{N+2}(1-\varepsilon)^{2-12/(N+4)} \ge \frac{4\varepsilon}{N+2}(1-2\varepsilon)$.

(b) The bound $C_N \le \frac{4\varepsilon}{N+2} N^{2\varepsilon} e^{5200\varepsilon}$ holds whenever $0 < \varepsilon(1+\log N) \le \frac{1}{100}$.

In particular $\log(1-2\varepsilon) \le \log C_N - \log \frac{4\varepsilon}{N+2} \le \varepsilon(2\log N + 5200)$ in the range of (b).

The last, weaker bound in (a) also holds for $\frac{1}{2} \le \varepsilon < 1$, because its right side is not positive there.

Idea of the proof. Appendix A represents M at time s, after its first flip, as Pólya's urn with αs extra balls of each color, where $\alpha = \varepsilon/(1-2\varepsilon)$ (Lemma A.1). These pull the minority fraction toward $\frac{1}{2}$; dropping them gives (a). For (b), use an urn with $\kappa(s+10)$ extra balls, where $\kappa = 2\varepsilon/(1-4\varepsilon)$. Each step multiplies its hitting probability by at most about $e^{\kappa/s}$, whose product is about $e^{\kappa H_N} \approx N^{2\varepsilon}$.

2.3 The crossing

By Theorem 2.6, $C_N \approx 4x/N^2$ when $\varepsilon = x/N$ and x is of order $\log N$. Also $E_N = \frac{1}{2}(1-x/N)^{N-1} \approx \frac{1}{2}e^{-x}$. Setting them equal gives $x + \log x \approx 2\log N - \log 8$. So let

$$\Lambda_N = 2L - \log 8, \qquad G(x) = x + \log x,$$

and let g_N be the root of $G(g_N) = \Lambda_N$. It exists and is unique, since G increases from $-\infty$ to ∞ on $(0, \infty)$.

Theorem 2.7. *As $N \to \infty$ over even N, the following hold.*

(a) The crossing satisfies $x_N + \log x_N = \Lambda_N + O(L^2/N)$, so $x_N = g_N + O(L^2/N)$.

(b) The crossing satisfies $x_N = 2\log N - \log\log N - \log 16 + O(\log\log N/\log N)$.

Table 1 Elephant crossings x_N and the root g_N of $G(g) = \Lambda_N$. These values are numerical and are not used in any proof. We computed x_N in double precision from the recursion of Lemma 2.2, by Brent's method with an absolute tolerance of 10^{-12} in x, and we found g_N with a tolerance of about 10^{-15}.

N	x_N	g_N	$x_N/\log N$
50	3.940487	4.288636	1.007
400	7.629217	7.843768	1.273
1600	10.240565	10.340051	1.388
3200	11.546834	11.610464	1.431
12800	14.158845	14.182921	1.497
51200	16.778468	16.786946	1.547

Proof. (a) *Step 1. Both sides near the crossing.* Put $F(x) = \log C_N(x/N) - \log E_N(x/N)$ for $0 < x < N$, where $C_N(\varepsilon)$ and $E_N(\varepsilon)$ are the center and the end at flip probability ε. By Proposition 2.3, F is strictly increasing and x_N is its only zero. We show that

$$|F(x) - G(x) + \Lambda_N| \le \rho_N \quad (1 \le x \le 2\Lambda_N), \qquad \text{with } \rho_N = O(L^2/N). \tag{$*$}$$

Let $1 \le x \le 2\Lambda_N$ and $\varepsilon = x/N$. Then $\varepsilon \le 4L/N$, so for large N both parts of Theorem 2.6 apply, and together they say that $\log C_N = \log \frac{4\varepsilon}{N+2} + O(\varepsilon L)$. Here $\log \frac{4\varepsilon}{N+2} = \log x + \log 4 - 2L - \log(1 + 2/N)$, and $\log(1 + 2/N) \le 2/N \le 2xL/N$ because $x \ge 1$. For the end we use $\log(1 - y) = -y + O(y^2)$ for $0 \le y \le \frac{1}{2}$ in $\log E_N = -\log 2 + (N - 1)\log(1 - \frac{x}{N})$. So

$$\log C_N = \log x + \log 4 - 2L + O\Big(\frac{xL}{N}\Big), \qquad \log E_N = -\log 2 - x + O\Big(\frac{x^2}{N}\Big).$$

We subtract. Since $\log 4 + \log 2 = \log 8$, the main terms give $x + \log x - 2L + \log 8 = G(x) - \Lambda_N$, and both errors are $O(L^2/N)$ because $x \le 2\Lambda_N \le 4L$. This is $(*)$.

Keeping the constants gives an explicit ρ_N. Suppose that $400L(1+L) \le N$. Then $\varepsilon(1+\log N) \le 4L(1+L)/N \le \frac{1}{100}$, so the inequality after part (b) of Theorem 2.6 applies. With $\log(1-y) \ge -y-y^2$ for $0 \le y \le \frac{1}{2}$ and $x \le 4L$, the error term for $\log C_N$ above lies between $-(3 + 8L)/N$ and $(8L^2 + 20800L)/N$, and the one for $\log E_N$ lies between $-16L^2/N$ and $4L/N$. So $(*)$ holds with $\rho_N = (24L^2 + 20800L)/N$.

Step 2. Trap the zero. We have $G' = 1 + 1/x \ge 1$ and $G(g_N) = \Lambda_N$. So $G(x) - \Lambda_N < -\rho_N$ for $x < g_N - \rho_N$, and $G(x) - \Lambda_N > \rho_N$ for $x > g_N + \rho_N$. By $(*)$, $F(x) < 0$ for $1 \le x < g_N - \rho_N$ and $F(x) > 0$ for $g_N + \rho_N < x \le 2\Lambda_N$. Both ranges are nonempty for large N, because $\Lambda_N - \log \Lambda_N \le g_N \le \Lambda_N$ (note that $G(\Lambda_N - \log \Lambda_N) \le \Lambda_N \le G(\Lambda_N)$). Since F is increasing, its zero x_N lies between the two ranges, so $|x_N - g_N| \le \rho_N$. Then x_N lies in $[1, 2\Lambda_N]$, and $F(x_N) = 0$ in $(*)$ gives $|G(x_N) - \Lambda_N| \le \rho_N$. With the explicit ρ_N above, these two bounds hold for every even N with $L \ge 30$. Indeed $e^L \ge L^{12}/12!$ gives $24L^4 + 20800L^3 \le N$, so $400L(1 + L) \le N$ and $\rho_N \le 1/L^2$, and both ranges are nonempty because $g_N \ge \Lambda_N - \log \Lambda_N > 50$.

(b) We solve $g_N = \Lambda_N - \log g_N$ by one substitution. Since $\Lambda_N - \log \Lambda_N \le g_N \le \Lambda_N$, the ratio g_N/Λ_N is $1 + O(\log \Lambda_N/\Lambda_N)$, so $\log g_N = \log \Lambda_N + O(\log \Lambda_N/\Lambda_N)$. Also $\Lambda_N = 2 \log N - \log 8$ gives $\log \Lambda_N = \log 2 + \log \log N + O(1/\log N)$. So $g_N = 2 \log N - \log 8 - \log 2 - \log \log N + O(\log \log N/\log N)$, which is (b) for g_N, and (a) gives it for x_N. □

Table 1 gives crossings for six values of N of the form $50 \cdot 2^k$. The limit 2 of $x_N/\log N$ is approached slowly, because of the term $-\log \log N$.

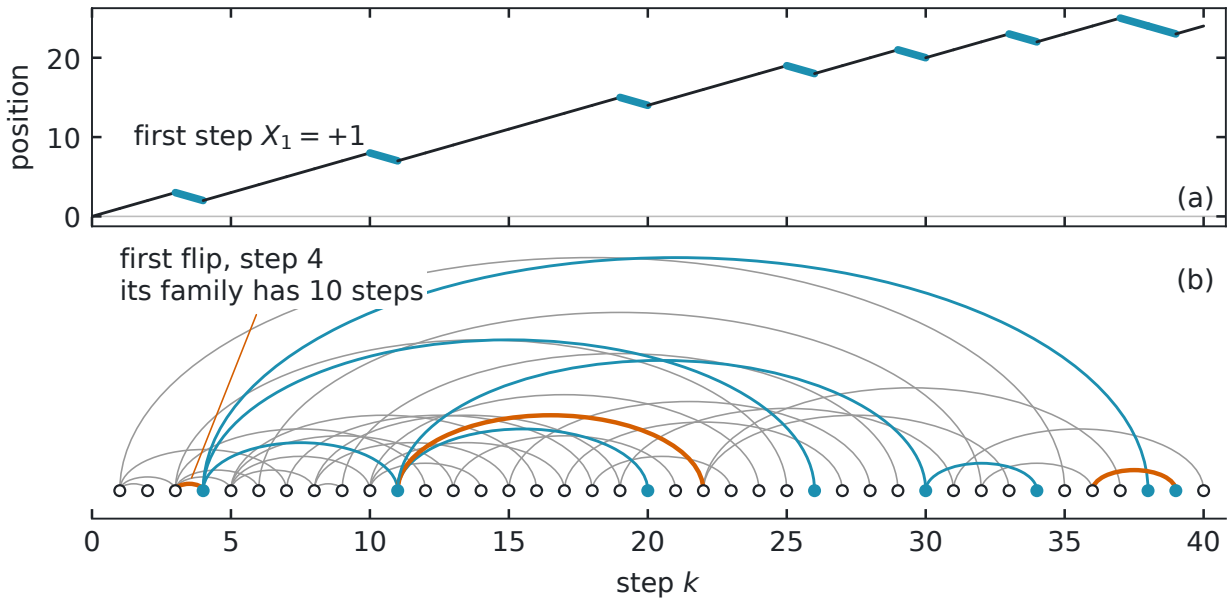


Fig. 1 One elephant walk with $N = 40$ steps and $\varepsilon = 0.07$. (a) The position against the step. The steps that disagree with the first step are drawn heavier. (b) The memory tree of the same walk. Step $k+1$ copies the step at the left end of its arc, and the arcs drawn heavier are flips. A step disagrees with the first step exactly when the path from it to step 1 crosses an odd number of flips (Lemma 3.1). The first flip, at step 4, turns its family of ten steps. The flip at step 22, inside that family, turns its own family of three steps back, and the flip at step 39 turns only that step

In Section S1, a lower bound for C_N matches Theorem 2.6(b) up to $1 + O(\varepsilon)$, and a further crossing term gives $x_N < g_N$ for large N. If z_N is the crossing value of T for the persistent walk, then $x_N - (2z_N - \log(2\pi))$ is $c_*/\log N$ to first order, with $c_* = \frac{1}{2}\log(2\pi) - \frac{1}{4}$.

3 The shape of the law at the crossing

In this section and Appendices B and C, "Assume (H)" means

$$N \geq 64, \qquad 0 < \varepsilon \leq \tfrac{1}{4}, \qquad x = N\varepsilon \geq 1. \tag{H}$$

Every other statement holds under its own hypotheses. Recall that $P_N(N-m) = \frac{1}{2}[f_N(m) + f_N(N-m)]$. Some statements also use odd N. There x_N is the value of x at which the two central values are as likely as the ends (it exists and is unique by Proposition 3.21). For a mode $a^* \geq N/2$ of P_N we put $m^* = N - a^*$, and m_h is the largest mode of f_N.

3.1 Families and the first flip

For $k_0 \geq 1$, the chain $(M_k^{[k_0]})_{k \geq k_0}$ starts at $M_{k_0}^{[k_0]} = 0$ with transitions $r_k(b)$. It counts disagreeing steps when the first k_0 steps are forced to agree; $M^{[1]}$ has the law of M. Put $Y_d = M_d^{[1]}$ and $Z_t^{(j)} = M_t^{[j-1]}$ for $j \geq 2$, $t \geq j-1$. In a random recursive tree on $[N] = \{1, \ldots, N\}$, vertex $k+1$ joins a uniform vertex of $[k]$, independently for $k = 1, \ldots, N-1$. The family $\mathcal{T}_v$ comprises v and its descendants.

Lemma 3.1 (tree and marks)**.** *Let $\mathcal{T}$ be the tree on $[N]$ in which the parent of $k+1$ is U_k. For $2 \le v \le N$ let $\mu_v = 1$ if step v flips the step it copies, and $\mu_v = 0$ otherwise. Then $\mathcal{T}$ is a random recursive tree, the marks μ_v are i.i.d. Bernoulli(ε) and independent of $\mathcal{T}$, and $X_v = X_1(-1)^{\sigma(v)}$, where $\sigma(v)$ is the number of marks on the path from v to the root, the root excluded. So $M_N = \#\{v : \sigma(v) \text{ odd}\}$. If the marks on $2, \dots, k_0$ are set to 0, then the number of odd vertices in $[k]$, $k \ge k_0$, is a copy of the chain $M^{[k_0]}$.*

Proof. The parent U_k of $k+1$ is uniform on $[k]$ and independent of the earlier choices, and the marks are the flip indicators. The path from $k+1$ to the root is $k+1$ followed by the path from U_k, so $\sigma(k+1) = \mu_{k+1} + \sigma(U_k)$ and, by induction on k, $X_{k+1} = X_{U_k}(-1)^{\mu_{k+1}} = X_1(-1)^{\sigma(k+1)}$. Now set $\mu_2 = \cdots = \mu_{k_0} = 0$. If b of the vertices in $[k]$ are odd, $k \ge k_0$, then $k+1$ is odd with conditional probability $(1-\varepsilon)\frac{b}{k} + \varepsilon(1-\frac{b}{k}) = r_k(b)$, whatever else is known about $[k]$. □

A single flip at v turns its family (Figure 1). For $k \ge v$, a family $\mathcal{T}_v$ containing s vertices of $[k]$ gains vertex $k+1$ with probability s/k. Lemma 2.4 gives $|\mathcal{T}_v|$ law β_v, mean N/v, and an average $N/(d(d+1))$ families of size d. If v alone is marked, $\mathcal{T}_v$ is the odd set. At fixed N, multiple marks have probability $O(\varepsilon^2)$ and no mark gives $M_N = 0$. Thus the first-order part of $f_N(m)$ in ε is $\varepsilon \sum_v \beta_v(m) = x/(m(m+1))$ for $1 \le m \le N-1$: [27, Eq. (10)] with one initial cell, also proved in Lean for M (Section 5).

To go beyond first order we cut the tree at the first flip.

Lemma 3.2 (splitting)**.** *Let $\mathcal{T}$ be a random recursive tree on $[N]$, let $2 \le v \le N$ and let $S \subseteq [N]$ with $\min S = v$. Given that $\mathcal{T}_v$ has vertex set S, the restrictions of $\mathcal{T}$ to S and to $[N] \setminus S$, relabeled in increasing order, are independent random recursive trees. They are independent of the parent of v, which is uniform on $[v-1]$.*

Proof. The tree $\mathcal{T}$ is uniform over the $(N-1)!$ increasing trees on $[N]$. Its family $\mathcal{T}_v$ has vertex set S exactly when every $u \in S \setminus \{v\}$ has its parent in S and every $u \notin S$ with $u \ge 2$ has its parent outside S. So, given this event, each element of S other than v picks its parent freely among the smaller elements of S, v picks any element of $[v-1]$, and each element of $[N] \setminus S$ other than 1 picks freely among the smaller elements of $[N] \setminus S$. This is the claim. □

Lemma 3.3 (the first flip)**.** *Let $V = \min\{v \ge 2 : \mu_v = 1\}$, with $V = \infty$ and $|\mathcal{T}_V| = 0$ if there is no mark. If $V < \infty$, let $\bar{Y}$ be the number of vertices u of $\mathcal{T}_V$ with $\sigma(u)$ even, and let $\bar{Z}$ be the number of odd vertices outside $\mathcal{T}_V$. Then for $2 \le j \le N$, $1 \le d \le N-j+1$ and integers y, z,*

$$\mathbb{P}\big(V = j,\, |\mathcal{T}_V| = d,\, \bar{Y} = y,\, \bar{Z} = z\big) = \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)\,\mathbb{P}(Y_d = y)\,\mathbb{P}\big(Z^{(j)}_{N-d} = z\big).$$

In particular $\mathbb{P}(V = j,\, |\mathcal{T}_V| = d) = \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)$, and for $m \ge 1$

$$f_N(m) = \sum_{j=2}^{N} \varepsilon(1-\varepsilon)^{j-2} \sum_{d=1}^{N-j+1} \beta_j(d)\,\mathbb{P}\big(d - Y_d + Z^{(j)}_{N-d} = m\big),$$

where Y_d and $Z^{(j)}_{N-d}$ are independent.

Proof. Marks and tree are independent, and $|\mathcal{T}_j|$ has the law β_j, which gives the law of $(V, |\mathcal{T}_V|)$. Condition on $V = j$ and on the vertex set S of $\mathcal{T}_j$, with $|S| = d$. The marks on $j+1, \dots, N$ are still i.i.d., and by Lemma 3.2 the trees on S and on $[N] \setminus S$ are independent random recursive trees. The vertices $1, \dots, j-1$ carry no mark, so j is odd, and $u \in S$ is even exactly when the

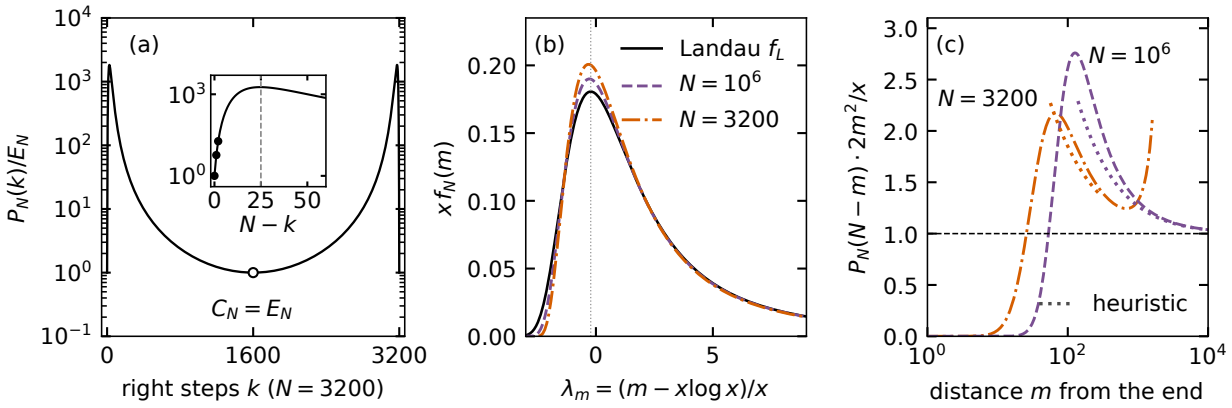


Fig. 2 The elephant walk at its crossing. (a) The law P_N for $N = 3200$ at its crossing $x = 11.5468$, divided by the end E_N. The center equals the end, and the law rises from each end to a mode and falls to the center. The inset shows the last 60 values, with dots at N, $N-1$ and $N-2$. The end is below its neighbor, and the modes are at distance 25 from each end (dashed). (b) The law of M_N, scaled as $xf_N(m)$, against $\lambda_m = (m - x\log x)/x$, with the Landau density f_L. The dotted line is its mode λ_{L}. (c) The tail $P_N(N-m)$ divided by $x/(2m^2)$. The dashed line is the limit 1 of the tail theorem in Theorem S2.1, and the dotted curves are the heuristic correction $1 + x(2\log m + 2\gamma - 3)/m$ for $2x\log x \le m \le N/x$. The rise of the $N = 3200$ curve near $m = N/2$ is the mirror term $f_N(N-m)$. At $N = 10^6$ we use $x = g_N = 22.4407$

path from u up to j, with j excluded, has an odd number of marks. So by Lemma 3.1 for the tree on S rooted at j, the count $\bar{Y}$ has the law of Y_d. Paths from outside S stay outside S, and the tree on $[N] \setminus S$ has the unmarked vertices $1, \dots, j-1$ first. So by Lemma 3.1 the count $\bar{Z}$ has the law of $Z^{(j)}_{N-d}$. The two counts are functions of the two trees and of the marks on the two disjoint sets S and $[N] \setminus S$, so they are independent, and their joint law depends on S only through d. This gives the joint law. On $\{V = j\}$ we have $M_N = d - \bar{Y} + \bar{Z}$, and $M_N = 0$ if there is no mark. Summing over j, d and the pairs (y, z) with $d - y + z = m$ gives the formula for $f_N(m)$. □

Only the law of each pair $(Y_d, Z^{(j)}_{N-d})$ enters the lemma. Henceforth $(Y_d)_{d\ge1}$ is a copy of $M^{[1]}$, and $(Z^{(j)}_t)_{t\ge j-1}$ is a copy of $M^{[j-1]}$, independent of Y and $(V, |\mathcal{T}_V|)$. The latter walk starts with $j-1$ steps agreeing with X_1.

Lemma 3.4 (unimodality given the first step [10, Section 2, proof of Theorem 1.1])**.** *The function f_N is unimodal.*

The proof of [10, Theorem 1.1] treats every N and $p = 1 - \varepsilon \in [0, 1]$, for $N - 2M_N$ given $X_1 = +1$; a Lean proof is recorded in Section 5. Thus f_N is nondecreasing on $\{0, \dots, m_h\}$ and nonincreasing on $\{m_h, \dots, N\}$. Its fair mixture P_N can have two humps when the mode of f_N is far from the center.

3.2 The end and its neighbor

The heuristic threshold is $x > 2$ (Figure 2(a), inset). Given $X_1 = +1$, the end needs no flip, whereas $N-1$ needs one flip that is never copied. A flip at step j is never copied with probability about j/N; summing $\varepsilon j/N$ over j gives the factor $x/2$.

Theorem 3.5 (endpoint dip). *Let $N \geq 2$ and $0 < \varepsilon < 1$, and put $\rho_1 = P_N(N-1)/E_N$. Then*

$$\rho_1 > \frac{x}{2(1-\varepsilon)}.$$

If moreover $N \geq 3$ and $\varepsilon \leq \frac{1}{2}$, then

$$\rho_1 \leq \frac{x(1+1/N)^2}{2-3\varepsilon} + \frac{\varepsilon}{1-\varepsilon}.$$

Hence the end is strictly below its neighbor if $x \geq 2(1-\varepsilon)$. If $N \geq 3$, $\varepsilon \leq \frac{1}{2}$ and the right side of the second bound is below 1, the end is strictly above its neighbor.

Proof. By Lemma 2.2, $\rho_1 = \big(f_N(1) + f_N(N-1)\big)/f_N(0)$ with $f_N(0) = (1-\varepsilon)^{N-1}$. We bound the two terms in turn.

The term $f_N(1)$. On the event $M_N = 1$ the chain moves up exactly once, say from time $j-1$ to time j with $2 \leq j \leq N$. Before that it stays at 0 for $j-2$ steps, each with probability $1 - r_t(0) = 1 - \varepsilon$. The move up has probability ε. After it the chain stays at 1 at the times $k = j, \dots, N-1$, each with probability $1 - r_k(1) = (1-\varepsilon)(1-c'/k)$, where $c' = (1-2\varepsilon)/(1-\varepsilon) \leq 1$. Multiplying and summing over j gives

$$f_N(1) = \varepsilon(1-\varepsilon)^{N-2} \sum_{j=2}^{N} \vartheta_j, \qquad \vartheta_j = \prod_{k=j}^{N-1} \Big(1 - \frac{c'}{k}\Big).$$

For the lower bound we use $c' \leq 1$. Then $\vartheta_j \geq \prod_{k=j}^{N-1}(1-1/k) = (j-1)/(N-1)$, and these lower bounds sum to $N/2$ over $2 \leq j \leq N$. So $f_N(1)/f_N(0) \geq x/(2(1-\varepsilon))$. For the upper bound let $N \geq 3$ and $\varepsilon \leq \frac{1}{2}$, so that $c' \geq 0$. Then $1 - c'/k \leq e^{-c'/k}$ and $\sum_{k=j}^{N-1} 1/k \geq \log(N/j)$ give $\vartheta_j \leq (j/N)^{c'}$. As $(t/N)^{c'}$ is nondecreasing in t, we can compare the sum with an integral, namely

$$\sum_{j=2}^{N} \vartheta_j \leq \int_2^{N+1} \Big(\frac{t}{N}\Big)^{c'} dt \leq \frac{N(1+1/N)^{c'+1}}{c'+1} \leq \frac{N(1+1/N)^2(1-\varepsilon)}{2-3\varepsilon},$$

using $c' \leq 1$ and $1/(c'+1) = (1-\varepsilon)/(2-3\varepsilon)$. So $f_N(1)/f_N(0) \leq x(1+1/N)^2/(2-3\varepsilon)$.

The term $f_N(N-1)$. The event $M_N = N-1$ needs an up move at every time. The first has probability $r_1(0) = \varepsilon$, and the one at time $k \geq 2$ has probability $r_k(k-1) = 1 - \varepsilon - (1-2\varepsilon)/k$. This lies strictly between 0 and 1, as every $r_k(b)$ with $0 < b < k$ does, and it lies in $[\frac{1}{2}, 1-\varepsilon]$ when $\varepsilon \leq \frac{1}{2}$. So $f_N(N-1) > 0$, and $f_N(N-1) \leq \varepsilon(1-\varepsilon)^{N-2}$ when $\varepsilon \leq \frac{1}{2}$. This makes the lower bound strict and gives the term $\varepsilon/(1-\varepsilon)$ in the upper bound. (For $N = 2$ the values 1 and $N-1$ coincide, and the argument is the same.) □

Corollary 3.6 (the dip at the crossing). *Let $N \geq 2$ be even and $0 < \varepsilon < 1$, and suppose that $x \geq 2(1-\varepsilon)$ and $C_N \leq E_N$. Then $C_N \leq E_N < P_N(N-1)$, and P_N is not unimodal. In particular, if $x_N \geq 2(1-\varepsilon_N)$, then at the crossing $E_N = C_N < P_N(N-1)$, and P_N is not unimodal.*

Proof. Theorem 3.5 gives $P_N(N-1) > E_N \geq C_N$. Suppose that P_N is nondecreasing on $\{0, \dots, a_0\}$ and nonincreasing on $\{a_0, \dots, N\}$. It is symmetric about h, so we may replace a_0 by $N - a_0$ and assume $a_0 \leq h$. Then $P_N(a) \leq C_N$ for every $a \geq h$, including $a = N-1$, which is a contradiction. □

By Proposition 2.3, $C_N \le E_N$ holds exactly when $\varepsilon \le \varepsilon_N$, and $x \ge 2(1-\varepsilon)$ says that $\varepsilon \ge 2/(N+2)$. So the corollary covers the ε with $2/(N+2) \le \varepsilon \le \varepsilon_N$. In particular the condition $x_N \ge 2(1-\varepsilon_N)$ says that $\varepsilon_N \ge 2/(N+2)$. By Proposition 2.3 it holds exactly when $C_N \le E_N$ at $\varepsilon = 2/(N+2)$. We now show that this is so for every even $N \ge 8$.

Proposition 3.7 (the hypothesis holds from $N = 8$ on, computer-assisted)**.** *For every even $N \ge 8$ we have $C_N < E_N$ at $\varepsilon = 2/(N+2)$, and hence $x_N > 2(1-\varepsilon_N)$. So for every even $N \ge 8$, at the crossing $E_N = C_N < P_N(N-1)$, and P_N is not unimodal. For $N = 2$, 4 and 6 the condition $x_N \ge 2(1-\varepsilon_N)$ fails, and no ε meets the hypotheses of Corollary 3.6.*

Proof. The range $N \ge 1688$. Take $\varepsilon = 2/N$. Then $\varepsilon(1+\log N) \le \frac{1}{100}$, so Theorem 2.6(b) gives $C_N \le \frac{8}{N(N+2)} N^{4/N} e^{10400/N}$. The right side decreases in N and is below $1.4 \cdot 10^{-3}$ at $N = 1688$. Since $\log(1-y) \ge -y - y^2$ for $0 \le y \le \frac{1}{2}$, we also have $E_N = \frac{1}{2}(1-\frac{2}{N})^{N-1} \ge \frac{1}{2} e^{-2-2/N} > 0.06$. So $C_N < E_N$ at $\varepsilon = 2/N$, and Proposition 2.3 gives $\varepsilon_N > 2/N$. Hence $C_N < E_N$ at $\varepsilon = 2/(N+2)$ as well.

The range $8 \le N < 1688$. For $N = 8$ the ratio C_N/E_N at $\varepsilon = 2/(N+2)$ is exactly $\frac{67903}{71680} < 1$. For even $10 \le N \le 2000$ the script `certify_center.py` encloses ε_N in an interval, by computing $C_N - E_N$ in ball arithmetic at its two ends and using Proposition 2.3, and it checks by an exact rational comparison that the lower end of the enclosure lies above $2/(N+2)$. So $\varepsilon_N > 2/(N+2)$, and Proposition 2.3 gives $C_N < E_N$ at $\varepsilon = 2/(N+2)$. The ratio of the lower end to $2/(N+2)$ is at least 1.107, with its minimum at $N = 10$.

The three smallest N. At $\varepsilon = 2/(N+2)$ the ratio C_N/E_N is exactly 2, $\frac{5}{3}$ and $\frac{508}{405}$ for $N = 2$, 4 and 6, so $\varepsilon_N < 2/(N+2)$ there. These ratios and the one for $N = 8$ are computed in exact rational arithmetic by `certify_elephant.py`. The last claim of the proposition about $N \ge 8$ is Corollary 3.6. □

3.3 The bulk

The steps that disagree with X_1 lie in the flipped families. Recall the compound Poisson law CP_N of Section 1.2. On average $x/(d(d+1))$ flipped vertices have a family of size d. If no flipped step has a flipped descendant, which is likely when $x^2 \log N/N$ is small, then M_N is the total size of the flipped families, and this is close to the compound Poisson sum.

The CP law and Landau limit are known (Section 1.3). We give explicit total variation and local bounds, the latter with jumps truncated at $N-1$. Kölbig and Schorr [37] compute f_L and its mode; Lemma 3.10 adds a certified enclosure.

Theorem 3.8 (compound Poisson bulk)**.** *For $N \ge 1$ and $0 < \varepsilon < 1$,*

$$\mathrm{TV}(f_N, \mathrm{CP}_N) \le \delta_\Sigma := \varepsilon^2(NH_N - 2N + 1) + 2\varepsilon^2(N-1) + 4\varepsilon^2 \sum_{k=0}^{N-2} (H_k^2 + H_k).$$

Also $\delta_\Sigma \le \delta_B := \frac{x^2 \log N}{N} + \frac{4x^2}{N} e^{4x^2/N} + \frac{4x^2(H_N^2 + H_N)}{N}$.

Here $\mathrm{TV}(P,Q) = \frac{1}{2}\sum_k |P(k) - Q(k)|$, so the bound holds for every mass and event, and $\delta_\Sigma = O(x^2 \log^2 N/N)$. Appendix C.1 couples marks with Poisson variables and reveals parents successively. Changing one parent affects family sizes along only two root paths, whose depth second moments have order $\log^2 N$.

Families of size up to x contribute nearly $x \log x$; those of order x give fluctuations of order x. Centering M_N at $x \log x$ and dividing by x gives λ_m and the index-one stable density f_L.

Appendix C shows that e^B is the characteristic function of a law of class L, so f_L is unimodal by [38].

Theorem 3.9 (Landau limit). *(a) Let $x \geq 1$ and $N \geq 4x$. For every integer m, with $\mathrm{CP}_N(m) = 0$ for $m < 0$,*

$$\bigl|x\,\mathrm{CP}_N(m) - f_L(\lambda_m)\bigr| \leq E_0 := \frac{889 + 165\log x}{x} + \frac{6.6\,x}{N} \leq 889\Bigl(\frac{1+\log x}{x} + \frac{x}{N}\Bigr).$$

(b) Let $0 < \varepsilon \leq \frac{1}{4}$ and $x \geq 1$. For every integer $m \geq 0$,

$$\bigl|x f_N(m) - f_L(\lambda_m)\bigr| \leq E_0 + x\delta_\Sigma \leq 889\Bigl(\frac{1+\log x}{x} + \frac{x}{N}\Bigr) + x\delta_B.$$

(c) Let $N_n \geq 1$ and $0 < \varepsilon_n < 1$ for $n \geq 1$, and put $x_n = N_n\varepsilon_n$. If $x_n \to \infty$ and $x_n^2 \log^2 N_n/N_n \to 0$, then $(M_{N_n} - x_n \log x_n)/x_n$ converges in law to the law with density f_L.

At the crossing, the disagreement count is about $x \log x$, with Landau fluctuations of order x (Figure 2(b)); Theorem 2.7 supplies both conditions of (c). For the untruncated law, (c) is Pakes' theorem [28, p. 993]. Bertoin proves an analogue for vertices cut from a random recursive tree's root in another regime [31, Proposition 1]. Thus (c) is new only in its transfer to the walk; we know no earlier form of (a) or (b).

Appendix C proves the local bound by expanding the compound Poisson exponent at zero and applying Fourier inversion. The expansion gives the centering $x \log x$, and its remainder gives the rate.

To locate the modes we need to know that f_L is curved at its maximum.

Lemma 3.10 (the mode of the Landau density, computer-assisted). *(a) The density f_L has exactly one mode λ_{L}, and $-0.22278299 < \lambda_{\mathrm{L}} < -0.22278297$. It is nondecreasing on $(-\infty, \lambda_{\mathrm{L}}]$ and nonincreasing on $[\lambda_{\mathrm{L}}, \infty)$.*

(b) The derivative vanishes at the mode, $f_L'(\lambda_{\mathrm{L}}) = 0$, and $-0.0792 \leq f_L''(\lambda) \leq -0.0786$ for $-0.2229 \leq \lambda \leq -0.2227$.

(c) For every real λ, $f_L(\lambda_{\mathrm{L}}) - f_L(\lambda) \geq \kappa_L \min\{1, (\lambda - \lambda_{\mathrm{L}})^2\}$ with $\kappa_L = 2.6 \cdot 10^{-10}$.

The proof is in Appendix C.3, and it is computer-assisted. Lean proves the lemma as well, with the unimodality of f_L that [38] gives as its only input (Section 5), so the computation in ball arithmetic is an independent check.

Corollary 3.11 (where the modes are). *(a) Assume (H). Let κ_L be as in Lemma 3.10(c), let $E_1 = 889\bigl(\frac{1+\log x}{x} + \frac{x}{N}\bigr) + x\delta_B$ be the bound of Theorem 3.9(b), and put*

$$E' = E_1 + \frac{16x^2 H_N}{N^2} + \frac{\|f_L''\|_\infty}{8x^2}.$$

If $E' < \kappa_L/2$ and $m_h \leq N/4$, then every mode $a^* \geq N/2$ of P_N satisfies $|m^* - x\log x - \lambda_{\mathrm{L}} x| \leq x\sqrt{2E'/\kappa_L}$, and so does every mode of f_N.

(b) *There are K and N_2 such that the following holds for every $N \ge N_2$, even or odd, and every ε with $\log N \le x \le 2\log N$. Every mode $a^* \ge N/2$ of P_N has*

$$\left|m^* - x\log x - \lambda_{\mathrm{L}} x\right| \le K\sqrt{x\log x},$$

and so has every mode of f_N in place of m^*. In particular this holds at the crossing, with $x = x_N$, for every large even N (Theorem 2.7(b)).

In (a) every constant is explicit. (Here $\|f_L''\|_\infty \le 16/\pi^4$, since $|e^{B(t)}| = e^{-\pi|t|/2}$.) We do not compute K and N_2 in (b).

So each hump of the law at the crossing sits about $x\log x - 0.2228\,x$ steps from its end. The proof is in Appendix C.4. It compares f_N at a mode with f_N at the integer nearest to $x\log x + \lambda_{\mathrm{L}} x$, and uses Theorem 3.9(b) and Lemma 3.10(c).

Remark 3.12 (how good the bulk approximation is). The bounds of this subsection are far from sharp at the sizes we can compute. At $N = 10^6$ and $x = 22.4407$ the bound δ_B is still about 0.46, but there the computed law has $|f_N(m)/\mathrm{CP}_N(m) - 1| \le 3.6 \cdot 10^{-4}$ for $1 \le m \le 10^4$. At every tested crossing m^* equals the mode of CP_N. The proof of Corollary 3.11(b) needs the bound of Theorem 3.9(b) to be below $\kappa_L/2$, where $\kappa_L = 2.6 \cdot 10^{-10}$ is the constant of Lemma 3.10(c). With the constant 889 this needs $x > 2 \cdot 10^{14}$, and numerically the best κ_L is about 0.026. These numbers are computed in double precision and are not used in any proof.

3.4 The tail

Beyond the humps the law falls like $x/(2m^2)$. We prove in Theorem S2.1 that $P_N(N-m) = \frac{x}{2m^2}(1+o(1))$ for even N with $x \ge 1$ and $x\log^2 N/N \to 0$, uniformly for $\omega_N x\log^2(x+1) \le m \le \delta'_N N$, for every $\omega_N \to \infty$ and every $\delta'_N \to 0$.

At $N = 10^6$, $x = 22.4407$ and $m = 10^4$ the computed ratio $P_N(N-m)/(x/(2m^2))$ is 1.03777. Heuristically, at moderate m the law CP_N suggests $f_N(m)\,m(m+1)/x \approx 1 + x(2\log m + 2\gamma - 3)/m$, which we do not prove. This factor is still about 2 near $m = 70$ when $N = 3200$ (Figure 2(c)).

3.5 The shape at every even crossing

This subsection proves that the second difference of P_N changes sign at most four times. At every even crossing with $N \ge 8$ this gives the unique interior minimum and the two intervals of modes, with no asymptotic input.

Throughout this subsection and the next, $0 < \varepsilon < \frac{1}{2}$ and $c = 1 - 2\varepsilon$, so that $r_t(b) = \varepsilon + cb/t$. For $t \ge 1$ and every integer a we write $P_t(a) = \mathbb{P}(A_t = a)$, which is 0 unless $0 \le a \le t$, and

$$D_t(a) = P_t(a+1) - 2P_t(a) + P_t(a-1).$$

We also put $s_t = D_t(1) = P_t(0) - 2P_t(1) + P_t(2)$ for $t \ge 2$.

Lemma 3.13 (recurrence for the second difference)**.** *For every $t \ge 1$ and every integer a,*

$$P_{t+1}(a) = \big(1 - r_t(a)\big)P_t(a) + r_t(a-1)\,P_t(a-1), \tag{1}$$

$$D_{t+1}(a) = \big(1 - r_t(a+1)\big)D_t(a) + r_t(a-2)\,D_t(a-1). \tag{2}$$

Also $P_t(a) = P_t(t-a)$ and $D_t(a) = D_t(t-a)$. The same two recurrences hold with $f_t(m)$ in place of $P_t(a)$.

Proof. Given $A_t = a$, the copied step is $+1$ with probability a/t, so step $t+1$ is $+1$ with probability $(1-\varepsilon)\frac{a}{t} + \varepsilon(1-\frac{a}{t}) = r_t(a)$. This gives (1) for $0 \le a \le t+1$, where at $a = 0$ and at $a = t+1$ the formula has one extra term, namely $r_t(-1)P_t(-1)$ or $(1 - r_t(t+1))P_t(t+1)$, and this term is 0 because $P_t(-1) = P_t(t+1) = 0$. For other a every term vanishes. The symmetry of P_t is Lemma 2.2, and it gives the symmetry of D_t. The chain M of Lemma 2.2 has the same transition probabilities, so (1) holds for f_t as well.

For (2) put $\varsigma = c/t$, so that $r_t(b+1) = r_t(b) + \varsigma$. We insert (1) at $a+1$, a and $a-1$ into the definition of $D_{t+1}(a)$ and collect the coefficients of $P_t(a+1)$, $P_t(a)$, $P_t(a-1)$ and $P_t(a-2)$. They are

$$1 - r_t(a+1), \qquad 3r_t(a) - 2, \qquad 1 - 3r_t(a-1), \qquad r_t(a-2).$$

The right side of (2) has the coefficients $1-r_t(a+1)$, $-2+2r_t(a+1)+r_t(a-2)$, $1-r_t(a+1)-2r_t(a-2)$ and $r_t(a-2)$. The second ones agree because $2r_t(a+1) + r_t(a-2) = 3r_t(a)$, and the third ones agree because $r_t(a+1) + 2r_t(a-2) = 3r_t(a-1)$. □

Trinas and Valle [12, (2.2)] use (1), the forward equation of Lemma 2.2, for a superdiffusive local limit theorem. Their higher-difference recurrence [12, (4.10)] supports a certified fixed-memory study of log-concavity. We use (2) for signs: its coefficients are positive for $2 \le a \le t-1$, so sign changes enter only through the two edges. The next lemma gives the required variation diminishing property.

For a finite list z of real numbers let $\mathrm{sp}(z)$ be its sign pattern. We delete the zeros of z, replace each entry by its sign, and merge equal neighbors. So $\mathrm{sp}(z)$ is a list of alternating signs. For two such lists we write $p \preceq q$ if p is a subsequence of q. We put $\Pi_5 = (+,-,+,-,+)$ and $\Pi_3 = (+,-,+)$.

Lemma 3.14 (one step)**.** *Let $\zeta = (\zeta_1, \dots, \zeta_n)$ be real, and let $\eta_i = A_i\zeta_i + B_i\zeta_{i+1}$ for $1 \le i \le n-1$ with $A_i \ge 0$ and $B_i \ge 0$. Let $w = (\zeta_1, \eta_1, \zeta_2, \eta_2, \dots, \eta_{n-1}, \zeta_n)$. Then $\mathrm{sp}(w) = \mathrm{sp}(\zeta)$ and $\mathrm{sp}(\eta) \preceq \mathrm{sp}(\zeta)$.*

We prove this in Appendix E.1.

Lemma 3.15 (the sequence next to the end)**.** *For every $t_1 \ge 2$ we have $\mathrm{sp}(s_2, s_3, \dots, s_{t_1}) \preceq \Pi_3$. Also, for every $0 \le \varepsilon \le 1$, the numbers s_3 and s_4 do not both vanish.*

We prove this in Appendix E.2.

Theorem 3.16 (at most five blocks)**.** *Let $N \ge 3$ and $0 < \varepsilon < \frac{1}{2}$. Then*

$$\mathrm{sp}\big(D_N(1), D_N(2), \dots, D_N(N-1)\big) \preceq (+,-,+,-,+).$$

So the second difference of P_N changes sign at most four times on $\{1, \dots, N-1\}$. The same holds for the second difference of f_N, and for f_N it holds for $N = 2$ as well.

We prove this in Appendix E.3.

A negative-to-positive change at $a_0 < a_1$ and its reflection give $-,+,+,-$ at $a_0 < a_1 < N - a_1 < N - a_0$. A negative second difference between them would give the fifth alternating sign forbidden by Theorem 3.16. For a zero, trace (2) backwards: positivity of its coefficients forces two zero parents, unless the parents have opposite signs, again giving five alternating signs. Reaching $t = 5$ forces $s_3 = s_4 = 0$, contrary to Lemma 3.15. This proves a whole convex stretch.

Theorem 3.17 (no holes)**.** *Let $N \ge 5$ and $0 < \varepsilon < \frac{1}{2}$. Let $1 \le a_0 < a_1 \le N/2$ be integers with $D_N(a_0) < 0 < D_N(a_1)$. Then $D_N(a) > 0$ for every $a_1 \le a \le N - a_1$.*

We prove this in Appendix E.4.

Lemma 3.18 (the center from no holes)**.** *Let $N \geq 6$ be even and $0 < \varepsilon < \frac{1}{2}$. If $P_N(1) > \max\{E_N, C_N\}$, then $P_N(a) > C_N$ for $1 \leq a \leq N-1$, $a \neq h$, and $D_N(h) > 0$.*

Proof. Suppose that $P_N(a) \leq C_N$ for some $1 \leq a < h$, and let k be the first minimizer of P_N on $\{1, \ldots, h\}$. Then $2 \leq k < h$ and $P_N(k-1) > P_N(k) \leq P_N(k+1)$, so $D_N(k) > 0$. The first maximizer a_0 on $\{0, \ldots, k\}$ has value at least $P_N(1)$, which exceeds both endpoint values of this interval. Thus $1 \leq a_0 < k$ and $D_N(a_0) < 0$. Theorem 3.17 gives $D_N(a) > 0$ for $k \leq a \leq N-k$.

Put $\delta(a) = P_N(a+1) - P_N(a)$. These first differences are strictly increasing on $\{k-1, \ldots, N-k\}$, and symmetry gives $\delta(h) = -\delta(h-1)$. Hence $\delta(h-1) < 0$, and every difference from k to $h-1$ is negative. This makes $P_N(k) > P_N(h)$, a contradiction. Reflection proves the other half, and $D_N(h) = 2(P_N(h-1) - P_N(h)) > 0$. □

Corollary 3.19 (the shape at every even crossing)**.** *For every even $N \geq 8$, at the crossing,*

$$P_N(a) > C_N = E_N \quad (1 \leq a \leq N-1,\ a \neq h), \qquad D_N(h) > 0.$$

There are integers $1 \leq l \leq r < h$ such that P_N strictly increases from 0 to l, is constant from l to r, and strictly decreases from r to h. Its two intervals of global maximizers are $\{l, \ldots, r\}$ and $\{N-r, \ldots, N-l\}$.

Proof. At the crossing $P_N(1) > E_N = C_N$ by Proposition 3.7, so Lemma 3.18 gives the minimum and central curvature. For the humps, let $p = \mathrm{sp}(D_N(1), \ldots, D_N(h))$. It ends in $+$, and symmetry makes the full pattern p followed by its reverse, with the central positive blocks merged. If p has j blocks, the full pattern has $2j - 1 \leq 5$ by Theorem 3.16. Thus p is a subsequence of $(+, -, +)$.

The first differences $\delta(a) = P_N(a+1) - P_N(a)$ on $0 \leq a < h$ therefore have at most three successive monotonic stretches, namely nondecreasing, nonincreasing, nondecreasing. Their first value is positive, and their last is $-D_N(h)/2 < 0$. The first stretch stays positive and the last stays negative. The middle stretch passes through zero at most once, possibly with consecutive zeros. This gives the claimed rise, plateau and fall. Symmetry gives the other hump. □

Corollary 3.20 (a window from the maximum)**.** *Let $N \geq 8$ be even and $x = x_N$. Let $[l, r]$ be the left interval of maximizers in Corollary 3.19, and put*

$$M = P_N(r), \qquad B = r - 1 + M^{-1}, \qquad b = \min\{h, \lceil B \rceil - 1\}.$$

Then $D_N(a) > 0$ for every integer $b \leq a \leq N-b$. This window is nonempty.

Proof. Let q be the first integer after l with $D_N(q) > 0$. Corollary 3.19 gives $D_N(l) < 0 < D_N(h)$, so $q \leq h$. The constant plateau gives $q > r$. By its definition P_N is concave on $[r, q]$. Hence

$$\frac{M(q-r+1)}{2} \leq \sum_{a=r}^{q-1} P_N(a) \leq \frac{1 - P_N(h)}{2} < \frac{1}{2},$$

where the first inequality sums the chord from $P_N(r) = M$ to the nonnegative value $P_N(q)$. Thus $q < B$, and consequently $q \leq b \leq h$. Theorem 3.17, applied at $l < q$, gives positivity on $[q, N-q]$, which contains $[b, N-b]$. □

Both corollaries use the finite endpoint-dip certificates of Proposition 3.7, but no mode-location or convex-window estimate. The height bound supplies a window at every even length, possibly just the center. Heuristically, Theorem 3.9(b) and Corollary 3.11 give $M = P_N(r) = \frac{f_L(\lambda_L)}{2x}(1 + o(1))$, hence, with $m^* = r$, $B = m^* + \big(2/f_L(\lambda_L) + o(1)\big)x$. Since $f_L(\lambda_L) \approx 0.1807$, this is about $m^* + 11x$, far below h for large N; at $N = 4000$, computed $B - m^*$ is about $9.9x$. This is not a bound: the $o(1)$ is unevaluated and the constants of Theorem 3.9 are not sharp. For a stronger prescribed window, see Section S5.

3.6 Odd N

Proposition 3.21 (odd N)**.** *Let $N \geq 3$ be odd, and write $\widetilde{C}_N = P_N(\frac{N-1}{2}) = P_N(\frac{N+1}{2})$ for the two central values. The ratio $\widetilde{C}_N/E_N$ is strictly increasing in ε, and it equals 1 at exactly one $\varepsilon_N \in (0, \frac{1}{2})$. Put $x_N = N\varepsilon_N$.*

(a) For every $0 < \varepsilon < \frac{1}{2}$,

$$\widetilde{C}_N = \Big(1 - \frac{c}{N}\Big)^{-1} C_{N+1},$$

where C_{N+1} is the center of the walk with $N + 1$ steps and the same ε.

The two central values at odd N reach the center at $N + 1$ with equal probabilities, giving the factor $1 - c/N$. By Proposition S3.1, their crossing satisfies Theorem 2.7; $P_N(0) < P_N(1)$ there for odd $N \geq 1689$, and the convexity window of Section S5 holds at every sufficiently large odd N.

Proof. The crossing. By Proposition 2.3, $P_N(k)/E_N$ is strictly increasing in ε for $k = \frac{N+1}{2}$. It tends to 0 as $\varepsilon \to 0$. At $\varepsilon = \frac{1}{2}$ the steps are independent and fair, so $\widetilde{C}_N = \binom{N}{k} 2^{-N} > 2^{-N} = E_N$. So the ratio equals 1 at exactly one ε_N, and $\varepsilon_N < \frac{1}{2}$.

(a) Put $k = \frac{N+1}{2}$. Since $\varepsilon + \frac{c}{2} = \frac{1}{2}$, we have $r_N(k) = \frac{1}{2} + \frac{c}{2N}$ and $r_N(k-1) = \frac{1}{2} - \frac{c}{2N}$. So (1) with $t = N$ and $a = k$ gives

$$C_{N+1} = P_{N+1}(k) = \Big(\frac{1}{2} - \frac{c}{2N}\Big)\big(P_N(k) + P_N(k-1)\big) = \Big(1 - \frac{c}{N}\Big)\widetilde{C}_N.$$

□

4 The fixed-memory limit law

In this section the reversal probability $\varepsilon \in (0, \frac{1}{4})$ is fixed. So the memory parameter $p = 1 - \varepsilon$ is fixed too, and $a = 2p - 1 = 1 - 2\varepsilon$ lies in $(\frac{1}{2}, 1)$. We write $S_k = \sum_{j \leq k} X_j$, so that $S_N = 2A_N - N$. If $X_1 = 1$, then S_k/k^a converges almost surely to a random variable $\mathcal{L}_a$ [8, Theorem 3.7]. With a fair first step the walk given $X_1 = -1$ is the mirror image of the walk given $X_1 = 1$. So S_k/k^a converges almost surely to a random variable $\mathcal{L}_a^\circ$ with the law of $\xi\mathcal{L}_a$, where ξ is a fair sign independent of $\mathcal{L}_a$ [11, (1.3) and (1.6)]. Guérin, Laulin and Raschel prove that both limits have positive, bounded and smooth densities [11, Theorem 1.3]. We write g_a for the density of $\mathcal{L}_a^\circ$, which is even. Guérin, Laulin, Raschel and Simon prove that $\mathcal{L}_a$ is unimodal [10, Theorem 1.1].

4.1 Memory near one

Corollary 4.1 (the fixed-memory limit near one)**.** *For the elephant walk with $X_1 = 1$ and fixed memory $p = (1+a)/2$, $a \in (\frac{1}{2}, 1)$, let $\mathcal{L}_a = \lim_{k\to\infty} S_k/k^a$ as in [8, Theorem 3.7]. If Z has characteristic function $\mathbb{E}e^{itZ} = \exp\{-it\log|t| - \pi|t|/2\}$, continuously extended at $t = 0$, then*

$$\frac{1-\mathcal{L}_{1-2\varepsilon}}{2\varepsilon} + \log(1/\varepsilon) \implies Z \qquad (\varepsilon \downarrow 0).$$

Proof. Fix $0 < \varepsilon < 1/8$, put $a = 1-2\varepsilon$, and use one infinite walk with $X_1 = 1$. By [8, (2.4), Appendix A], $\mathcal{M}_k = \Gamma(k)S_k/\Gamma(k+a)$ is a martingale with limit $\mathcal{L}_a$. It is Bercu's martingale divided by $\Gamma(1+a)$. Its increments are

$$\mathcal{M}_{k+1} - \mathcal{M}_k = \frac{\Gamma(k+1)}{\Gamma(k+1+a)}\left(X_{k+1} - \frac{aS_k}{k}\right).$$

Log-convexity of gamma gives $1 \le k^a\Gamma(k)/\Gamma(k+a) \le (1+a/k)^{1-a} \le 1+1/k$. Since $S_k = k - 2M_k$, Lemma B.2 gives

$$\mathbb{E}\big[1 - a^2(S_k/k)^2\big] \le 4\varepsilon + \frac{4\mathbb{E}M_k}{k} \le 4\varepsilon(1+\log k).$$

Orthogonality of the increments therefore gives square-mean convergence and

$$\mathbb{E}|\mathcal{L}_a - \mathcal{M}_n|^2 \le C\varepsilon\sum_{k\ge n} k^{-2a}(1+\log k) \le C\varepsilon n^{1-2a}(1+\log n),$$

with C independent of $a \in [3/4, 1)$, since $2a - 1 \ge 1/2$.

Take $n = \lfloor \varepsilon^{-4/3} \rfloor$ and $x = n\varepsilon$. Then $x \to \infty$ and $x^3\log^2 n/n = n^2\varepsilon^3\log^2 n \to 0$, so Theorem 3.9(c) gives $Z_\varepsilon = (M_n - x\log x)/x \Rightarrow Z$. The tail bound gives $\mathbb{E}|(\mathcal{L}_a - \mathcal{M}_n)/\varepsilon|^2 = O(\varepsilon^{1/3}(1+\log(1/\varepsilon))) = o(1)$, using $n^{4\varepsilon} \to 1$. Also $|\mathcal{M}_n - n^{-a}S_n| \le n^{-a} = o(\varepsilon)$. Finally,

$$n^{-a}S_n = e^{2\varepsilon\log n}[1 - 2\varepsilon(\log x + Z_\varepsilon)],$$

$$\frac{1-n^{-a}S_n}{2\varepsilon} + \log(1/\varepsilon) = Z_\varepsilon + o_{\mathbb{P}}(1).$$

The second equality uses $\log x = \log n + \log\varepsilon$, tightness of Z_ε, and $\varepsilon\log^2 n \to 0$. Replacing $n^{-a}S_n$ by $\mathcal{L}_a$ proves the claim. □

Theorem 4.7 below upgrades this convergence to a uniform density approximation and total variation convergence. Trinas and Valle find $\operatorname{Var}\mathcal{L}_a = 2(1-a) + o(1-a)$ [12, Eq. (4.5)]. Both results say that $\mathcal{L}_a$ concentrates at 1 as $a \to 1$. Theorem 4.3 below uses neither result. It needs only the second and fourth exact moments.

4.2 The number of modes

A probability density h on $\mathbb{R}$ is *unimodal* if for some m it is nondecreasing on $(-\infty, m]$ and nonincreasing on $[m, \infty)$. Guérin, Laulin, Raschel and Simon ask for a unique local maximum [10], which also rules out a flat top. Our g_a mixes the unimodal density of $\mathcal{L}_a$ with its mirror image, and unimodality is not preserved by mixtures [12, Section 4]. Writing L_q for the limit with $\mathbb{P}(X_1 = 1) = q$, Guérin, Laulin, Raschel and Simon conjecture that for each a there is $q(a) \in [0, 1/2]$ such that L_q is bimodal when $|q - 1/2| < q(a)$ and unimodal otherwise [10, Remark 2.1(b)]. At $q = 1/2$, this asks whether $\mathcal{L}_a^\circ$ is unimodal or bimodal. The next theorem proves this for every a. Lemma F.4 excludes flat intervals, so its modes are points.

Theorem 4.2 (at most two modes)**.** *Let $a \in (\frac{1}{2}, 1)$. Exactly one of the following holds.*

(a) The density g_a is nondecreasing on $(-\infty, 0]$ and nonincreasing on $[0, \infty)$.

(b) There is $c > 0$ such that g_a is nondecreasing on $[0, c]$ and nonincreasing on $[c, \infty)$, and $g_a(0) < g_a(c)$.

In case (b), g_a is nondecreasing on $(-\infty, -c]$ and nonincreasing on $[-c, 0]$, and it is not unimodal. All these monotonicities are strict. The unique local maximum is 0 in case (a), and the only local maxima are $\pm c$ in case (b). In case (a), there is $d > 0$ such that g_a is strictly concave on $(-d, d)$ and strictly convex on each tail. In case (b), there are $0 < d_1 < c < d_2$ such that g_a is strictly convex on $(-d_1, d_1)$ and on each tail beyond $\pm d_2$, and strictly concave on $(-d_2, -d_1)$ and (d_1, d_2). Thus it has exactly two inflection points in case (a) and four in case (b).

So g_a has at most two modes, and in case (b) we say that it has exactly two. In the proof a Gaussian kernel turns the second differences of P_N into a smooth function with at most their sign pattern, by Descartes' rule of signs for exponential sums. Then $N \to \infty$, and then the width of the kernel tends to 0. The lemmas are proved in Appendix F. A real function v has *values at most* Π if $\mathrm{sp}(v(y_1), \dots, v(y_j)) \preceq \Pi$ for all $y_1 < \cdots < y_j$, and *increments at most* Π if $\mathrm{sp}(v(y_2) - v(y_1), \dots, v(y_j) - v(y_{j-1})) \preceq \Pi$ for all such points. We put $\Pi_4 = (+, -, +, -)$.

Proof. Fix a, so that $\varepsilon = (1 - a)/2 \in (0, \frac{1}{4})$.

Step 1, smoothing. Fix $\sigma > 0$ and let $\varphi(y) = e^{-y^2/(2\sigma^2)}/(\sigma\sqrt{2\pi})$. For $N \geq 3$ put $\eta = 2N^{-a}$ and $z_k = (2k - N)N^{-a}$, so that $S_N/N^a = z_{A_N}$, and let

$$V_N(y) = \eta^{-2} \sum_{k=1}^{N-1} D_N(k)\, \varphi(y - z_k).$$

Since $\varphi(y - z_k) = \varphi(y) e^{-z_k^2/(2\sigma^2)} e^{y z_k/\sigma^2}$ and the z_k increase with k, Lemma F.1 and Theorem 3.16 show that V_N has values at most Π_5. (If every $D_N(k)$ vanishes, then $V_N = 0$ and every pattern is empty.)

Step 2, the limit $N \to \infty$. Let $u(y) = \mathbb{E}\varphi(y - \mathcal{L}_a^\circ)$. The derivatives of φ are bounded, so u is smooth and $u^{(j)}(y) = \mathbb{E}\varphi^{(j)}(y - \mathcal{L}_a^\circ)$. We show that $V_N(y) \to u''(y)$ for every y. Put $P_N(k) = 0$ for $k \notin \{0, \dots, N\}$. Then $D_N(k) = 0$ unless $-1 \leq k \leq N + 1$, and summation by parts gives

$$\sum_{k=-1}^{N+1} D_N(k)\, \varphi(y - z_k) = \sum_{k=0}^{N} P_N(k)\, \Delta\varphi(y - z_k), \qquad \Delta\varphi(t) = \varphi(t + \eta) - 2\varphi(t) + \varphi(t - \eta).$$

Taylor's formula gives $|\eta^{-2}\Delta\varphi(t) - \varphi''(t)| \leq \eta^2 \|\varphi''''\|_\infty/12$. By symmetry the four terms with $k \in \{-1, 0, N, N + 1\}$ on the left are at most $\varphi(0)\,(6P_N(0) + 2P_N(1))$ in absolute value. Here $P_N(0) \leq (1 - \varepsilon)^{N-1}$, and the proof of Theorem 3.5 gives $f_N(1) \leq \varepsilon N(1 - \varepsilon)^{N-2}$ and $f_N(N - 1) \leq \varepsilon(1 - \varepsilon)^{N-2}$, so that $P_N(1) \leq \varepsilon N(1 - \varepsilon)^{N-2}$. As $\eta^{-2} = N^{2a}/4$, these four terms times η^{-2} tend to 0. Hence

$$V_N(y) = \mathbb{E}\varphi''\big(y - S_N/N^a\big) + o(1) \longrightarrow \mathbb{E}\varphi''(y - \mathcal{L}_a^\circ) = u''(y),$$

because $S_N/N^a \to \mathcal{L}_a^\circ$ in law and φ'' is bounded and continuous. By Lemma F.2, u'' has values at most Π_5.

Step 3, one derivative fewer. By dominated convergence, $u'(y) \to 0$ as $y \to \pm\infty$. So u has increments at most Π_4, by Lemma F.3.

Step 4, the limit $\sigma \to 0$. Write $u = u_\sigma$. As g_a is bounded and continuous, $u_\sigma(y) = \int g_a(y - t)\varphi(t)\,dt$ tends to $g_a(y)$ for every y. So the increments of u_σ tend to those of g_a, and Lemma F.2 shows that g_a has increments at most Π_4. Lemma F.4 also makes g_a'' bounded and continuous, so $u_\sigma'' = g_a'' * \varphi \to g_a''$ pointwise. Hence g_a'' has values at most Π_5.

Step 5, symmetry. Let $0 \le y_1 < \cdots < y_j$, let $d_i = g_a(y_{i+1}) - g_a(y_i)$, and let p be the pattern of the d_i. As g_a is even, the points $-y_j < \cdots < -y_1 < y_1 < \cdots < y_j$, with $-y_1$ left out if $y_1 = 0$, have the increments $-d_{j-1}, \ldots, -d_1$, then 0 if $y_1 > 0$, then $d_1, \ldots, d_{j-1}$. Their pattern is the reverse of $-p$ followed by p. It has $2|p|$ entries and is $\preceq \Pi_4$. So p is empty, $(+)$, $(-)$ or $(+,-)$. In particular there are no $0 \le y < y' < y''$ with $g_a(y) > g_a(y') < g_a(y'')$.

If no increment on $[0,\infty)$ is positive, (a) holds by evenness. Otherwise $g_a(y_1) < g_a(y_2)$ for some $0 \le y_1 < y_2$. Some increment on $[y_2,\infty)$ is negative, since otherwise $g_a \ge g_a(y_2) > 0$ there, against $\int g_a = 1$. If $g_a(y_4) < g_a(y_3)$ with $y_2 \le y_3 < y_4$, then g_a cannot rise after y_4, so it attains its maximum over $[0,\infty)$ at some $c \in [0, y_4]$. If $0 \le y < y' \le c$ and $g_a(y) > g_a(y')$, then $g_a(y') < g_a(c)$, and (y, y', c) is a fall followed by a rise. If $c \le y < y'$ and $g_a(y) < g_a(y')$, then (c, y, y') is one. So g_a is nondecreasing on $[0, c]$ and nonincreasing on $[c,\infty)$. If $g_a(0) = g_a(c)$, then g_a would be nonincreasing on $[0,\infty)$, against $g_a(y_1) < g_a(y_2)$. So (b) holds.

As $g_a(0) < g_a(c)$ in (b), the two cases exclude each other. In (b), symmetry gives the monotonicity on $(-\infty, 0]$. If g_a were unimodal with mode m, then $m \ge 0$ would give $g_a(-c) \le g_a(0)$ and $m < 0$ would give $g_a(0) \ge g_a(c)$, while $g_a(\pm c) > g_a(0)$. Finally a local maximum y other than the points named would make g_a constant on a one-sided neighborhood of that point. Lemma F.4 excludes this. The same lemma makes each stated monotonicity strict: equality at two distinct points of a monotone interval would force constancy between them.

Step 6, curvature. The Fourier bound in Lemma F.4 gives $g_a(y), g_a'(y) \to 0$ as $y \to \infty$. Evenness and the Π_5 bound give at most three sign blocks for g_a'' on $(0,\infty)$. Its last sign must be positive: otherwise g_a' would be eventually nonincreasing with limit zero, and g_a eventually nondecreasing, contrary to $g_a > 0$ and $g_a \to 0$. Analyticity and strict monotonicity imply that the first nonconstant term in the even Taylor series of g_a at zero is negative in case (a) and positive in case (b). The same signs therefore hold for g_a'' near zero. In case (b), g_a'' must also be negative somewhere, since g_a' is positive near zero and negative somewhere beyond c. The positive-half-line sign patterns are consequently $(-,+)$ in case (a) and $(+,-,+)$ in case (b). Analyticity excludes an interval of zero curvature and makes the resulting convexities strict. In case (b), $g_a'(d_1) > 0 > g_a'(d_2)$, so $d_1 < c < d_2$. □

Theorem 4.3 (two modes for memory near one)**.** *If $\frac{4}{5} \le a < 1$, that is $0 < \varepsilon \le \frac{1}{10}$, then case (b) of Theorem 4.2 holds. So g_a has exactly two modes.*

Proof. In case (a), Khinchine's representation [41] gives $\mathcal{L}_a^\circ \overset{d}{=} UZ$, with U uniform on $(0,1)$ and independent of Z. Hence Cauchy–Schwarz gives $\mathbb{E}(\mathcal{L}_a^\circ)^4 \ge \frac{9}{5}[\mathbb{E}(\mathcal{L}_a^\circ)^2]^2$. The moment recursion of [11, Theorem 1.4] instead gives

$$\frac{\mathbb{E}\mathcal{L}_a^4}{(\mathbb{E}\mathcal{L}_a^2)^2} = \frac{6(2a^2+2a-1)}{4a-1}\,B(2a,2a),$$

where B is the beta function. For $a \ge \frac{3}{4}$, its integral and $t(1-t) \le \frac{1}{4}$ give $B(2a,2a) \le 4^{-2a+3/2}B(\frac{3}{2},\frac{3}{2}) = \pi 2^{-4a}$. Put $h(a) = (2a^2+2a-1)/(4a-1)$. On $[\frac{3}{4}, 1]$ we have $(\log h)' \le 2 < 4\log 2$; the first inequality reduces to $4a(4a^2+a-2) \ge 0$. Thus $6\pi h(a) 2^{-4a}$ decreases there, and for $a \ge \frac{4}{5}$ the displayed moment ratio is at most

$$\frac{282\pi}{55\,2^{16/5}} < \frac{141}{80} < \frac{9}{5},$$

using $\pi < 22/7$ and $2^{1/5} > 8/7$. Even moments are unchanged by symmetrization, so this contradicts case (a). Theorem 4.2 then gives case (b). □

The next theorem proves that the fair-start law changes shape exactly once. The proof combines the shape alternative with a certified positive parameter derivative of the central curvature throughout the transition bracket.

Theorem 4.4 (the threshold, computer-assisted)**.** *Put*

$$z_- = 1838418790980431/2748779069440000 = 0.668812860014597754\ldots,$$
$$z_+ = 45960469774511/68719476736000 = 0.668812860014601028\ldots,$$

so that $z_+ - z_- < 3.3 \cdot 10^{-15}$.

(a) There is exactly one $a_* \in (\frac{1}{2}, 0.6712]$ *with* $g''_{a_*}(0) = 0$*, and* $z_- < a_* < z_+$.

(b) For $\frac{1}{2} < a \le a_*$*, case (a) of Theorem 4.2 holds. For* $a_* < a < 1$*, case (b) holds. Thus* a_* *is the unique transition from one mode to two, and the critical density is unimodal.*

(c) We have $g''_a(0) < 0$ *for* $\frac{1}{2} < a < a_*$ *and* $g''_a(0) > 0$ *for* $a_* < a \le 0.6712$.

The bracket measures uncertainty in the location of a single transition, not an unclassified set of possible shapes. On $[\frac{4}{5}, 1)$, Theorem 4.3 gives a second proof of bimodality without computation. The results for arbitrary bias below use first differences; the symmetric curvature bound need not persist under bias.

Remark 4.5 (the method). Appendix G bounds the Fourier integral for $g''_a(0)$ by Taylor models in a and a tail estimate from the fixed-point equation. It uses scaled moments near $a = \frac{1}{2}$ and a Hankel test for $a \ge 0.6705$. The original certificate takes about 2.8 hours of CPU time. On the exact finite law, $D_N(N/2)$ changes sign at $a \approx 0.6563$ for $N = 4000$ and at $a \approx 0.6592$ for $N = 8000$. These two numbers are computed and are not used in any proof.

4.3 Arbitrary initial bias

Write h_a for the density of $\mathcal{L}_a$ and $g_{a,q}(x) = qh_a(x) + (1-q)h_a(-x)$ for initial bias $\mathbb{P}(X_1 = 1) = q$. Thus $g_{a,1/2} = g_a$.

Theorem 4.6 (the interval of bimodal biases)**.** *For every* $a \in (1/2, 1)$ *and* $q \in [0, 1]$*, the density* $g_{a,q}$ *has at most two local maxima. There is* $b_a \in [0, 1/2]$ *such that it has exactly two local maxima if and only if*

$$b_a < q < 1 - b_a.$$

Outside this open interval, including its endpoints, the density is unimodal with a unique local maximum. The interval is empty exactly when $a \le a_*$*, in which case* $b_a = 1/2$*. At finite time, for every* $0 < \varepsilon < 1$ *and* $N \ge 1$*, the count law has at most two modal plateaus, including its endpoints.*

Proof. The mode bound. Let P_N here denote the count law with first-step bias q. Proposition F.5 proves the finite assertion and, more precisely, excludes two successive strict valleys. The local limit theorem of Trinas and Valle [12, Theorem 3.1] gives

$$\frac{N^a}{2} P_N\left(\left\lfloor \frac{N + N^a x}{2} \right\rfloor\right) \longrightarrow g_{a,q}(x) \qquad (x \in \mathbb{R}).$$

The lattice point has the correct parity and differs from x after scaling by less than $2N^{-a}$. Continuity therefore justifies this form of the local limit, including the span-two factor. Lemma F.4 excludes constant intervals for every bias. If $x < y < z$ were local maxima, there would be $x < u < y < v < z$ with

$$g_{a,q}(u) < \min\{g_{a,q}(x), g_{a,q}(y)\}, \qquad g_{a,q}(v) < \min\{g_{a,q}(y), g_{a,q}(z)\}.$$

For each pair choose a point below the smaller local maximum in its one-sided maximizing neighbourhood. The absence of a constant interval makes that decrease strict. The five lattice indices are ordered for all large N, and the local limit preserves the four strict inequalities. This contradicts Proposition F.5.

The bias interval. Fix a, write $f_q = g_{a,q}$ and $h = h_a$, and call $x < y < z$ a strict valley if $f_q(y) < \min\{f_q(x), f_q(z)\}$. Proposition F.6 and the same local limit show that, for $q \geq 1/2$, f_q has no strict valley on $[0, \infty)$ and $f_q(-x) \leq f_q(x)$ for $x \geq 0$. Consequently it has a strict valley if and only if $f_q(x) > f_q(y)$ for some $x < y \leq 0$. Such a descent gives the valley $(x, y, -x)$ by reflection dominance. Conversely a valley must start at $x < 0$; if its middle point y is positive, the no-valley property at $(0, y, z)$ gives $f_q(0) \leq f_q(y) < f_q(x)$.

At $q = 1$, conditional unimodality [10] excludes a valley, so h is nondecreasing on $(-\infty, 0]$. For fixed $x < y \leq 0$, the difference $f_q(x) - f_q(y)$ is affine in q and nonpositive at $q = 1$. If positive at $q \geq 1/2$, it remains positive for every $r \in [1/2, q]$. The valley biases in $[1/2, 1]$ are therefore downward closed. They are relatively open, since the strict inequalities at three fixed points persist under a small change of bias, and they exclude 1. Reflection gives an open interval $(b_a, 1 - b_a)$, possibly empty.

A strict valley gives two local maxima by maximizing on compact intervals on its two sides; integrability and positivity supply the outer points below the two witness heights. Conversely, two local maxima give a strict valley because there is no flat interval. The mode bound excludes a third maximum. A continuous integrable density without a strict valley is unimodal, and absence of flat intervals makes its maximum unique. Finally the bias interval is nonempty exactly when it contains $1/2$, so Theorem 4.4 identifies the empty case. □

This proves the initial-bias conjecture of [10, Remark 2.1(b)], with its half-width $q(a) = 1/2 - b_a$.

4.4 Uniform profiles as memory tends to one

Theorem 4.7 (The two peaks near memory one)**.** *Let h_a denote the density of $\mathcal{L}_a$, and put*

$$A_\varepsilon = 1 + 2\varepsilon \log(1/\varepsilon), \qquad K_\varepsilon(z) = 2\varepsilon h_{1-2\varepsilon}(A_\varepsilon - 2\varepsilon z),$$
$$H_\varepsilon(z) = 4\varepsilon g_{1-2\varepsilon}(A_\varepsilon - 2\varepsilon z)\mathbf{1}\{z < A_\varepsilon/(2\varepsilon)\}.$$

As $\varepsilon \downarrow 0$, both profiles converge uniformly to f_L, and $\|K_\varepsilon - f_L\|_1 \to 0$. If $c_a > 0$ is the positive mode of g_a, then

$$c_{1-2\varepsilon} = 1 + 2\varepsilon \log(1/\varepsilon) - 2\varepsilon\lambda_{\mathrm{L}} + o(\varepsilon), \qquad 4\varepsilon g_{1-2\varepsilon}(c_{1-2\varepsilon}) \longrightarrow f_L(\lambda_{\mathrm{L}}).$$

For $q \in [0, 1]$, define the two-component density

$$\widetilde{g}_{\varepsilon,q}(x) = \frac{q}{2\varepsilon} f_L\left(\frac{A_\varepsilon - x}{2\varepsilon}\right) + \frac{1-q}{2\varepsilon} f_L\left(\frac{A_\varepsilon + x}{2\varepsilon}\right).$$

Then both approximation bounds are uniform over the initial bias:

$$\sup_{q\in[0,1]} 2\varepsilon \|g_{1-2\varepsilon,q} - \widetilde{g}_{\varepsilon,q}\|_\infty \longrightarrow 0, \qquad \sup_{q\in[0,1]} \|g_{1-2\varepsilon,q} - \widetilde{g}_{\varepsilon,q}\|_1 \longrightarrow 0.$$

Proof. Write $f = f_L$, $m = \lambda_{\mathrm{L}}$ and $B_\varepsilon = A_\varepsilon/(2\varepsilon) \to \infty$. The inputs about f are that it is continuous, nonnegative, integrates to one, tends to zero at both infinities, has a unique strict maximum at m, and is strictly concave on a neighbourhood of m. The certified mode lemma supplies the last two assertions; the Fourier representation and the Riemann–Lebesgue lemma supply the tail limits.

The fixed-memory bridge gives convergence of every bounded interval mass of K_ε to the corresponding interval mass of f. The density K_ε is unimodal, since $h_{1-2\varepsilon}$ is unimodal [10]. Its modes tend to m. To see this without assuming pointwise convergence, fix $b < m$. Choose a short interval about m and an equal-length interval strictly between b and m. Continuity and the strict maximum make the limiting mass of the former larger. A mode to the left of b would make the approximating density nonincreasing on both intervals and force the reverse mass ordering. The argument to the right is identical. On any fixed interval avoiding m, the approximating densities are consequently monotone in the correct direction for all sufficiently small ε. Their values are squeezed between adjacent interval averages. Letting the interval length decrease to zero proves $K_\varepsilon(z) \to f(z)$ for every $z \neq m$. The identity

$$\int |K_\varepsilon - f| = 2 - 2\int \min(K_\varepsilon, f)$$

and dominated convergence now give L^1 convergence.

If k is a nonnegative unimodal density with mode b, then

$$k(y) \leq \frac{1}{y-b} \qquad (y > b),$$

because its integral over $[b, y]$ is at least $(y-b)k(y)$. Since the modes of K_ε remain bounded, $\sup_{y \geq B_\varepsilon} K_\varepsilon(y) \to 0$. The fair-density identity gives, for $z < B_\varepsilon$,

$$H_\varepsilon(z) = K_\varepsilon(z) + K_\varepsilon(2B_\varepsilon - z).$$

It follows that $H_\varepsilon(z) \to f(z)$ for every $z \neq m$.

For small enough ε, the strict-shape theorem gives one positive hump of $g_{1-2\varepsilon}$ with a concavity interval between two convex intervals. After the affine change of variables, H_ε is unimodal and has the same convex–concave–convex partition up to B_ε. We make no convexity assertion across the truncation point. The preceding mode argument, or its pointwise version, shows that its mode m_ε tends to m.

Choose two fixed triples, one on each side of m, in the neighbourhood where f is strictly concave. Each strict midpoint-concavity inequality persists for H_ε. Neither convex part of H_ε can contain its corresponding triple. Hence its concavity interval contains a fixed neighbourhood $[L, R]$ of m for all sufficiently small ε. For fixed $L < \ell < x < m$ in this neighbourhood, concavity gives

$$H_\varepsilon(m_\varepsilon) \leq H_\varepsilon(x) + \frac{m_\varepsilon - x}{x - \ell}\{H_\varepsilon(x) - H_\varepsilon(\ell)\}.$$

First let $\varepsilon \downarrow 0$ and then $x \uparrow m$. This bounds the limiting upper peak height by $f(m)$. The inequality $H_\varepsilon(m_\varepsilon) \geq H_\varepsilon(x)$ gives the reverse bound. Thus $H_\varepsilon(m_\varepsilon) \to f(m)$. This secant argument is necessary: weak convergence and unimodality alone need not bound the height at the mode.

Monotonicity and a finite grid give uniform convergence on compact intervals away from m. Near m, the converging peak height and the values at two nearby endpoints give the same conclusion. Beyond a large fixed interval, monotonicity bounds H_ε by its endpoint values, and f

tends to zero. Thus $H_\varepsilon \to f$ uniformly on $\mathbb{R}$. The displayed folding identity and the uniform tail bound then give $K_\varepsilon \to f$ uniformly below B_ε; above B_ε both functions vanish uniformly in the limit.

Finally, $m_\varepsilon = (A_\varepsilon - c_{1-2\varepsilon})/(2\varepsilon)$ and $H_\varepsilon(m_\varepsilon) = 4\varepsilon g_{1-2\varepsilon}(c_{1-2\varepsilon})$. These identities give the stated mode and height limits. Reflection and mixing do not increase either the supremum error or the density L^1 error: for $0 \le q \le 1$,

$$\big\|q(K_\varepsilon - f) + (1-q)(K_\varepsilon - f)(2B_\varepsilon - \cdot)\big\|_r \le \|K_\varepsilon - f\|_r, \qquad r = 1, \infty.$$

An affine change of variables gives the two assertions uniform in q. □

Corollary 4.8 (both peaks under a biased start)**.** *Fix $0 < \eta \le 1/2$. For all sufficiently small ε, uniformly over $q \in [\eta, 1-\eta]$, the density $g_{1-2\varepsilon,q}$ has exactly two local maxima $x^-_{\varepsilon,q} < x^+_{\varepsilon,q}$. Their positions and heights satisfy*

$$\begin{aligned} x^+_{\varepsilon,q} &= A_\varepsilon - 2\varepsilon\lambda_{\mathrm{L}} + o(\varepsilon), \qquad & 2\varepsilon g_{1-2\varepsilon,q}(x^+_{\varepsilon,q}) &= q f_L(\lambda_{\mathrm{L}}) + o(1), \\ x^-_{\varepsilon,q} &= -A_\varepsilon + 2\varepsilon\lambda_{\mathrm{L}} + o(\varepsilon), & 2\varepsilon g_{1-2\varepsilon,q}(x^-_{\varepsilon,q}) &= (1-q) f_L(\lambda_{\mathrm{L}}) + o(1). \end{aligned}$$

All four errors are uniform over this interval of biases.

Proof. Fix a small window about λ_{L}. In the positive peak's scaled coordinates, Theorem 4.7 gives uniform convergence to qf_L on this window: the other component is evaluated at $2B_\varepsilon - z$ and tends uniformly to zero there. The strict gap between the value at λ_{L} and the two endpoint values persists uniformly because $q \ge \eta$. Continuity gives a local maximum in the interior. Reflection gives a second local maximum in the negative window, using $1 - q \ge \eta$. The windows are disjoint for small ε. Theorem 4.6 excludes any further local maximum. Shrinking the fixed windows proves the position estimates, and the uniform density approximation gives the height estimates. □

The uniform approximation of Theorem 4.7 allows biases approaching 0 or 1, but the two-mode conclusion above does not. An absolute approximation error need not resolve a component whose weight tends to zero. The interval classification nevertheless gives $b_a \to 0$ as $a \uparrow 1$: for every fixed $\eta > 0$, the corollary puts η inside the bimodal interval for all sufficiently large a, so $0 \le b_a < \eta$. No rate for this convergence is asserted.

5 Verification

Apart from short exact ratios such as Proposition 3.7, the proofs use four numerical certificates. First, the 60-term sum in Theorem 2.6(b) lies in $[102.586396, 102.586398]$, as checked by `certify_elephant.py`. With the analytic tail, it is below 102.59, giving the constant 5200. This value fixes the explicit thresholds in Proposition 3.7 and the second-order results. Second, the same script certifies the Landau derivative and curvature enclosures of Lemma 3.10. Lean checks these independently.

Third, `certify_center.py` supplies the finite endpoint-dip input of Proposition 3.7 for $10 \le N \le 2000$. It brackets each crossing and checks by an exact rational comparison that the lower end of the bracket exceeds $2/(N+2)$. It also checks the center minimum independently, which Corollary 3.19 no longer needs. The script `certify_convex.py` supports only the finite convexity window of Section S5 (Proposition S5.5). The dated outputs and rational witnesses in `data/` record script hashes and software versions. Certification scripts reject failed checks and, where applicable, empty ranges.

Fourth, `certify_threshold.py` proves the classification outside the bracket in Theorem 4.4 (Appendix G). It imports `threshold_taylor.py`, `threshold_hankel.py` and `threshold_poin twise.py`. Its 9 runs are recorded in `data/threshold_*.txt`, with source hashes, versions, exit status and signed accepted intervals. The mode `check` verifies that the records join into the cover used in the proof, with the sign $-$ on all of $(\frac{1}{2}, z_-]$ and $+$ on all of $[z_+, 0.6712]$. The soundness checks raise exceptions and are not Python `assert` statements, so `python -O` does not remove them. Theorems 4.2 and 4.3 use no computation. The script `verify_elephant_modes.py` checks the rational arithmetic in the fourth-moment proof of Theorem 4.3, retains checks of the earlier two-moment bound for $a \geq 39/40$, and computes the finite-N numbers of Remark 4.5, which are not proof inputs. Its versioned output is `data/verify_modes_output.txt`.

The extension `certify_transition.py` closes the unresolved bracket by certifying $0.68 < \partial_a g_a''(0) < 0.70$ and $-0.09 < g_a^{(4)}(0) < -0.07$ on a larger interval containing it. It bounds the actual parameter derivative through the differentiated smoothing equation. The source hashes, interval enclosures and checks are in `data/transition_derivative_certificate.json`; the independent rerun at higher precision is recorded in `data/transition_derivative_run.txt`. All checks remain active under `python -O`.

The Lean 4.33.1 library in `kagey-problems/problem131/lean/PaperD`, with Mathlib revision `0df444a360ea`, checks the finite identities, crossing bounds, complete modal intervals, maximum-height window and sign arguments. It uses only `propext`, `Classical.choice` and `Quot.sound`. Generated integer certificates use exact outward rounding, without `native_decide`. The bulk and eventual-shape proofs retain six explicit classical inputs: Fourier inversion, the compound Poisson characteristic function, Efron–Stein, Scheffé [44], class-L unimodality and Hoeffding's hypergeometric bound. The enlarged weak-Landau range uses only the first three, with Mathlib's Lévy theorem; the Jensen bound has no additional classical input. The statement-by-statement map, including these dependencies, is in `MAP.md` and `coverage_tables.tex`.

The entire extension and strict shape theorem are formalized from the limit density, convergence in law, the conditional characteristic function's smoothing inequality and the density's Fourier representation. Its continuity, unit bound, symmetry and decay follow from a probability measure with density. The fourth-moment proof for $a \geq 4/5$ uses the raw moment formulas of [11]; Lean proves the unimodal kurtosis inequality directly. The all-bias first-difference theorem and Proposition F.6 use the actual finite law, including both boundary histories. The lattice local limit [12], conditional unimodality and Fourier representation then give the centred bias interval in Theorem 4.6. Its no-flatness, half-line shape and mode-count bounds are conclusions, not hypotheses.

For Theorem 4.7, Portmanteau supplies interval masses from the stated weak bridge. Lean then proves localization, peak-height convergence, uniform profiles and L^1 convergence, using the fair curvature theorem, conditional unimodality and Landau density facts. Reflection and mixing give uniformity in the bias, the two peak windows of Corollary 4.8, and $b_a \to 0$.

For the transition, Lean checks the inhomogeneous maximum argument, dominated Fourier differentiation, rational slack inequalities, and the passage from the derivative bounds to a unique transition with unimodal critical density. The detailed differentiated-smoothing integral estimates and Arb transcendental enclosures remain separate written and numerical proofs. Their bounds are explicit inputs to the formal interface. The total variation upgrade of Theorem 3.8 and the martingale proof of the fixed-memory weak bridge also remain unformalized; the earlier pointwise compound Poisson bound is formalized. The current source inventory and full axiom audit are in `reports/paperD-transition-2026-11-01`.

Each numerical record identifies its software versions. The derivative extension was rerun with

Python 3.12.14 and python-flint 0.8.0, at 1600-bit precision with 1100 moments. Figures and other numerical illustrations are not proof inputs. The repository gives rerun commands.

A note on the use of AI

I wrote the original draft of this paper and its mathematics. I used Claude Opus 5.5 (Anthropic, through Claude Code) for literature searches, citation checks and referee-style reviews that informed the framing and abstract. I read every source cited in the paper. I proposed and drafted the following additions: Lemma F.4; the strictness and inflection-point clauses of Theorem 4.2; the fourth-moment extension of Theorem 4.3 to $a \geq 4/5$; the Jensen lower bound in Theorem 2.6(a); and the weaker weak-convergence hypothesis in Theorem 3.9(c). I used GPT-6 Astra (OpenAI, through Codex) to revise the manuscript and to revise and extend the proofs of these additions. The associated Lean files in `kagey-problems/problem131/lean/PaperD` were edited using GPT-6 Astra through Codex. Section 5 distinguishes the formal proofs, their named classical hypotheses and the separate numerical checks. I take responsibility for the claims, proofs and computations in the paper.

Statements and Declarations

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Ethics and consent. Not applicable.

Data availability. The outputs of the computations described in Section 5 are in the repository's archived certificate data and the accompanying `paperD/data` folder. Sections S1 to S6 of this version hold results and proofs that the journal version omits.

Code availability. Code, data and Lean files are in `kagey-problems/problem131` of the public repository. The proof-producing certificate code, its records, the verification scripts and the Lean source files are pinned by the release tag `paperD-arxiv-v2`, including the certificate for Theorem 4.4. The run records identify the source hashes. Run `python -B verify_all.py --paper d` there for the numerical checks. The `paperD/README.md` gives separate commands for the endpoint-dip certificate, the finite convexity window in Section S5, and the full threshold certificate. It records their dependencies and run times; `SUPPLEMENT.md` maps scripts to results.

Author contributions. I am the sole author and am responsible for the whole paper.

A Proofs for Section 2

We prove Theorem 2.6. Throughout, $N \geq 2$ is even and we condition on $X_1 = +1$. By Proposition 2.5(a) we need to bound $V_{t+1}(1)$, and we compare V with explicit hitting probabilities of Pólya urns. Let $\mathcal{L}_s F(b) = (1-r_s(b))F(b)+r_s(b)F(b+1)$, so that $V_s = \mathcal{L}_s V_{s+1}$ and $V_N(b) = \mathbf{1}\{b = h\}$. Both M_t and $t - M_t \geq 1$ are nondecreasing. So from $b \geq 1$ the chain stays at states ≥ 1, and for

$1 \le b \le s-1$ we have $V_s(b) = 0$ unless b is feasible, that is, $b \in \mathcal{S}_s = \{b \in \mathbb{Z} : 1 \le b \le h,\ 1 \le s-b \le h\}$. Note that $\mathcal{S}_N = \{h\}$.

For $c \ge 0$ the Pólya(c) chain moves at time s from b to $b+1$ with probability $r_s^c(b) = \frac{b+c}{s+2c}$. It is Pólya's urn with weights $b+c$ and $s-b+c$ for the two colors, that is, with c extra balls of each color. For $b+c>0$ and $s-b+c>0$ let $\Phi_s^c(b) = \binom{N-s}{h-b}\mathrm{B}(h+c, h+c)/\mathrm{B}(b+c, s-b+c)$, where B is the beta function and the binomial coefficient is 0 unless $0 \le h-b \le N-s$. Theorem 2.6(a) will follow from Lemma A.1(d), and (b) from Proposition A.4.

Lemma A.1. *Let $c \ge 0$, $1 \le s \le N$, $b \in \mathbb{Z}$, $b+c>0$ and $s-b+c>0$.*

(a) The number $\Phi_s^c(b)$ is the probability that the Pólya(c) chain at b at time s is at h at time N. So $\Phi_N^c(b) = \mathbf{1}\{b=h\}$, and $\Phi_s^c(b) = r_s^c(b)\Phi_{s+1}^c(b+1) + (1-r_s^c(b))\Phi_{s+1}^c(b)$ for $s<N$.

(b) If $s<N$ and $\Phi_s^c(b)>0$, then $r_s^c(b)\Phi_{s+1}^c(b+1)/\Phi_s^c(b) = \bar{r}_s(b) := \frac{h-b}{N-s}$.

(c) If $0<\varepsilon<\frac{1}{2}$ and $\alpha = \frac{\varepsilon}{1-2\varepsilon}$, then $r_s(b) = r_s^{\alpha s}(b)$ for $0 \le b \le s$.

(d) If $0<\varepsilon<\frac{1}{2}$ and $0 \le c \le \alpha s$, then $V_s(b) \ge \Phi_s^c(b)$ for $1 \le b \le s-1$. In particular $V_{t+1}(1) \ge \Phi_{t+1}^0(1) = \pi_t$ for $1 \le t \le h$.

Proof. (a) Let $x = b+c$, $y = s-b+c$ and $(x)_k = x(x+1)\cdots(x+k-1)$. As in the proof of Lemma 2.4, each order of the $N-s$ draws with $h-b$ draws of the first color has probability $(x)_{h-b}(y)_{N-s-h+b}/(x+y)_{N-s} = \mathrm{B}(h+c, h+c)/\mathrm{B}(x,y)$, and there are $\binom{N-s}{h-b}$ such orders. The recursion is the Markov property.

(b) Use $\binom{N-s-1}{h-b-1}/\binom{N-s}{h-b} = \frac{h-b}{N-s}$ and $\mathrm{B}(x,y)/\mathrm{B}(x+1,y) = \frac{x+y}{x} = 1/r_s^c(b)$.

(c) Since $1+2\alpha = 1/(1-2\varepsilon)$, we have $\frac{b+\alpha s}{s+2\alpha s} = (1-2\varepsilon)(\frac{b}{s}+\alpha) = r_s(b)$.

(d) We check the hypotheses of Lemma A.2(b) below for $F_u = \Phi_u^c$ on $\{1,\dots,u-1\}$, where $s \le u \le N$. The reason is that M has $\alpha u \ge c$ extra balls at time u, so it is pulled toward $\frac{u}{2}$ at least as strongly as the Pólya(c) chain, and from a state closer to $\frac{u}{2}$ it is easier to end at h.

By (a), $F_N(b) = \mathbf{1}\{b=h\}$, and F_u vanishes off $\mathcal{S}_u$ because the binomial coefficient does. Let $s \le u < N$ and $b \in \mathcal{S}_u$. Subtracting the recursion in (a) from the definition of $\mathcal{L}_u$ gives

$$\mathcal{L}_u F_{u+1}(b) - F_u(b) = \big(r_u(b) - r_u^c(b)\big)\big(F_{u+1}(b+1) - F_{u+1}(b)\big).$$

We show that the two factors on the right never have opposite signs. For the first factor, $r_u^c(b) = \frac{1}{2} - (\frac{u}{2}-b)/(u+2c)$, and by (c) the same formula with αu in place of c gives $r_u(b)$. So the first factor is $\frac{u}{2}-b$ times $\frac{1}{u+2c} - \frac{1}{u+2\alpha u}$. The second of these is nonnegative because $c \le \alpha s \le \alpha u$. Hence the first factor has the sign of $\frac{u}{2}-b$ or vanishes. For the second factor there are three cases. If $b=h$ then $b \ge \frac{u}{2}$ and $F_{u+1}(b+1) = 0$, so the factor is ≤ 0. If $u-b=h$ then $b \le \frac{u}{2}$ and $F_{u+1}(b) = 0$, so the factor is ≥ 0. Otherwise both values are positive, and the formula for Φ gives $F_{u+1}(b+1)/F_{u+1}(b) = \frac{(h-b)(u-b+c)}{(h-u+b)(b+c)}$. With $b = \frac{u}{2} - d$ the numerator minus the denominator is $2d(h+c)$, so the ratio is ≥ 1 if $b \le \frac{u}{2}$ and ≤ 1 if $b \ge \frac{u}{2}$. In all three cases the second factor has the sign of $\frac{u}{2}-b$ or vanishes. Hence $\mathcal{L}_u F_{u+1} \ge F_u$ on $\mathcal{S}_u$, and Lemma A.2(b) with $s_0 = s$ gives $V_s \ge F_s$. Finally $\Phi_{t+1}^0(1) = \beta_{t+1}(h) = \pi_t$ by (a) and Lemma 2.4. □

Lemma A.2. *Let $1 \le s_0 \le N$, and for $s_0 \le s \le N$ let $F_s \ge 0$ be a function on $\{1,\dots,s-1\}$.*

(a) If $F_N(h) \ge 1$ and $\mathcal{L}_s F_{s+1} \le K_s F_s$ on $\mathcal{S}_s$ for $s_0 \le s < N$, with constants $K_s \ge 0$, then $V_s \le \big(\prod_{u=s}^{N-1} K_u\big) F_s$ on $\mathcal{S}_s$.

(b) If $F_N(b) \le \mathbf{1}\{b = h\}$, each F_s vanishes off $\mathcal{S}_s$, and $\mathcal{L}_s F_{s+1} \ge F_s$ on $\mathcal{S}_s$ for $s_0 \le s < N$, then $V_s \ge F_s$.

Proof. We use backward induction on s, from $V_N(b) = \mathbf{1}\{b = h\}$ and $V_s = \mathcal{L}_s V_{s+1}$. Note that $\mathcal{L}_s$ is an average of two values, so it keeps inequalities between functions. In (a), suppose that $V_{s+1} \le \prod_{u>s} K_u\, F_{s+1}$ on $\mathcal{S}_{s+1}$. This also holds off $\mathcal{S}_{s+1}$, because there $V_{s+1} = 0$ and $F_{s+1} \ge 0$. So on $\mathcal{S}_s$ we get $V_s = \mathcal{L}_s V_{s+1} \le \prod_{u>s} K_u\, \mathcal{L}_s F_{s+1} \le \prod_{u \ge s} K_u\, F_s$. In (b), suppose that $V_{s+1} \ge F_{s+1}$ on $\{1, \dots, s\}$. Then $V_s = \mathcal{L}_s V_{s+1} \ge \mathcal{L}_s F_{s+1} \ge F_s$ on $\mathcal{S}_s$, and $F_s = 0 \le V_s$ off $\mathcal{S}_s$. □

Idea of the proof of the upper bound. We compare M with an urn whose number of extra balls, $\kappa(s+10)$ of each color at time s, grows more than twice as fast as the αs of M. Its hitting probability $\Phi_s^{\kappa_s}$ is explicit. One step of M applied to it gives it back times a factor Δ_s, and Proposition A.4 shows that $\log \Delta_s \le \kappa/s$ up to a smaller term. Two effects enter Δ_s. Raising the extra balls from κ_s to κ_{s+1} changes Φ, and we bound this through the derivative of Φ_s^τ in τ (Lemma A.3). Also M pulls toward the middle less than the comparison urn, which helps. The product of the factors is about $e^{\kappa H_N}$. Let ψ be the digamma function.

Lemma A.3. *Let $2 \le s \le N$, $b \in \mathcal{S}_s$ and $\tau \ge 0$. Put $X = b + \tau$, $Y = s - b + \tau$, $\Sigma = X + Y$ and $z = (s - 2b)^2/\Sigma^2$. Then*

$$\partial_\tau \log \Phi_s^\tau(b) < -\log(1 - z) + \frac{1 + z}{(1 - z)\Sigma} + \frac{2(1 + z)}{3\Sigma^2(1 - z)^2}.$$

Proof. We differentiate the beta functions and bound the digamma function by the first terms of its expansion. Only the two beta functions in $\Phi_s^\tau(b)$ depend on τ, and $\log \mathrm{B}(x, y) = \log \Gamma(x) + \log \Gamma(y) - \log \Gamma(x + y)$. So the derivative is $2\psi(\Sigma) - \psi(X) - \psi(Y)$ minus the difference $2\psi(N + 2\tau) - 2\psi(h + \tau)$. The duplication formula [45, 5.5.5] gives $2\psi(2y) = \psi(y) + \psi(y + \frac{1}{2}) + 2\log 2$, and ψ is increasing. With $y = h + \tau$ this shows that the difference we subtract is more than $2 \log 2$. For $x > 0$, [45, 5.9.15] writes $\log x - \frac{1}{2x} - \psi(x)$ as $2\int_0^\infty \frac{t\,dt}{(t^2 + x^2)(e^{2\pi t} - 1)}$, which lies in $(0, \frac{1}{12x^2})$ since $\int_0^\infty \frac{t\,dt}{e^{2\pi t} - 1} = \frac{1}{24}$. So $\psi(\Sigma) < \log \Sigma - \frac{1}{2\Sigma}$, and $\psi(x) > \log x - \frac{1}{2x} - \frac{1}{12x^2}$ for $x = X$ and $x = Y$. Hence the derivative is less than $\log \frac{\Sigma^2}{4XY} + (\frac{1}{2X} + \frac{1}{2Y} - \frac{1}{\Sigma}) + \frac{1}{12}(\frac{1}{X^2} + \frac{1}{Y^2})$. Now $X - Y = 2b - s$, so $4XY = \Sigma^2 - (X - Y)^2 = \Sigma^2(1 - z)$ and $X^2 + Y^2 = \Sigma^2(1 + z)/2$. So the three terms are $-\log(1 - z)$, $\frac{X^2 + Y^2}{2XY\Sigma} = \frac{1 + z}{(1 - z)\Sigma}$ and $\frac{X^2 + Y^2}{12X^2Y^2} = \frac{2(1 + z)}{3\Sigma^2(1 - z)^2}$. □

Proposition A.4. *Let $0 < \varepsilon < \frac{1}{4}$, $\kappa = \frac{2\varepsilon}{1 - 4\varepsilon}$, $\kappa_s = \kappa(s + 10)$ and $\Sigma_s = s + 2\kappa_s$. For $1 \le s \le N - 1$ and $b \in \mathcal{S}_s$ let $\Delta_s(b) = \mathcal{L}_s \Phi_{s+1}^{\kappa_{s+1}}(b) \big/ \Phi_s^{\kappa_s}(b)$.*

(a) Let $c = \kappa_s$, $c' = \kappa_{s+1}$, $d = \frac{s}{2} - b$, $u = \frac{d}{s + 2c'}$, $v = \frac{d}{N - s}$ and $\rho = \frac{s + 2c'}{s(1 + 2\alpha)}$. Then $|u| < \frac{1}{2}$ and

$$\Delta_s(b) = \mathcal{J}\, \frac{\Phi_s^{c'}(b)}{\Phi_s^{c}(b)}, \qquad \mathcal{J} = 1 - (\rho - 1)\frac{u(u + v)}{\frac{1}{4} - u^2}.$$

(b) The factor satisfies $\log \Delta_s(b) \le \frac{\kappa}{\Sigma_s} + \frac{2\kappa}{3\Sigma_s^2}$.

(c) For $1 \le t \le h$, $V_{t+1}(1) \le \pi_t \exp\big(\kappa\big[H_{N-1} - H_t + \frac{2}{3t}\big] + \kappa_{t+1}\big[\log \frac{(t+1)^2}{4t} + \frac{2}{3}\big]\big)$.

(d) The center satisfies $C_N \le U_N := \varepsilon c_1 e^{\kappa H_{N-1}} \sum_{t=1}^h w_t (1 - \varepsilon)^{t-1} e^{\kappa \mathcal{G}(t)}$, where $\mathcal{G}(t) = (t + 11)\big(\log \frac{(t+1)^2}{4t} + \frac{2}{3}\big) - H_t + \frac{2}{3t}$.

The slope κ solves $2(\kappa-\alpha)/(1+2\alpha)=\kappa$, so that in (b) the gain in $\mathcal{J}$ cancels the main loss from raising the extra balls from c to c'.

Proof. (a) Write $r'=r_s^{c'}(b)$, $\bar r=\bar r_s(b)$ and $r=r_s(b)$. By Lemma A.1(b) with c', and the recursion in Lemma A.1(a), we have $r'\Phi_{s+1}^{c'}(b+1)=\bar r\,\Phi_s^{c'}(b)$ and $(1-r')\Phi_{s+1}^{c'}(b)=(1-\bar r)\Phi_s^{c'}(b)$. So the numerator of $\Delta_s(b)$ divided by $\Phi_s^{c'}(b)$ is $\frac{(1-r)(1-\bar r)}{1-r'}+\frac{r\bar r}{r'}$. Here $r'=\frac12-u$, $\bar r=\frac12+v$ and $r=\frac12-\rho u$, since $\frac12-r=(1-2\varepsilon)\frac{d}{s}=\rho u$. Over the denominator $r'(1-r')=\frac14-u^2$ the numerator is $f(u,v)+f(-u,-v)$ with $f(u,v)=(\frac12+\rho u)(\frac12-v)(\frac12-u)$, and this equals $\frac14-\rho u^2-(\rho-1)uv$. Also $|d|\le\frac{s}{2}-1$, so $|u|<\frac12$.

(b) *Step 1. The factor $\mathcal{J}$.* Write $c=\kappa_s$, $c'=c+\kappa$ and $\Sigma=\Sigma_s$. Since $c'-\alpha s=(\kappa-\alpha)s+11\kappa$, the choice of κ gives $\rho-1=\kappa+\nu_s$ with $\nu_s=\frac{22\kappa}{s(1+2\alpha)}$. As u and v have the sign of d, $u(u+v)\ge u^2$. So $\mathcal{J}\le 1-(\kappa+\nu_s)W'$ with $z'=4u^2$ and $W'=\frac{z'}{1-z'}$. Also $\mathcal{J}>0$ by (a), since $0<r<1$ and $\bar r$, $1-\bar r$ are not both 0. So $\log\mathcal{J}\le-(\kappa+\nu_s)W'$.

Step 2. The ratio $\Phi_s^{c'}/\Phi_s^c$. The logarithm $\log(\Phi_s^{c'}(b)/\Phi_s^c(b))$ is the integral over $\tau\in[c,c']$ of the derivative in Lemma A.3. Its bound increases in z and decreases in Σ, while z decreases and Σ increases in τ. So we may take the bound at $\tau=c$, where $X+Y=\Sigma$, and multiply it by the length κ of the interval. Put $W=\frac{z}{1-z}$. Then $\frac{1}{1-z}=1+W$, $\frac{1+z}{1-z}=1+2W$ and $-\log(1-z)\le W$, so

$$\log\Delta_s(b)\le\kappa\Big[W+\frac{1+2W}{\Sigma}+\frac{2(1+2W)(1+W)}{3\Sigma^2}\Big]-(\kappa+\nu_s)W'. \tag{3}$$

Step 3. What is left to show. The terms of (3) without W and W' are $\frac{\kappa}{\Sigma}+\frac{2\kappa}{3\Sigma^2}$, which is the bound we want. Since $(1+2W)(1+W)=1+3W+2W^2$, the other terms are

$$\kappa(W-W')+\frac{2\kappa W}{\Sigma}+\frac{2\kappa W}{\Sigma^2}+\frac{4\kappa W^2}{3\Sigma^2}-\nu_sW',$$

and we must show that they add up to at most 0. If $W=0$ then $d=0$, so $W'=0$ and there is nothing to do. So let $W>0$. The first four terms are positive, and we show that each is a small multiple of $\frac{\kappa}{s}W$. The last term is negative (it comes from the shift 10 in κ_s), and it is a larger multiple of $\frac{\kappa}{s}W$. We need three facts, with $X=b+c$ and $Y=s-b+c$ as in Step 2.

(i) We have $1+W\le\frac{\Sigma}{2(1+c)}$. Indeed $X,Y\ge1+c$, since $b\ge1$ and $s-b\ge1$, and the larger of X and Y is at least $\frac{\Sigma}{2}$. So $XY\ge(1+c)\Sigma/2$, and $1+W=\frac{1}{1-z}=\frac{\Sigma^2}{4XY}$.

(ii) We have $W-W'\le\frac{2\kappa}{1+c}W$, and so $W'\ge\frac{9}{11}W$. Indeed z and z' have the same numerator $(s-2b)^2$, and their denominators are Σ^2 and $(s+2c')^2=(\Sigma+2\kappa)^2$. So $z'=\lambda z$ with $\lambda=(\frac{\Sigma}{\Sigma+2\kappa})^2\ge1-\frac{4\kappa}{\Sigma}$. Then

$$W-W'=\frac{(1-\lambda)z}{(1-z)(1-\lambda z)}\le(1-\lambda)W(1+W)\le\frac{4\kappa}{\Sigma}\,W\,\frac{\Sigma}{2(1+c)}=\frac{2\kappa}{1+c}W,$$

where we used $1-\lambda z\ge1-z$ and then (i). Since $c=\kappa(s+10)\ge11\kappa$, we have $\frac{2\kappa}{1+c}\le\frac{2}{11}$.

(iii) We have $\Sigma\ge s$, $1+c>\kappa s$ and $\Sigma^2\ge3s$. The first two are clear. For the third, $W>0$ means $s\ne2b$, and with $b\ge1$ and $s-b\ge1$ this forces $s\ge3$.

Step 4. The terms with W. We bound the four positive terms one by one.

$$\begin{aligned}
\kappa(W-W') &\le \frac{2\kappa^2}{1+c}W \le \frac{2\kappa}{s}W && \text{by (ii) and } 1+c>\kappa s,\\
\frac{2\kappa W}{\Sigma} &\le \frac{2\kappa}{s}W && \text{by } \Sigma \ge s,\\
\frac{2\kappa W}{\Sigma^2} &\le \frac{2\kappa}{3s}W && \text{by } \Sigma^2 \ge 3s,\\
\frac{4\kappa W^2}{3\Sigma^2} &\le \frac{4\kappa W(1+W)}{3\Sigma^2} \le \frac{2\kappa W}{3\Sigma(1+c)} \le \frac{2\kappa}{3s}W && \text{by (i) and } \Sigma \ge s.
\end{aligned}$$

The four bounds add up to $\frac{16\kappa}{3s}W$. For the negative term, (ii) gives $-\nu_s W' \le -\frac{9}{11}\nu_s W = -\frac{18\kappa}{s(1+2\alpha)}W$. So

$$\log \Delta_s(b) \le \frac{\kappa}{\Sigma} + \frac{2\kappa}{3\Sigma^2} + W\Big[\frac{16\kappa}{3s} - \frac{18\kappa}{s(1+2\alpha)}\Big],$$

and the bracket is negative because $1+2\alpha<2$ for $\varepsilon<\frac14$, so $18/(1+2\alpha)>9>16/3$.

(c) We first compare $V_{t+1}(1)$ with $\Phi_{t+1}^{\kappa_{t+1}}(1)$, and then this with $\pi_t=\Phi_{t+1}^0(1)$. For the first step apply Lemma A.2(a) with $s_0=t+1$, $F_s=\Phi_s^{\kappa_s}$ and $K_s=\exp(\frac{\kappa}{s}+\frac{2\kappa}{3s^2})$. This is allowed, since $F_N(h)=1$, $F_s>0$ on $\mathcal{S}_s$, and $\mathcal{L}_sF_{s+1}=\Delta_sF_s\le K_sF_s$ on $\mathcal{S}_s$ by (b) and $\Sigma_s\ge s$. The product of the K_s over $t<s<N$ is the exponential of $\kappa(H_{N-1}-H_t)+\frac{2\kappa}{3}\sum_{s>t}s^{-2}$, and $\sum_{s>t}s^{-2}\le\frac1t$. Since $1\in\mathcal{S}_{t+1}$, we get $V_{t+1}(1)\le e^{\kappa[H_{N-1}-H_t+2/(3t)]}\,\Phi_{t+1}^{\kappa_{t+1}}(1)$.

For the second step, $\log(\Phi_{t+1}^{\kappa_{t+1}}(1)/\pi_t)$ is the integral over $\tau\in[0,\kappa_{t+1}]$ of the derivative in Lemma A.3 with $s=t+1$ and $b=1$. As in Step 2 of (b), the bound of that lemma is largest at $\tau=0$. There $X=1$, $Y=t$ and $\frac{1}{1-z}=\frac{(t+1)^2}{4t}$, and the bound is $\log\frac{(t+1)^2}{4t}+\frac{t^2+1}{2t(t+1)}+\frac{t^2+1}{12t^2}\le\log\frac{(t+1)^2}{4t}+\frac23$, because the two fractions are at most $\frac12$ and $\frac16$ for $t\ge1$. Multiplying by the length κ_{t+1} of the interval gives (c).

(d) Insert (c) into Proposition 2.5(a), and use $\kappa_{t+1}=\kappa(t+11)$ and $\pi_t=c_1w_t$. □

Proof of Theorem 2.6. (a) By Proposition 2.5(a) and Lemma A.1(d), $C_N\ge\varepsilon c_1\sum_t w_t(1-\varepsilon)^{t-1}$. Since $u\mapsto(1-\varepsilon)^u$ is convex, Jensen's inequality and $\sum_t w_t(t-1)=2-\frac{6}{h+2}$ give the stronger bound

$$C_N \ge \frac{4\varepsilon}{N+2}(1-\varepsilon)^{2-6/(h+2)} \ge \frac{4\varepsilon}{N+2}(1-2\varepsilon). \tag{4}$$

The last step uses $0\le 2-6/(h+2)\le 2$ and $(1-\varepsilon)^2\ge1-2\varepsilon$.

(b) *Step 1. The constants.* Let $0<\varepsilon(1+\log N)\le\frac{1}{100}$. Since $\log N\ge0$, we have $\varepsilon\le\frac{1}{100}$. So $1-4\varepsilon\ge0.96$ and $\kappa\le2\varepsilon/0.96\le2.1\varepsilon$. Also $\kappa\le2\varepsilon(1+8\varepsilon)$, because $(1+8\varepsilon)(1-4\varepsilon)=1+4\varepsilon(1-8\varepsilon)\ge1$. Since $\log\frac{(t+1)^2}{4t}\ge0$ and $\frac23(t+11)>1+\log t\ge H_t$, we have $\mathcal{G}(t)>0$. Since $\frac{(t+1)^2}{4t}\le\frac{t+1}{2}$, $\frac23<\log2$ and $H_t\ge\frac{2}{3t}$, also $\mathcal{G}(t)\le(t+11)\log(t+1)\le12t\log(t+1)$. So $\kappa\mathcal{G}(t)\le25.2\,\varepsilon t\log(t+1)$, and this is at most $\frac t2\log2$ for $t\le h$, since then $25.2\,\varepsilon\log(t+1)\le25.2\,\varepsilon\log N\le0.252<\frac12\log2$.

Step 2. The sum over t. Since $e^y-1\le ye^y$ for $y\ge0$, Step 1 gives $e^{\kappa\mathcal{G}(t)}-1\le\kappa\mathcal{G}(t)2^{t/2}$. By Proposition A.4(d), $(1-\varepsilon)^{t-1}\le1$, $\sum_t w_t=1$ and $w_t\le2t2^{-t}$ (Proposition 2.5(c)),

$$C_N \le \varepsilon c_1 e^{\kappa H_{N-1}}\Big(1+50.4\,\varepsilon\sum_{t\ge1}t^2\log(t+1)2^{-t/2}\Big),$$

and we bound the series as follows. The ratio of its consecutive terms is $(1+\frac1t)^2\frac{\log(t+2)}{\log(t+1)}2^{-1/2}$, which decreases in t and is below $\frac34$ at $t=60$. So the terms with $t>60$ add up to less than four

times the term at $t = 61$, and this is below 10^{-4}. The first 60 terms add up to 102.5864 to four decimals (enclosed in ball arithmetic by `certify_elephant.py`), so the series is below 102.59.

Step 3. The factor $e^{\kappa H_{N-1}}$. Finally $H_{N-1} \le 1 + \log N$, so $\kappa H_{N-1} \le 2\varepsilon \log N + 2\varepsilon + 16\varepsilon^2(1 + \log N) \le 2\varepsilon \log N + 3\varepsilon$. With $1 + y \le e^y$ this proves (b), since $3 + 50.4 \cdot 102.59 < 5200$. Any upper bound for the series below $(5200 - 3)/50.4$, which exceeds 103.1, would give the same constant. □

B The families and the mode

This appendix states the family estimates used for the mode bound in Corollary 3.11. Their proofs are in Appendix D, in the order stated here, and also support the tail theorem in Theorem S2.1. We keep the conventions of Section 3. All constants in this appendix are explicit. Appendix D writes C for an absolute constant whose value may change from line to line. Since M is independent of X_1, $f_N(m) = \mathbb{P}(M_N = m)$. Every path of a chain $M^{[k_0]}$ is nondecreasing in time, and we use the paths of Y and $Z^{(j)}$ fixed after Lemma 3.3. The size $|\mathcal{T}_v|$ of the family has the law β_v of Lemma 2.4. Let $\bar\beta_v(s) = \mathbb{P}(|\mathcal{T}_v| \ge s)$, with $\bar\beta_v(s) = 1$ for $s \le 1$. By Pascal's rule $\binom{N-s}{v-1} - \binom{N-s-1}{v-1} = \binom{N-s-1}{v-2}$ the tail of β_v telescopes. So, for $1 \le s \le N-1$,

$$\bar\beta_v(s) = \binom{N-s}{v-1} \Big/ \binom{N-1}{v-1} = \prod_{i=1}^{v-1} \Big(1 - \frac{s-1}{N-i}\Big) \le e^{-(v-1)(s-1)/(N-1)}, \qquad \beta_v(s) = \frac{v-1}{N-s}\,\bar\beta_v(s), \tag{5}$$

where the product vanishes as soon as one factor does. Where they are defined, the ratios are

$$\frac{\beta_v(d)}{\beta_v(d-1)} = 1 - \frac{v-2}{N-d}, \qquad \frac{\bar\beta_v(s-1)}{\bar\beta_v(s)} = 1 + \frac{v-1}{N-s-v+2}. \tag{6}$$

Lemma B.1 (monotone sandwich)**.** *Let $0 \le a \le b \le N$. If $m_h \le a$, then $(b-a+1)f_N(b) \le \mathbb{P}(a \le M_N \le b)$. If $b \le m_h$, then $(b-a+1)f_N(a) \le \mathbb{P}(a \le M_N \le b)$.*

Lemma B.2 (tail)**.** *Let $0 < \varepsilon < 1$, $k_0 \ge 1$ and $t \ge k_0$. For every integer $z \ge 1$,*

$$\mathbb{P}\big(M_t^{[k_0]} \ge z\big) \le \frac{\varepsilon t H_z}{z}, \qquad \mathbb{E}M_t^{[k_0]} \le \varepsilon t \log \frac{t}{k_0}.$$

In particular $\mathbb{P}(M_N \ge z) \le xH_z/z$, and $\mathbb{P}(Z_s^{(j)} \ge z) \le xH_z/z$ for $j - 1 \le s \le N$.

Lemma B.3 (the first flipped family)**.** *For $1 \le d \le N-1$,*

$$\frac{x}{d(d+1)}\Big(1 - \frac{2x(N-d-1)}{N(d+2)}\Big) \le \mathbb{P}\big(|\mathcal{T}_V| = d\big) \le \frac{x}{d(d+1)}, \qquad \mathbb{P}\big(|\mathcal{T}_V| \ge d\big) \le x\Big(\frac{1}{d} - \frac{1}{N}\Big).$$

Proposition B.4 (local lower bound)**.** *Let $N \ge 64$, $0 < \varepsilon < 1$ and $2 \le m \le N/4$. Then $f_N(m) \ge \frac{x}{m(m+1)}(1 - e_1'(m)) \ge \frac{x}{m(m+1)}(1 - e_1(m))$, where*

$$e_1'(m) = 2\varepsilon H_{m+1} + \frac{xH_m}{m} + \frac{2x}{m+2} + 8\varepsilon \log N + 20me^{-m/9}$$

and

$$e_1(m) = 20\Big(\frac{x \log(m+1)}{m} + \varepsilon \log N + me^{-m/9}\Big).$$

Idea of the proof. We use the formula of Lemma 3.3. Given the time j of the first flip, the count M_N is $d - Y_d + Z^{(j)}_{N-d}$, where d is the size of the first flipped family. As d grows by one this quantity moves by at most one, so it hits m at some d close to m unless Y or Z is large. (If no step of the family flips back and the rest of the tree has no flip, then $Y = Z = 0$ and the quantity is d itself, which hits m at $d = m$.) The law β_j of d is nearly flat near m, and this gives almost the full weight $\beta_j(m)$. Then we sum over j. The error e_1 collects the chance that Y or Z is large, the small slope of β_j, and the first flips that come too late to be treated this way.

Lemma B.5 (mode bound)**.** *Let $N \geq 64$ and $x \geq 1$, and put $\bar{c} = \frac{1}{40}$ and $A_0 = 15400$. If $\varepsilon \log N \leq \bar{c}$ and $804x \log(x+2) \leq N$, then $m_h \leq A_0 x \log^2(x+1)$. In particular this holds if $\varepsilon \log N \leq \bar{c}$ and $A_0 x \log(x+2) \leq N/4$.*

C Proofs for the bulk

We prove Theorems 3.8 and 3.9, Lemma 3.10 and Corollary 3.11. We keep the notation of Section 3 and of Appendix B. So $\mathcal{T}$ is the tree of Lemma 3.1, $\mathcal{T}_v$ is the family of v, and μ_v is the mark of v. By Lemma 2.4 and the discussion after Lemma 3.1, $|\mathcal{T}_v|$ has the law β_v, with mean N/v, and $\sum_{v=2}^{N} \beta_v(d) = N/(d(d+1))$ for $1 \leq d \leq N-1$. Let

$$\Psi_N(\theta) = \sum_{d=1}^{N-1} \frac{e^{i\theta d} - 1}{d(d+1)}, \qquad \Psi_\infty(\theta) = \sum_{d \geq 1} \frac{e^{i\theta d} - 1}{d(d+1)}.$$

The law CP_N has the characteristic function $\widehat{\mathrm{CP}}_N(\theta) = e^{x\Psi_N(\theta)}$.

C.1 Proof of Theorem 3.8

We use $\mathrm{TV}(P, Q) = \frac{1}{2}\sum_k |P(k) - Q(k)|$. The case $N = 1$ is immediate. Assume $N \geq 2$ and $0 < \varepsilon < 1$. Put $Z_+ = \sum_{v=2}^{N} \mu_v |\mathcal{T}_v|$. Unless two marked vertices are strictly nested, $M_N = Z_+$. The family means therefore give

$$\mathbb{P}(M_N \neq Z_+) \leq \varepsilon^2 \sum_{v=2}^{N} (N/v - 1) = \varepsilon^2 (N H_N - 2N + 1).$$

Given the tree, replace the independent Bernoulli marks by independent Poisson variables of mean ε. The scalar total variation is $\varepsilon(1 - e^{-\varepsilon}) \leq \varepsilon^2$, since the only positive Bernoulli-minus-Poisson mass is at 1. Product coupling and the weighted sum cost at most $(N-1)\varepsilon^2$. The resulting conditional law is $\mathrm{CP}(\Lambda)$, where $\Lambda = \varepsilon \sum_{v=2}^{N} \delta_{|\mathcal{T}_v|}$ has deterministic mass $\lambda = (N-1)\varepsilon$. By the family identity, $\mathbb{E}\Lambda\{d\} = x/[d(d+1)]$, so $\mathrm{CP}(\mathbb{E}\Lambda) = \mathrm{CP}_N$.

Use the L1 norm on signed measures and convolution $*$, for which $\|\sigma * \tau\|_1 \leq \|\sigma\|_1 \|\tau\|_1$. On intensities of mass λ, the CP map and its derivatives in a zero-mass direction D are

$$F(\nu) = e^{-\lambda} \sum_{r \geq 0} \frac{\nu^{*r}}{r!}, \qquad F'(\nu)[D] = F(\nu) * D, \qquad F''(\nu)[D, D] = F(\nu) * D * D,$$

where $\nu^{*0} = \delta_0$. The series and derivatives converge in L1. Let $\mathcal{F}_j = \sigma(U_1, \ldots, U_j)$, $\Lambda_j = \mathbb{E}(\Lambda \mid \mathcal{F}_j)$ and $D_j = \Lambda_j - \Lambda_{j-1}$. Each Λ_j is nonnegative of mass λ. Along the segment between consecutive

intensities F has norm 1, so Taylor's remainder has norm at most $\|D_j\|_1^2/2$. The linear term has conditional mean zero given $\mathcal{F}_{j-1}$. Telescoping expectations and dividing by two gives

$$\mathrm{TV}(\mathbb{E}F(\Lambda), F(\mathbb{E}\Lambda)) \le \frac{1}{4}\sum_{j=1}^{N-1}\mathbb{E}\|D_j\|_1^2.$$

If $\Lambda^{(j)}$ resamples only U_j, independence gives $D_j = \mathbb{E}(\Lambda - \Lambda^{(j)} \mid \mathcal{F}_j)$. Conditional Jensen thus bounds its squared norm in expectation by that of $\Lambda - \Lambda^{(j)}$. Moving the subtree of $j+1$ changes family sizes only along the old and new parent paths, excluding the root. Each altered atom contributes at most 2ε in L1. Writing dp for depth, and using $(a+b)^2 \le 2(a^2+b^2)$, gives

$$\mathbb{E}\|\Lambda - \Lambda^{(j)}\|_1^2 \le 16\varepsilon^2\mathbb{E}\mathrm{dp}(U_j)^2 \le 16\varepsilon^2(H_{j-1}^2 + H_{j-1}).$$

For the last inequality, the depth pgf satisfies $g_1(s) = 1$ and $jg_{j+1}(s) = s\sum_{v\le j} g_v(s)$. Subtraction gives $g_v(s) = \prod_{i=1}^{v-1}(i-1+s)/i$. Thus the depth has the law of a sum of independent Bernoulli$(1/i)$ variables, and its second moment is at most $H_{v-1}^2 + H_{v-1}$. Average over the independent uniform parent $U_j \in \{1,\dots,j\}$.

Adding the three errors proves the stronger bound

$$\mathrm{TV}(\mathcal{L}(M_N), \mathrm{CP}_N) \le \varepsilon^2\left\{NH_N - N + 4\sum_{j=0}^{N-2}(H_j^2 + H_j)\right\} \le \delta_\Sigma.$$

Finally $H_N \le 1 + \log N$ and $H_j \le H_N$ give $\delta_\Sigma \le (x^2/N)\{\log N + 2 + 4(H_N^2 + H_N)\} \le \delta_B$. Every singleton discrepancy is at most total variation. □

C.2 Proof of Theorem 3.9

We write $B(t) = -it\log|t| - \frac{\pi}{2}|t|$ for real $t \ne 0$, and $B(0) = 0$. For $t > 0$ this is the value of $u\log u$ at $u = -it$, with the principal logarithm, and $B(-t) = \overline{B(t)}$. We need three facts about Ψ_N and Ψ_∞.

Fact 1. For real θ, $|\Psi_N(\theta) - \Psi_\infty(\theta)| \le \sum_{d\ge N}\frac{2}{d(d+1)} = \frac{2}{N}$.

Fact 2. For $0 < |\theta| \le \pi$ we have $\Psi_\infty(\theta) = -i\theta\log(-i\theta) + R(\theta)$ with $|R(\theta)| \le \theta^2\big(0.631\,|\log|\theta|| + 1.53\big)$. The reason is that Ψ_∞ has a closed form in which θ enters only through $1 - e^{i\theta}$, and $1 - e^{i\theta}$ is close to $-i\theta$ for small θ. The term R is what this replacement changes. We prove it in the next four paragraphs.

We start with the closed form. For $0 < |s| \le 1$, $s \ne 1$,

$$\sum_{d\ge1}\frac{s^d - 1}{d(d+1)} = \chi(s) := \frac{(1-s)\log(1-s)}{s},$$

because $\frac{1}{d(d+1)} = \frac{1}{d} - \frac{1}{d+1}$, $\sum_d s^d/d = -\log(1-s)$, $\sum_d s^d/(d+1) = (-\log(1-s) - s)/s$ and $\sum_d \frac{1}{d(d+1)} = 1$. So $\Psi_\infty(\theta) = \chi(e^{i\theta})$.

Next we compare $1 - e^{i\theta}$ with $-i\theta$. Write $1 - e^{i\theta} = (-i\theta)\,\mathsf{u}$ with $\mathsf{u} = \mathrm{sinc}(\theta/2)\,e^{i\theta/2}$, where $\mathrm{sinc}\,y = \sin y/y$ is positive for $|y| \le \frac{\pi}{2}$. So u is close to 1 when θ is small. The argument of $-i\theta$ is $\mp\frac{\pi}{2}$ for $\pm\theta > 0$ and that of u is $\theta/2$. They add up to a number in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\log(1 - e^{i\theta}) = \log(-i\theta) + \log\mathsf{u}$ for principal logarithms.

Now we get a formula for R. By the two steps above, $\chi(e^{i\theta}) = (-i\theta)\,\mathsf{u}e^{-i\theta}\big(\log(-i\theta) + \log \mathsf{u}\big)$. Subtracting $-i\theta\log(-i\theta)$ gives

$$R(\theta) = -i\theta\log(-i\theta)\big(\mathsf{u}e^{-i\theta} - 1\big) - i\theta\,\mathsf{u}e^{-i\theta}\log \mathsf{u}.$$

It remains to bound the three quantities in this formula. We show that

$$|\mathsf{u}e^{-i\theta} - 1| \le 0.631\,|\theta|, \qquad |\log \mathsf{u}| \le 0.53\,|\theta|, \qquad |\log(-i\theta)| \le |\log|\theta|| + \frac{\pi}{2}.$$

For the first, put $y = \theta/2$. Then $|\mathsf{u}e^{-i\theta} - 1| = |\operatorname{sinc} y - e^{iy}| \le (1 - \operatorname{sinc} y) + |1 - e^{iy}| \le \frac{y^2}{6} + |y| \le (\frac{1}{2} + \frac{\pi}{24})|\theta|$, where the last step uses $|\theta| \le \pi$, and $\frac{1}{2} + \frac{\pi}{24} \le 0.631$. For the second, $|\log \mathsf{u}|^2 = \log^2 \operatorname{sinc}(\theta/2) + \theta^2/4$, and $-\log \operatorname{sinc} y = \sum_{k\ge1} \frac{\zeta(2k)}{k\pi^{2k}} y^{2k}$ has positive coefficients. So $|\log \operatorname{sinc}(\theta/2)|/|\theta|$ increases in $|\theta|$, and its value at $|\theta| = \pi$ gives $|\log \mathsf{u}| \le \big(\frac{1}{4} + \frac{\log^2(\pi/2)}{\pi^2}\big)^{1/2}|\theta| \le 0.53\,|\theta|$. The third holds because $-i\theta$ has modulus $|\theta|$ and argument $\pm\frac{\pi}{2}$. Since $|\mathsf{u}e^{-i\theta}| \le 1$, the formula for R now gives

$$|R(\theta)| \le |\theta|\Big(|\log|\theta|| + \frac{\pi}{2}\Big) \cdot 0.631\,|\theta| + |\theta| \cdot 0.53\,|\theta| \le \theta^2\big(0.631\,|\log|\theta|| + 1.53\big),$$

as $0.631 \cdot \frac{\pi}{2} + 0.53 \le 1.53$.

Fact 3. For $|\theta| \le \pi$, $-\operatorname{Re}\Psi_\infty(\theta) = \sum_d \frac{1-\cos\theta d}{d(d+1)} \ge \frac{|\theta|}{2\pi}$. We may take $\theta \ne 0$. Keep only the terms with $d \le D := \lfloor \pi/|\theta| \rfloor$ and use $1 - \cos v \ge 2v^2/\pi^2$ for $|v| \le \pi$. This gives at least $\frac{2\theta^2}{\pi^2}\sum_{d\le D} \frac{d}{d+1} \ge \frac{\theta^2 D}{\pi^2} \ge \frac{|\theta|}{2\pi}$, since $D \ge \frac{\pi}{2|\theta|}$.

The function e^B is a characteristic function of class L. Let S_x have the law CP_∞ with parameter x, namely the law of $\sum_{d\ge1} d\,\mathcal{N}_d$. Its characteristic function is $e^{x\Psi_\infty(\theta)}$. By Fact 2, for $t \ne 0$ and $x \ge \max(1, |t|/\pi)$,

$$x\Psi_\infty(t/x) - it\log x = B(t) + xR(t/x), \qquad |xR(t/x)| \le \frac{t^2}{x}\big(0.631(|\log|t|| + \log x) + 1.53\big). \quad (7)$$

So the characteristic function of $(S_x - x\log x)/x$ tends to $e^{B(t)}$ as $x \to \infty$, for every real t. The limit is continuous at 0, so by Lévy's continuity theorem it is the characteristic function of a probability law. It is integrable, so this law has the bounded continuous density f_L of Theorem 3.9. For $0 < c < 1$ and real t we have $B(ct) = cB(t) - ict\log c$, and hence

$$e^{B(t)} = e^{B(ct)} \cdot e^{B((1-c)t)}\, e^{it[(1-c)\log(1-c) + c\log c]}.$$

The last two factors form the characteristic function of a scaled and shifted copy of the law. So the law is of class L (self-decomposable), and f_L is unimodal by [38].

Proof of (a). By Fourier inversion on $\mathbb{Z}$ [42] and the substitution $\theta = t/x$,

$$x\,\mathrm{CP}_N(m) = \frac{1}{2\pi}\int_{-\pi x}^{\pi x} e^{-i\lambda_m t} e^{A(t)}\,dt, \qquad A(t) = x\Psi_N(t/x) - it\log x.$$

Let $|t| \le \pi x$. By (7) and Fact 1,

$$|A(t) - B(t)| \le \frac{t^2}{x}\big(0.631(|\log|t|| + \log x) + 1.53\big) + \frac{2x}{N},$$

and by Facts 1 and 3, $\max(\operatorname{Re} A(t), \operatorname{Re} B(t)) \le -\frac{|t|}{2\pi} + \frac{2x}{N}$. Now $|e^A - e^B| \le |A-B|\, e^{\max(\operatorname{Re} A, \operatorname{Re} B)}$, and $\frac{2x}{N} \le \frac{1}{2}$ because $N \ge 4x$. So for $|t| \le \pi x$

$$|e^{A(t)} - e^{B(t)}| \le e^{1/2} e^{-|t|/2\pi} \Big[\frac{t^2}{x} \big(0.631(|\log|t|| + \log x) + 1.53 \big) + \frac{2x}{N} \Big].$$

For $s \ge 1$ the tangent of the logarithm at 16 gives $\log s \le s/16 + \log 16 - 1$, and for $0 < s < 1$ we have $s|\log s| \le 1/e$. So $s^2|\log s| \le s^3/16 + (4\log 2 - 1)s^2 + s/e$ for every $s \ge 0$. We insert this with $s = |t|$ and integrate, using $\int_0^\infty t^k e^{-t/2\pi}\, dt = k!\,(2\pi)^{k+1}$. This gives

$$\frac{1}{2\pi} \int_{-\pi x}^{\pi x} |e^{A(t)} - e^{B(t)}|\, dt \le e^{1/2} \Big[\frac{K_0 + 16\pi^2 \cdot 0.631 \log x}{x} + \frac{4x}{N} \Big],$$

where $K_0 = 6\pi^3 \cdot 0.631 + 16\pi^2 \big(0.631(4\log 2 - 1) + 1.53\big) + 4\pi \cdot 0.631/e \le 538.6$. Beyond πx we use $|e^{B(t)}| = e^{-\pi|t|/2}$, which gives $\frac{1}{2\pi} \int_{|t|>\pi x} |e^{B(t)}|\, dt = \frac{2}{\pi^2} e^{-\pi^2 x/2} \le \frac{4}{\pi^4 x} \le \frac{0.042}{x}$, since $e^{-y} \le 1/y$ for $y > 0$. Since $f_L(\lambda_m) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-i\lambda_m t} e^{B(t)}\, dt$, the error is at most the sum of the two integrals. With $e^{1/2} \le 1.6488$ and $16\pi^2 \cdot 0.631 \le 99.66$ this sum is at most

$$\frac{1.6488 \cdot 538.6 + 0.042 + 1.6488 \cdot 99.66 \log x}{x} + \frac{6.5952\, x}{N} \le \frac{889 + 165 \log x}{x} + \frac{6.6\, x}{N}.$$

The second bound in (a) holds because $165 \le 889$. This proves (a). Fourier inversion on $\mathbb{Z}$ holds at every integer m, and $\mathrm{CP}_N(m) = 0$ for $m < 0$, so the proof covers every integer m.

Proof of (b). Since $\varepsilon \le \frac{1}{4}$, we have $N \ge 4x$. By Theorem 3.8, $|x f_N(m) - x\, \mathrm{CP}_N(m)| \le x\delta_\Sigma \le x\delta_B$. Add (a).

Proof of (c). Drop the index n and let S_x have law CP_∞. Adding the independent Poisson jumps of sizes $d \ge N$ couples CP_N to S_x at cost at most $1 - e^{-x \sum_{d \ge N} 1/(d(d+1))} = 1 - e^{-x/N}$. Thus

$$\mathrm{TV}(\mathcal{L}(M_N), \mathcal{L}(S_x)) \le \delta_\Sigma + 1 - e^{-x/N} \longrightarrow 0$$

under the weaker condition $x^2 \log^2 N / N \to 0$. Equation (7) and Lévy's continuity theorem give $(S_x - x\log x)/x \Rightarrow f_L$ as $x \to \infty$. Total variation contracts under this affine map, so the same weak limit holds for M_N. At the crossing $x_N \sim 2\log N$, and both conditions hold. □

C.3 Proof of Lemma 3.10

Step 1. A real integral. We claim that for every real λ and $k \ge 0$,

$$f_L^{(k)}(\lambda) = \frac{1}{\pi} \int_0^\infty (-s)^k\, e^{-s\log s - \lambda s} \sin(\pi s)\, ds. \tag{8}$$

For $k = 0$ this is [29, Eq. (4.3)], and we include a short proof. Since $B(-t) = \overline{B(t)}$, the definition of f_L gives $f_L(\lambda) = \frac{1}{\pi} \operatorname{Re} \int_0^\infty \Phi(-it)\, dt$ with $\Phi(u) = \exp(u\log u + \lambda u)$. The function Φ is holomorphic off the negative real axis. Put $u = \rho e^{i\varphi}$ with $-\pi < \varphi \le -\frac{\pi}{2}$. Then $\operatorname{Re}(u\log u) = \rho(\log\rho\cos\varphi - \varphi\sin\varphi)$, where $\cos\varphi \le 0$ and $\varphi\sin\varphi \ge 0$. For $\log\rho \ge |\lambda| + 1$ this gives $\operatorname{Re}(u\log u + \lambda u) \le -\rho(|\cos\varphi| + \varphi\sin\varphi) \le -\rho$, because $|\cos\varphi| + \varphi\sin\varphi \ge 1$ on $[-\pi, -\frac{\pi}{2}]$ (for $\psi = \varphi + \pi \in [0, \frac{\pi}{2}]$ it equals $\cos\psi + (\pi - \psi)\sin\psi \ge \cos\psi + \sin\psi \ge 1$). Also Φ is bounded near 0. So by Cauchy's theorem we may turn the ray $u = -it$ to the lower side of the negative real axis, where $u = -s$ and $\log u = \log s - i\pi$. There $u\log u = -s\log s + i\pi s$, and $dt = i\, du = -i\, ds$. Hence

$$f_L(\lambda) = \frac{1}{\pi} \operatorname{Re} \Big[-i \int_0^\infty e^{-s\log s - \lambda s}\, e^{i\pi s}\, ds \Big] = \frac{1}{\pi} \int_0^\infty e^{-s\log s - \lambda s} \sin(\pi s)\, ds.$$

The integrand decays faster than every exponential, so we may differentiate under the integral sign. This proves (8). Also f_L'' is bounded on the real line, because $t^2 e^{B(t)}$ is integrable and we may differentiate the Fourier formula for f_L twice.

Step 2. Certified enclosures. Let $-0.223 \le \lambda \le 0$ and $k \in \{1, 2\}$, and split the integral in (8) at $\delta = 2^{-20}$ and at 30. On $[0, \delta]$ we have $-s \log s \le 1/e$, $-\lambda s \le 0.223\, s$ and $|\sin \pi s| \le \pi s$, so this part is at most $\pi e^{1/e + 0.223\delta} \delta^{k+2}/(k+2)$ in absolute value. On $[30, \infty)$ we have $s \log s + \lambda s \ge cs$ with $c = \log 30 - 0.223 > 3$, so this part is at most $\int_{30}^{\infty} s^2 e^{-cs}\, ds \le 30^2 e^{-30c}$. Both are below 10^{-17}. On $[\delta, 30]$ the integrand is holomorphic in a neighborhood of the path, and we enclose the integral with the rigorous integration algorithm of [39], in ball arithmetic with 128 bits. The parameter λ enters as a ball, so one run encloses the integral for every λ in an interval. The script `certify_elephant.py` does this and reports

$$f_L'(-0.2229) \in [9.232 \cdot 10^{-6}, 9.234 \cdot 10^{-6}], \qquad f_L'(-0.2227) \in [-6.548 \cdot 10^{-6}, -6.546 \cdot 10^{-6}],$$

and $f_L''(\lambda) \in [-0.0792, -0.0786]$ for every $\lambda \in [-0.2229, -0.2227]$. In the same way it finds that f_L' is positive at -0.22278299 and negative at -0.22278297. The two enclosures are $6.8984026 \cdot 10^{-10}$ and $-8.8809315 \cdot 10^{-10}$ with radii below 10^{-17}, and these are the smallest margins in the proof.

Step 3. The mode. Write $[a, b] = [-0.2229, -0.2227]$. By Step 2, f_L' is strictly decreasing on $[a, b]$ with $f_L'(a) > 0 > f_L'(b)$. So f_L' has exactly one zero λ_{L} in $[a, b]$, and f_L is strictly increasing on $[a, \lambda_{\mathrm{L}}]$ and strictly decreasing on $[\lambda_{\mathrm{L}}, b]$. Since f_L is unimodal, there is a point λ_* such that f_L is nondecreasing on $(-\infty, \lambda_*]$ and nonincreasing on $[\lambda_*, \infty)$. If $\lambda_* > \lambda_{\mathrm{L}}$, then f_L would be nondecreasing just to the right of λ_{L}, which is false. In the same way $\lambda_* < \lambda_{\mathrm{L}}$ is impossible. So $\lambda_* = \lambda_{\mathrm{L}}$, and f_L is nondecreasing on $(-\infty, \lambda_{\mathrm{L}}]$ and nonincreasing on $[\lambda_{\mathrm{L}}, \infty)$. By Step 2 again, $\lambda_{\mathrm{L}} \in (-0.22278299, -0.22278297)$. This proves (a) and (b).

Step 4. The constant κ_L. Let $\delta_0 = \min(\lambda_{\mathrm{L}} - a, b - \lambda_{\mathrm{L}}) > 0$. For $|\lambda - \lambda_{\mathrm{L}}| \le \delta_0$, Taylor's formula with $f_L'(\lambda_{\mathrm{L}}) = 0$ and $f_L'' \le -0.078$ on $[a, b]$ gives $f_L(\lambda_{\mathrm{L}}) - f_L(\lambda) \ge 0.039(\lambda - \lambda_{\mathrm{L}})^2$. For $|\lambda - \lambda_{\mathrm{L}}| > \delta_0$ the monotonicity of Step 3 gives $f_L(\lambda) \le f_L(\lambda_{\mathrm{L}} \pm \delta_0) \le f_L(\lambda_{\mathrm{L}}) - 0.039\, \delta_0^2$. Since $\delta_0 < 1$, the bound in (c) holds with $0.039\, \delta_0^2$ in place of κ_L. By (a), $\lambda_{\mathrm{L}} - a > 0.2229 - 0.22278299 > 1.17 \cdot 10^{-4}$ and $b - \lambda_{\mathrm{L}} > 0.22278297 - 0.2227 = 8.297 \cdot 10^{-5}$, so $0.039\, \delta_0^2 > 0.039 \cdot (8.297 \cdot 10^{-5})^2 > 2.68 \cdot 10^{-10}$. So (c) holds with $\kappa_L = 2.6 \cdot 10^{-10}$. □

C.4 Proof of Corollary 3.11

The method is to compare f_N at a mode with f_N at the integer m' nearest to the predicted place $x \log x + \lambda_{\mathrm{L}} x$. By Theorem 3.9(b) both values are close to values of f_L, and f_L falls off quadratically away from λ_{L}. So a mode cannot be far from m'.

We first prove (a). So we assume (H), $E' < \kappa_L/2$ and $m_h \le N/4$, with E_1 and E' as in the statement.

Let m' be the integer nearest to $x \log x + \lambda_{\mathrm{L}} x$. This number increases in $x \ge 1$ and is at least $\lambda_{\mathrm{L}} > -\frac{1}{2}$, so $m' \ge 0$. Also $|\lambda_{m'} - \lambda_{\mathrm{L}}| \le \frac{1}{2x}$, and since $f_L'(\lambda_{\mathrm{L}}) = 0$, $f_L(\lambda_{m'}) \ge f_L(\lambda_{\mathrm{L}}) - \|f_L''\|_\infty/(8x^2)$. Let $\tilde{m}$ be a mode of f_N or $\tilde{m} = m^*$. We claim that $f_N(\tilde{m}) \ge f_N(m') - 16xH_N/N^2$. For a mode of f_N this is clear. For m^* we have $f_N(m^*) + f_N(N - m^*) = 2P_N(N - m^*) \ge 2P_N(N - m') \ge f_N(m')$. Here $N - m^* \ge \lceil N/2 \rceil \ge \lceil N/4 \rceil \ge m_h$. So f_N is nonincreasing on $\{\lceil N/4 \rceil, \dots, N\}$ by Lemma 3.4. There are at least $N/4$ integers from $\lceil N/4 \rceil$ to $\lceil N/2 \rceil$. So Lemma B.1 (with $a = \lceil N/4 \rceil \ge m_h$ and $b = \lceil N/2 \rceil$) gives the second inequality below, and Lemma B.2 gives the third,

$$f_N(N - m^*) \le f_N(\lceil N/2 \rceil) \le \frac{\mathbb{P}(M_N \ge \lceil N/4 \rceil)}{N/4} \le \frac{16xH_N}{N^2}.$$

Now Theorem 3.9(b), used at $\tilde{m}$ and at m', gives

$$f_L(\lambda_{\tilde{m}}) \geq x f_N(\tilde{m}) - E_1 \geq x f_N(m') - \frac{16x^2 H_N}{N^2} - E_1 \geq f_L(\lambda_{m'}) - 2E_1 - \frac{16x^2 H_N}{N^2} \geq f_L(\lambda_{\mathrm{L}}) - 2E'.$$

Since $2E' < \kappa_L$, Lemma 3.10(c) gives $(\lambda_{\tilde{m}} - \lambda_{\mathrm{L}})^2 \leq 2E'/\kappa_L$, which is the claim.

Now we prove (b). Let N be large and $\log N \leq x \leq 2\log N$. Then $x\delta_B$, x/N, $16x^2 H_N/N^2$ and $\|f_L''\|_\infty/(8x^2)$ are all at most $\log x/x$, because each is $O(\log^5 N/N)$ or $O(1/\log^2 N)$. So $E' \leq C_1 \log x/x$ with an absolute constant C_1. As $\log x/x \leq \log\log N/\log N$, for large N we have $E' < \kappa_L/2$. Also (H) holds, $\varepsilon \log N \leq 2\log^2 N/N \leq \frac{1}{40}$ and $804x\log(x+2) \leq N$, so Lemma B.5 gives $m_h \leq A_0 x \log^2(x+1) \leq N/4$. Hence by (a) $|m^* - x\log x - \lambda_{\mathrm{L}} x| \leq x\sqrt{2E'/\kappa_L} \leq K\sqrt{x\log x}$ with $K = \sqrt{2C_1/\kappa_L}$, and the same holds for every mode of f_N. At the crossing $x_N = 2\log N - \log\log N + O(1)$ for even N, by Theorem 2.7(b). So $\log N \leq x_N \leq 2\log N$ for large even N. □

D Proofs for the family estimates

These proofs establish the estimates stated in Appendix B. We retain its notation, including the pathwise construction of the chains.

Proof of Lemma B.1. By Lemma 3.4, $f_N(\ell) \geq f_N(b)$ for $a \leq \ell \leq b$ in the first case, and $f_N(\ell) \geq f_N(a)$ in the second. Sum over ℓ. □

Proof of Lemma B.2. We use the tree of Lemma 3.1 on $[t]$, with marks only on $v > k_0$. Every odd vertex lies in the family of a marked vertex, so $M_t^{[k_0]} \leq \sum_{v \text{ marked}} |\mathcal{T}_v|$. We call a family large if its size is at least z, and small otherwise. If $M_t^{[k_0]} \geq z$, then some marked family is large, or the small marked families have total size at least z. We use the union bound for the first event and Markov's inequality for the second. Marks and tree are independent, so

$$\mathbb{P}\big(M_t^{[k_0]} \geq z\big) \leq \varepsilon \sum_v \mathbb{P}(|\mathcal{T}_v| \geq z) + \frac{\varepsilon}{z} \sum_v \mathbb{E}\big[|\mathcal{T}_v|; |\mathcal{T}_v| < z\big].$$

Extend both sums to all nonroot vertices. By the first sum in Lemma 2.4, with t in place of N, on average $\frac{t}{d(d+1)}$ such vertices have a family of size d, for $1 \leq d < t$. The first sum is therefore at most $\sum_{d \geq z} \frac{t}{d(d+1)} = \frac{t}{z}$, and the second is at most $\sum_{d=1}^{z-1} \frac{t}{d+1} = t(H_z - 1)$. These bounds also hold when $z > t$. Together they give $\frac{\varepsilon t H_z}{z}$. For the mean, only the vertices $v > k_0$ can be marked and $\mathbb{E}|\mathcal{T}_v| = t/v$, so $\mathbb{E}M_t^{[k_0]} \leq \varepsilon \sum_{v=k_0+1}^{t} \frac{t}{v} \leq \varepsilon t \log \frac{t}{k_0}$. □

Proof of Lemma B.3. By Lemma 3.3, $\mathbb{P}(|\mathcal{T}_V| = d) = \sum_j \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)$, and $1 - (j-2)\varepsilon \leq (1-\varepsilon)^{j-2} \leq 1$. Apply the two sums of Lemma 2.4, and sum the upper bound over $d' \geq d$. □

Proof of Proposition B.4. We bound the inner sum of Lemma 3.3 from below for each early first flip, namely for $2 \leq j \leq J := \lfloor N/8 \rfloor$. The terms with $j > J$ are nonnegative and we drop them.

Step 1. One value of j. Fix $2 \leq j \leq J$ and put $\Gamma_d = d - Y_d + Z_{N-d}^{(j)}$ for $1 \leq d \leq N - j + 1$. The inner sum in Lemma 3.3 is $Q_j(m) = \mathbb{E}S_j$ with $S_j = \sum_d \beta_j(d)\mathbf{1}\{\Gamma_d = m\}$. Let $\mathcal{E}$ be the event $\{Z_{N-1}^{(j)} \leq m-1,\ Y_{2m} \leq m\}$. We claim that

$$S_j \geq \beta_j(m)\Big(\mathbf{1}_{\mathcal{E}} - \frac{2(j-2)}{N} Y_{2m}\Big).$$

Off $\mathcal{E}$ the right side is at most 0 and $S_j \ge 0$. So let $\mathcal{E}$ hold. As d grows by one, Y_d grows by 0 or 1 and $Z^{(j)}_{N-d}$ falls by 0 or 1. So every step of Γ lies in $\{-1,0,1\}$, and Γ cannot jump over m. It starts at $\Gamma_1 = 1 + Z^{(j)}_{N-1} \le m$ (note that $Y_1 = 0$). At $d^\circ = m + Y_{2m} \le 2m$ we have $\Gamma_{d^\circ} \ge d^\circ - Y_{d^\circ} \ge d^\circ - Y_{2m} = m$, since $Z^{(j)} \ge 0$ and Y is nondecreasing. (The index d° is allowed, because $2m + j \le N$.) So $\Gamma_d = m$ for some $d \le d^\circ$, and as β_j is nonincreasing, $S_j \ge \beta_j(d^\circ) = \beta_j(m + Y_{2m})$. It remains to compare $\beta_j(m+y)$ with $\beta_j(m)$ for $0 \le y \le m$. By (6),

$$\frac{\beta_j(m+y)}{\beta_j(m)} = \prod_{i=1}^{y}\Big(1 - \frac{j-2}{N-m-i}\Big) \ge 1 - \frac{2y(j-2)}{N}.$$

Here $N - m - i \ge N/2$, so each fraction is at most $\frac{2(j-2)}{N} \le \frac{1}{4}$, and a product of factors $1 - q_i$ with $0 \le q_i \le 1$ is at least $1 - \sum_i q_i$. This proves the claim. Now $\mathbf{1}_{\mathcal{E}} \ge 1 - \mathbf{1}\{Y_{2m} > m\} - \mathbf{1}\{Z^{(j)}_{N-1} \ge m\}$. Taking means in the claim gives

$$Q_j(m) \ge \beta_j(m)\big[1 - \mathbb{P}(Y_{2m} > m) - \mathbb{P}(Z^{(j)}_{N-1} \ge m)\big] - \frac{2(j-2)}{N}\,\mathbb{E}Y_{2m}\,\beta_j(m).$$

Step 2. The chance that Y or Z is large. Lemma B.2 gives

$$\mathbb{P}(Y_{2m} > m) \le 2\varepsilon H_{m+1}, \qquad \mathbb{P}(Z^{(j)}_{N-1} \ge m) \le \frac{xH_m}{m}, \qquad \mathbb{E}Y_{2m} \le 2\varepsilon m \log N.$$

None of these bounds depends on j. Put $p = 2\varepsilon H_{m+1} + xH_m/m$. If $p > 1$, then $e_1'(m) > 1$ and there is nothing to prove. So let $p \le 1$.

Step 3. Sum over j. By Lemma 3.3, $f_N(m) \ge \sum_{j=2}^{J} \mathbb{P}(V = j)\, Q_j(m)$. With Steps 1 and 2 this gives

$$f_N(m) \ge (1-p)\sum_{j=2}^{J} \mathbb{P}(V=j)\,\beta_j(m) - \frac{4\varepsilon m \log N}{N}\sum_{j=2}^{J}\mathbb{P}(V=j)\,(j-2)\,\beta_j(m).$$

In the second sum we use $\mathbb{P}(V = j) \le \varepsilon$ and the second sum in Lemma 2.4. So the second term is at most

$$\frac{4\varepsilon m \log N}{N} \cdot \frac{2x(N-m-1)}{m(m+1)(m+2)} \le \frac{x}{m(m+1)}\,8\varepsilon \log N.$$

The first sum is $\mathbb{P}(|\mathcal{T}_V| = m)$ without the terms $j > J$. So by Lemma B.3 it is at least $\frac{x}{m(m+1)}(1 - \frac{2x}{m+2}) - \varepsilon \sum_{j>J} \beta_j(m)$.

Step 4. The late first flips $j > J$. A family that starts after time $N/8$ rarely reaches size m. By (5), $\beta_j(m) \le \frac{j-1}{N-m} r^{j-1}$ with $r = e^{-a}$ and $a = \frac{m-1}{N-1} \le \frac{1}{4}$. We sum this over $j - 1 \ge J$. With $\sum_{i \ge J} i r^i \le r^J\big(\frac{J}{1-r} + \frac{1}{(1-r)^2}\big)$, $1 - e^{-a} \ge a/2$ and $N - m \ge \frac{3}{4}N$ we get $\varepsilon \sum_{j>J} \beta_j(m) \le \frac{4\varepsilon}{3N}\big(\frac{2J}{a} + \frac{4}{a^2}\big)e^{-aJ}$. Since $\lfloor N/8 \rfloor \ge \frac{N-1}{9}$ for $N \ge 64$, we have $aJ \ge \frac{m-1}{9}$. Also $J \le N/8$ and $a \ge \frac{m-1}{N}$. So

$$\varepsilon \sum_{j>J} \beta_j(m) \le x\Big(\frac{1}{3(m-1)} + \frac{16}{3(m-1)^2}\Big)e^{-(m-1)/9} \le \frac{x}{m(m+1)}\,20me^{-m/9}.$$

The last step uses $m + 1 \le 3(m-1)$ and $m + 1 \le 3(m-1)^2$, which hold for $m \ge 2$, and $e^{1/9} \le \frac{9}{8}$, since $17 \cdot \frac{9}{8} \le 20$.

Step 5. Collect. For $0 \le p \le 1$ and $q \ge 0$ we have $(1-p)(1-q) \ge 1-p-q$. So Steps 3 and 4 give

$$f_N(m) \ge \frac{x}{m(m+1)}\Big(1 - p - \frac{2x}{m+2} - 20me^{-m/9} - 8\varepsilon \log N\Big),$$

which is the bound with $e_1'(m)$. For $e_1' \le e_1$ we use $\log(m+1) \ge \log 3 > 1$ and $\log N \ge \log 64 > 4$. So $H_m \le 1 + \log m \le 2\log(m+1)$, $H_{m+1} \le 1 + \log N \le \frac{5}{4}\log N$ and $\frac{2x}{m+2} \le \frac{2x\log(m+1)}{m}$. Hence $e_1'(m) \le \frac{4x\log(m+1)}{m} + \frac{21}{2}\varepsilon \log N + 20me^{-m/9} \le e_1(m)$. □

Proof of Lemma B.5. We pick a point ℓ of order $x \log x$ at which $f_N(\ell)$ is at least half of $\frac{x}{\ell(\ell+1)}$. Up to its mode f_N is nondecreasing. So if the mode were far to the right of ℓ, the values from ℓ to the mode would carry more mass than the tail $\mathbb{P}(M_N \ge \ell)$ has.

Put $L_2 = \log(x+2) \ge \log 3 > 1$ and $\ell = \lceil 200xL_2 \rceil$, so that $200 \le \ell \le 201xL_2$ and $4\ell \le 804xL_2 \le N$. Since $\log N \ge 4$, we have $\varepsilon \le \frac{1}{160}$. We bound the five terms of $e_1'(\ell)$ in Proposition B.4. As $(1+\log s)/s$ decreases for $s \ge 1$, and $\log 200 \le 6$, $\log x \le L_2$ and $\log L_2 \le L_2$, we get $\frac{xH_\ell}{\ell} \le \frac{x(1+\log(200xL_2))}{200xL_2} \le \frac{7+2L_2}{200L_2} \le \frac{9}{200}$. Next $\frac{2x}{\ell+2} \le \frac{1}{100}$, $2\varepsilon H_{\ell+1} \le 2\varepsilon(1+\log N) \le \frac{1}{80} + \frac{1}{20}$ and $8\varepsilon \log N \le \frac{1}{5}$. Finally $se^{-s/9}$ decreases for $s \ge 9$, so $20\ell e^{-\ell/9} \le 4000e^{-200/9} < 10^{-5}$. The five terms add up to less than $\frac{1}{2}$, so $f_N(\ell) \ge \frac{x}{2\ell(\ell+1)}$. If $m_h \ge \ell$, Lemmas B.1 (with $a = \ell$, $b = m_h$) and B.2 give $(m_h - \ell + 1)f_N(\ell) \le \mathbb{P}(M_N \ge \ell) \le xH_\ell/\ell$. So $m_h \le (\ell+1)(1+2H_\ell)$ in every case. Here $\ell + 1 \le 202xL_2$, because $xL_2 \ge 1$. Also $H_\ell \le 1 + \log(201xL_2) \le 7 + 2L_2 \le 9L_2$, so $1 + 2H_\ell \le 19L_2$. Hence $m_h \le 3838\, xL_2^2$. Since $(x+1)^2 \ge x+2$, we have $L_2 \le 2\log(x+1)$, and so $m_h \le 15352\, x\log^2(x+1) \le A_0 x \log^2(x+1)$. The last claim holds because $804x\log(x+2) \le 4A_0 x \log(x+2)$. □

E Proofs for the second difference

We keep the notation of Section 3. This section proves the results of Section 3.5 on the signs of the second difference.

E.1 Proof of Lemma 3.14

We insert the numbers η_i into ζ one at a time. If $\eta_i = 0$, the sign pattern does not change. If $\eta_i > 0$, then $\zeta_i > 0$ or $\zeta_{i+1} > 0$, so η_i is placed next to an entry of its own sign, and the pattern does not change. The case $\eta_i < 0$ is the same. This proves the first claim. Deleting entries from a list replaces its sign pattern by a subsequence, and η is obtained from w by deleting the entries ζ_i. This proves the second claim. □

E.2 Proof of Lemma 3.15

Put $\xi_t = P_t(1)/P_t(0)$. By (1), $P_{t+1}(0) = (1-\varepsilon)P_t(0)$ and $P_{t+1}(1) = (1 - \varepsilon - \frac{c}{t})P_t(1) + \varepsilon P_t(0)$. So

$$\xi_{t+1} - \xi_t = \frac{1}{1-\varepsilon}\Big(\varepsilon - \frac{c\xi_t}{t}\Big), \qquad \xi_2 = \frac{2\varepsilon}{1-\varepsilon}.$$

We claim that $\xi_t < \varepsilon t/c$ for $t \ge 2$. For $t = 2$ this holds because $c < 1 - \varepsilon$. If it holds for t, then $0 < \xi_{t+1} - \xi_t \le \frac{\varepsilon}{1-\varepsilon} < \frac{\varepsilon}{c}$, and it holds for $t+1$. So ξ_t is strictly increasing for $t \ge 2$, and $\xi_t < 1$ whenever $t \le c/\varepsilon$.

Now (2) with $a = 1$, and $D_t(0) = P_t(1) - 2P_t(0)$, give for $t \geq 2$

$$s_{t+1} = \big(1 - r_t(2)\big)s_t + \sigma_t, \qquad \sigma_t = \Big(\varepsilon - \frac{c}{t}\Big)(\xi_t - 2)\, P_t(0).$$

Here $1 - r_t(2) \geq 1 - \varepsilon - c = \varepsilon > 0$. If $t < c/\varepsilon$, both factors of σ_t are negative and $\sigma_t > 0$. If $t > c/\varepsilon$, the first factor is positive, and the second is first negative and then positive, because ξ_t increases. So $\mathrm{sp}(\sigma_2, \sigma_3, \dots) \preceq \Pi_3$. Let $\Phi_t = \prod_{u=2}^{t-1}(1 - r_u(2)) > 0$. Then

$$\frac{s_t}{\Phi_t} = s_2 + \sum_{u=2}^{t-1} \frac{\sigma_u}{\Phi_{u+1}}, \qquad s_2 = 1 - 3\varepsilon.$$

If $\varepsilon < \frac{1}{3}$, the terms $s_2, \sigma_2, \sigma_3, \dots$ are first nonnegative, then nonpositive, then nonnegative. Their partial sums are nonnegative and nondecreasing in the first stretch, nonincreasing in the second, and nondecreasing in the third. So the sign pattern of the partial sums is a subsequence of Π_3. If $\varepsilon \geq \frac{1}{3}$, then $c/\varepsilon \leq 1$, so the terms are first nonpositive and then nonnegative, and the sign pattern of the partial sums is a subsequence of $(-, +)$. This proves the first claim.

For the second claim let $0 \leq \varepsilon \leq 1$. The proofs of (1) and (2) use only that step $t + 1$ is $+1$ with probability $r_t(a)$ given $A_t = a$, so both hold for this ε. We compute $P_3(0) = \frac{1}{2}(1 - \varepsilon)^2$ and $P_3(1) = P_3(2) = \frac{1}{2} - P_3(0)$, so $s_3 = P_3(0) - P_3(1) = (1 - \varepsilon)^2 - \frac{1}{2}$. Suppose that $s_3 = 0$. Then $\varepsilon = 1 - 2^{-1/2}$ and $P_3(0) = P_3(1) = \frac{1}{4}$. By (2) with $t = 3$ and $a = 1$, $s_4 = r_3(-1)D_3(0) = (\varepsilon - \frac{c}{3})(P_3(1) - 2P_3(0)) = -\frac{5\varepsilon - 1}{12}$. This is not 0, because $\varepsilon \neq \frac{1}{5}$. (With P_t defined by (1), the same computation works for every real ε, because the other root $1 + 2^{-1/2}$ of s_3 is not $\frac{1}{5}$ either. We need only $0 < \varepsilon < \frac{1}{2}$.) □

E.3 Proof of Theorem 3.16

For $3 \leq t \leq N$ let $\mathcal{Z}_t$ be the list

$$\mathcal{Z}_t = \big(s_N, s_{N-1}, \dots, s_{t+1},\ s_t, D_t(2), D_t(3), \dots, D_t(t-2), s_t,\ s_{t+1}, \dots, s_{N-1}, s_N\big).$$

It has $2N - t - 1$ entries. Its middle part is $(D_t(1), D_t(2), \dots, D_t(t-1))$, because $D_t(t-1) = D_t(1) = s_t$ by symmetry. The list $\mathcal{Z}_3$ is $(s_N, \dots, s_4, s_3, s_3, s_4, \dots, s_N)$, and $\mathcal{Z}_N$ is the list in the statement.

We check that $\mathcal{Z}_{t+1}$ is obtained from $\mathcal{Z}_t$ as in Lemma 3.14, for $3 \leq t \leq N - 1$. The first $N - t$ entries of $\mathcal{Z}_{t+1}$ are $s_N, \dots, s_{t+1}$, and they are copies of the first $N - t$ entries of $\mathcal{Z}_t$. So $A_i = 1$ and $B_i = 0$ there. The next $t - 2$ entries are $D_{t+1}(b)$ for $2 \leq b \leq t - 1$. By (2),

$$D_{t+1}(b) = r_t(b-2)\, D_t(b-1) + \big(1 - r_t(b+1)\big) D_t(b),$$

and $D_t(b-1)$ and $D_t(b)$ are neighbors in the middle part of $\mathcal{Z}_t$, at the right places. The coefficients satisfy $r_t(b-2) \geq \varepsilon$ and $1 - r_t(b+1) \geq 1 - \varepsilon - c = \varepsilon$, since $2 \leq b \leq t - 1$. The last $N - t$ entries of $\mathcal{Z}_{t+1}$ are $s_{t+1}, \dots, s_N$, and each is a copy of the entry of $\mathcal{Z}_t$ one place to its right. So $A_i = 0$ and $B_i = 1$ there.

By Lemma 3.14, $\mathrm{sp}(\mathcal{Z}_N) \preceq \mathrm{sp}(\mathcal{Z}_{N-1}) \preceq \cdots \preceq \mathrm{sp}(\mathcal{Z}_3)$. By Lemma 3.15 the sign pattern p of $(s_3, \dots, s_N)$ is a subsequence of Π_3, so it is one of the lists $()$, $(+)$, $(-)$, $(+, -)$, $(-, +)$ and $(+, -, +)$. The sign pattern of $\mathcal{Z}_3$ is p reversed, followed by p, with equal neighbors merged. In the six cases this is $()$, $(+)$, $(-)$, $(-, +, -)$, $(+, -, +)$ and $(+, -, +, -, +)$, and each is a subsequence of Π_5.

For f_N we argue in the same way with $D_t^f(m) = f_t(m+1) - 2f_t(m) + f_t(m-1)$ and the lists

$$\mathcal{Z}_t^f = \big(s_N^f, \dots, s_{t+1}^f,\ s_t^f, D_t^f(2), \dots, D_t^f(t-1),\ d_t, d_{t+1}, \dots, d_{N-1}\big), \qquad 2 \le t \le N,$$

where $s_t^f = D_t^f(1)$ and $d_u = (1 - r_u(u+1))D_u^f(u) = (\varepsilon - \frac{c}{u})f_u(u-1)$. The entry $D_{t+1}^f(t)$ of $\mathcal{Z}_{t+1}^f$ equals $r_t(t-2)D_t^f(t-1) + d_t$ by (2), and the other entries are as before. The proof of Lemma 3.15 applies to s_t^f without change, with $\xi_t = f_t(1)/f_t(0)$, $\xi_2 = \frac{\varepsilon}{1-\varepsilon}$ and $s_2^f = 1 - 3\varepsilon$. The numbers d_u are negative for $u < c/\varepsilon$ and positive for $u > c/\varepsilon$. So $\operatorname{sp}(\mathcal{Z}_2^f)$ is a subsequence of $(+,-,+)$ followed by $(-,+)$, and $\mathcal{Z}_N^f = (D_N^f(1), \dots, D_N^f(N-1))$. As before $\operatorname{sp}(\mathcal{Z}_N^f) \preceq \operatorname{sp}(\mathcal{Z}_2^f)$, and a subsequence of $(+,-,+)$ followed by $(-,+)$ is a subsequence of Π_5 once equal neighbors are merged. This argument needs only $N \ge 2$, and for $N = 2$ the lists $\mathcal{Z}_2^f$ and $\mathcal{Z}_N^f$ are the same. □

E.4 Proof of Theorem 3.17

By symmetry $D_N(N - a_1) > 0 > D_N(N - a_0)$. Suppose that $D_N(a) \le 0$ for some a with $a_1 < a < N - a_1$.

The case $D_N(a) < 0$. Then the entries of $\mathcal{Z}_N$ at $a_0 < a_1 < a < N - a_1 < N - a_0$ have the signs $-,+,-,+,-$. So $(-,+,-,+,-) \preceq \operatorname{sp}(\mathcal{Z}_N) \preceq \Pi_5$ by Theorem 3.16. This is impossible, because a subsequence of Π_5 with five terms is Π_5 itself.

The case $D_N(a) = 0$. We use the lists $\mathcal{Z}_t$ of the proof of Theorem 3.16 and number the entries of $\mathcal{Z}_t$ from 1 to $2N - t - 1$. So $D_t(b)$ stands at place $N - t + b$ for $1 \le b \le t - 1$. For $3 \le t \le N - 1$ let W_t be the list w of Lemma 3.14 for $\zeta = \mathcal{Z}_t$ and $\eta = \mathcal{Z}_{t+1}$. Then $\operatorname{sp}(W_t) = \operatorname{sp}(\mathcal{Z}_t) \preceq \Pi_5$. The entry of $\mathcal{Z}_{t+1}$ at place k is a combination of the entries of $\mathcal{Z}_t$ at the places k and $k+1$, with nonnegative coefficients. We call these two entries its parents. For $2 \le b \le t - 1$ the parents of $D_{t+1}(b)$ are $D_t(b-1)$ and $D_t(b)$, and both coefficients are positive.

For $4 \le t \le N$ let B_t be the set of integers b with $\max(2, a - N + t) \le b \le \min(a, t - 2)$. It is not empty, because $2 \le a \le N - 2$. We prove by downward induction on t that there are places $p_1 < p_2 < p_4 < p_5$ in $\mathcal{Z}_t$ such that the entries there have the signs $-,+,+,-$, that $D_t(b) = 0$ for every $b \in B_t$, and that all the places $N - t + b$ with $b \in B_t$ lie strictly between p_2 and p_4. For $t = N$ we have $B_N = \{a\}$, and we take the places a_0, a_1, $N - a_1$ and $N - a_0$.

Let the claim hold for $t+1$, where $4 \le t \le N - 1$, with the places $p_1' < p_2' < p_4' < p_5'$. Let $b \in B_{t+1}$. Then $D_{t+1}(b) = 0$ is a combination of its two parents with positive coefficients. So the parents both vanish, or they have strictly opposite signs. In the second case one parent is negative. In W_t it stands next to $D_{t+1}(b)$, hence between the positive entries of $\mathcal{Z}_{t+1}$ at the places p_2' and p_4', and these stand between the negative entries at p_1' and p_5'. So $(-,+,-,+,-) \preceq \operatorname{sp}(W_t) \preceq \Pi_5$, which is impossible. Hence $D_t(b-1) = D_t(b) = 0$ for every $b \in B_{t+1}$. Every element of B_t is of the form b or $b - 1$ with $b \in B_{t+1}$, so D_t vanishes on B_t.

The entry of $\mathcal{Z}_{t+1}$ at p_1' is negative, so one of its parents is negative. Let $p_1 \in \{p_1', p_1' + 1\}$ be its place in $\mathcal{Z}_t$, and choose p_2, p_4 and p_5 in the same way. Then $p_1 \le p_1' + 1 \le p_2' \le p_2$, and $p_1 \ne p_2$ because the signs differ. In the same way $p_4 < p_5$. Let q^- and q^+ be the smallest and the largest place $N - t - 1 + b$ with $b \in B_{t+1}$. The places $N - t + b$ with $b \in B_t$ lie between q^- and $q^+ + 1$. Since $p_2 \le p_2' + 1 \le q^-$ and $p_4 \ge p_4' \ge q^+ + 1$, and the entries at p_2 and p_4 are not zero, these places lie strictly between p_2 and p_4. This proves the claim for t.

The induction step with $t = 4$ shows that both parents of $D_5(b)$ vanish for every $b \in B_5 \subseteq \{2, 3\}$. The parents of $D_5(2)$ are s_4 and $D_4(2)$, and those of $D_5(3)$ are $D_4(2)$ and s_4. So $s_4 = D_4(2) = 0$. By (2) with $t = 3$ and $a = 2$, and since $D_3(1) = D_3(2) = s_3$, we have $D_4(2) = r_3(0)s_3 + \big(1 - r_3(3)\big)s_3 = 2\varepsilon s_3$. So $s_3 = 0$ as well, which contradicts Lemma 3.15. □

F Sign patterns and regularity

We prove the lemmas used for Theorem 4.2, with the notation of Sections 3.5 and 4.2. Deleting entries from a list replaces its pattern by a subsequence. An alternating list p satisfies $p \preceq q$ for every alternating list q with the same first sign and at least as many entries.

Lemma F.1 (Descartes' rule for exponential sums)**.** *Let $\lambda_1 < \cdots < \lambda_n$ and $b_1, \dots, b_n$ be real, with $b = (b_1, \dots, b_n) \neq 0$, and put $F(y) = \sum_{k=1}^{n} b_k e^{\lambda_k y}$. Then F has finitely many real zeros, and $\operatorname{sp}(F(y_1), \dots, F(y_j)) \preceq \operatorname{sp}(b)$ for all $y_1 < \cdots < y_j$. The same holds for ψF for every positive function ψ.*

Proof. Let $Z(F)$ be the number of distinct real zeros of F, and $S(b)$ the number of sign changes of b. We show $Z(F) \leq S(b)$ by induction on $S(b)$. If $S(b) = 0$, then F has the sign of the nonzero b_k everywhere. If $S(b) \geq 1$, choose i such that the last nonzero b_k with $k \leq i$ and the first nonzero b_k with $k > i$ have opposite signs, and choose $\lambda_i < \kappa < \lambda_{i+1}$. Then

$$G(y) = \frac{d}{dy}\big(e^{-\kappa y} F(y)\big) = \sum_{k=1}^{n} b_k (\lambda_k - \kappa) e^{(\lambda_k - \kappa) y}.$$

The coefficient $b_k(\lambda_k - \kappa)$ has the sign of $-b_k$ for $k \leq i$ and of b_k for $k > i$, so the coefficients of G have $S(b) - 1$ sign changes. By Rolle's theorem G vanishes between any two zeros of F, so $Z(F) \leq Z(G) + 1 \leq S(b)$.

If b_{k_0} is the first nonzero b_k, then $e^{-\lambda_{k_0} y} F(y) \to b_{k_0}$ as $y \to -\infty$. The zeros of F cut the line into at most $S(b) + 1$ open intervals, on each of which F has one sign. With equal neighbors merged, these signs form an alternating list that starts with the sign of b_{k_0} and has at most $S(b)+1$ entries, so it is $\preceq \operatorname{sp}(b)$. The nonzero values $F(y_1), \dots, F(y_j)$ lie in these intervals in order. A positive factor ψ changes no sign. □

Lemma F.2 (pointwise limits)**.** *Let v_n and v be real functions on $\mathbb{R}$ with $v_n(y) \to v(y)$ for every y, and let Π be an alternating list. If every v_n has values at most Π, then so has v. If every v_n has increments at most Π, then so has v.*

Proof. Let $y_1 < \cdots < y_j$, and let I be the set of indices i with $v(y_i) \neq 0$. For all large n, $v_n(y_i)$ has the sign of $v(y_i)$ for every $i \in I$. Then $\operatorname{sp}(v(y_1), \dots, v(y_j))$ is the pattern of the list $(v_n(y_i))_{i \in I}$, which is a subsequence of $\operatorname{sp}(v_n(y_1), \dots, v_n(y_j)) \preceq \Pi$. The increments of v_n at $y_1, \dots, y_j$ tend to those of v, and the same argument applies to them. □

Lemma F.3 (one derivative fewer)**.** *Let u be twice continuously differentiable on $\mathbb{R}$, with $u'(y) \to 0$ as $y \to \pm\infty$. If u'' has values at most Π_5, then u' has values at most Π_4, and u has increments at most Π_4.*

Proof. It is enough to treat points $y_1 < \cdots < y_j$ at which the values of u' are nonzero and alternate in sign. Let s be the sign of $u'(y_1)$. Choose $y_0 < y_1$ with $|u'(y_0)| < |u'(y_1)|$, and $y_{j+1} > y_j$ with $|u'(y_{j+1})| < |u'(y_j)|$. For $0 \leq i \leq j$ the mean value theorem gives $w_i \in (y_i, y_{i+1})$ with $u''(w_i)(y_{i+1} - y_i) = u'(y_{i+1}) - u'(y_i)$. The right side has the sign s for $i = 0$, the sign of $u'(y_{i+1})$ for $1 \leq i \leq j-1$, and the sign of $-u'(y_j)$ for $i = j$. In each case this is $(-1)^i s$. So u'' takes $j+1$ alternating signs, starting with s, at $w_0 < \cdots < w_j$. This list is $\preceq \Pi_5$, so $j + 1 \leq 5$, and $j + 1 \leq 4$ if s is $-$. In both cases the list of the j signs of $u'(y_1), \dots, u'(y_j)$ is $\preceq \Pi_4$.

For the increments at $y_1 < \cdots < y_j$, each nonzero $u(y_{i+1}) - u(y_i)$ has the sign of $u'(\xi_i)$ for some $\xi_i \in (y_i, y_{i+1})$, by the mean value theorem. So the pattern of the increments is a pattern of values of u'. □

Lemma F.4 (absence of flat intervals)**.** *For every $a \in (\frac{1}{2}, 1)$ and $1 < r < 1/a$, the characteristic function ϕ_a of $\mathcal{L}_a$ satisfies*

$$|\phi_a(t)| \le 2e^{-c|t|^r} \qquad (t \in \mathbb{R})$$

for some $c = c(a, r) > 0$. For every $q \in [0, 1]$, the density of the limit with first-step bias q extends to an entire function and is not constant on an interval of positive length.

Proof. The fixed-point equation of [11, Theorem 1.1 and (2.10)] gives

$$|\phi_a(t)| \le \int_0^1 |\phi_a(tv^a)|\,|\phi_a(t(1-v)^a)|\,dv.$$

Here ϕ_a is the characteristic function conditional on a positive first step. The random sign in the fixed-point equation causes no change in this inequality, since $|\phi_a(-s)| = |\phi_a(s)|$. Put $D(v) = v^{ar} + (1-v)^{ar} - 1$. Since $ar < 1$, we have $D(v) > 0$ for $0 < v < 1$. Dominated convergence therefore permits $K > 1$ such that

$$R := 2\int_0^1 e^{-K^r(\log 2)D(v)}\,dv < 1.$$

The law of $\mathcal{L}_a$ has a density [11, Theorem 1.3], so the Riemann–Lebesgue lemma gives $T > 0$ with $|\phi_a(t)| \le 2^{1-K^r}$ for $|t| \ge T$. Set $c = (\log 2)/T^r$ and $b(t) = 2e^{-ct^r}$ for $t \ge 0$. On $[0, T]$ we have $b \ge 1$, and on $[T, KT]$ we have $b \ge 2^{1-K^r}$. Thus $|\phi_a| \le b$ on $[0, KT]$.

For $t \ge KT$,

$$\frac{\int_0^1 b(tv^a)b(t(1-v)^a)\,dv}{b(t)} = 2\int_0^1 e^{-ct^rD(v)}\,dv \le R < 1.$$

If the bound $|\phi_a| \le b$ first ceased to hold at a finite supremum $t_* \ge KT$, continuity would give the bound on $[0, t_*]$. The fixed-point inequality would then give $|\phi_a(t_*)| \le Rb(t_*) < b(t_*)$, and continuity would extend the bound beyond t_*. This contradiction proves the bound on $[0, \infty)$; symmetry of its modulus gives the bound on $\mathbb{R}$.

Since $r > 1$, the Fourier integral for the density converges absolutely and locally uniformly for complex arguments: for every $B < \infty$, $\int_{\mathbb{R}} e^{B|t|}|\phi_a(t)|\,dt < \infty$. It therefore defines an entire function. At initial bias q, the characteristic function is $q\phi_a(t) + (1-q)\phi_a(-t)$ and obeys the same bound by the triangle inequality. Each resulting density is integrable with integral one, so it is not constant on $\mathbb{R}$. The identity theorem excludes a constant interval. □

F.1 First differences for a biased start

Proposition F.5 (Two modes for every initial bias)**.** *Let $0 < \varepsilon < 1$, let $q \in [0, 1]$ be the probability that the first step is positive, and let $P_n(k)$ be the law of the number of positive steps after $n \ge 1$ steps. Extend P_n by zero outside $\{0, \dots, n\}$. The first differences $d_n(k) = P_n(k+1) - P_n(k)$ satisfy*

$$\operatorname{sp}\big(d_n(-1), d_n(0), \dots, d_n(n)\big) \preceq (+, -, +, -).$$

In particular, P_n has at most two modal plateaus, including endpoint modes. Moreover, there are no integers $i_0 < \cdots < i_4$ for which

$$P_n(i_0) > P_n(i_1) < P_n(i_2) > P_n(i_3) < P_n(i_4).$$

Proof. Put $a = 1 - 2\varepsilon$ and $r_t(k) = \varepsilon + ak/t$. The forward equation is

$$P_{t+1}(k) = (1 - r_t(k))P_t(k) + r_t(k-1)P_t(k-1), \qquad P_1 = (1-q, q).$$

Subtracting adjacent equations gives

$$d_{t+1}(k) = r_t(k-1)d_t(k-1) + (1 - r_t(k+1))d_t(k), \qquad 1 \le k \le t-1. \tag{9}$$

Both coefficients are at least $\min(\varepsilon, 1-\varepsilon) > 0$: $r_t(j)$ is a convex combination of ε and $1-\varepsilon$ for $0 \le j \le t$. We treat the two boundaries separately.

Suppose first that $q < 1$. Since $P_t(0) = (1-q)(1-\varepsilon)^{t-1} > 0$, define

$$\xi_t = \frac{P_t(1)}{P_t(0)}, \qquad b = \frac{1-2\varepsilon}{1-\varepsilon} < 1, \qquad A = \frac{\varepsilon}{1-\varepsilon} > 0.$$

The boundary equation gives

$$\xi_{t+1} = (1 - b/t)\xi_t + A, \qquad \xi_1 = \frac{q}{1-q}.$$

Thus $\delta_t = \xi_{t+1} - \xi_t$ obeys

$$\delta_{t+1} = \frac{t-b}{t+1}\delta_t + \frac{A}{t+1}. \tag{10}$$

A nonnegative increment is followed by a positive increment. Consequently, ξ_t first decreases and then increases, allowing either stretch to be empty. For $\ell_t = d_t(0) = P_t(0)(\xi_t - 1)$ this proves

$$\mathrm{sp}(\ell_1, \dots, \ell_N) \preceq (+, -, +).$$

If $q \le 1/2$, then $\xi_1 \le 1$ and the sharper bound is $\mathrm{sp}(\ell_1, \dots, \ell_N) \preceq (-, +)$. In particular, a positive ℓ_i cannot be followed by a negative ℓ_j. Zeros do not affect either conclusion.

Write $r_t^* = d_t(t-1)$. Reflection of the walk gives $r_t^*(q) = -\ell_t(1-q)$ and $d_t^{(1-q)}(t-1-k) = -d_t^{(q)}(k)$. For $0 < q \le 1/2$, the right boundary history therefore has pattern contained in $(-, +, -)$. For $q = 0$ it is nonpositive throughout: the conditional count is at most $t-1$, so $r_t^* = -P_t(t-1) \le 0$.

Fix N and, for $1 \le t \le N$, form the augmented row

$$Z_t = (\ell_N, \dots, \ell_{t+1}, d_t(0), \dots, d_t(t-1), r_{t+1}^*, \dots, r_N^*).$$

For $q \le 1/2$, its first row is contained in $(+, -, +, -)$. Here is an explicit check that also handles zeros at $\ell_1 = r_1^* = 2q - 1$. If Z_1 contained four ordered signs $-, +, -, +$ and the second lay in the reversed left history, the first two would give a positive left edge followed in time by a negative one, which is impossible. Otherwise the last three lie in the right history and have signs $+, -, +$. For $q > 0$ their reflected negatives contradict the $(-, +, -)$ exclusion for the left history at bias $1 - q$; for $q = 0$ they contradict its nonpositive right history. This proves the asserted base pattern, including $N = 1$.

Every entry of Z_{t+1} is a nonnegative combination of adjacent entries of Z_t. The interior entries use (9); the two new boundary entries and the remaining future edge values are copied with coefficients $(1, 0)$ or $(0, 1)$. Such a step cannot introduce an alternating strict-sign pattern: each nonzero entry has a parent of the same sign, and parents of successive opposite signs are strictly ordered. Thus $\mathrm{sp}(Z_N) \preceq (+, -, +, -)$. Reflection and sign reversal preserve this pattern, proving the claim also for $q \ge 1/2$, including $q = 1$.

Finally, extension by zero adds $d_N(-1) = P_N(0) \geq 0$ on the left and $d_N(N) = -P_N(N) \leq 0$ on the right; neither enlarges the pattern. Every modal plateau contributes a positive-to-negative change, so there are at most two. If the last displayed inequalities in the statement held, each of their four intervals would contain an adjacent difference of the same strict sign, by telescoping. These four differences would have signs $-, +, -, +$, a contradiction. □

Proposition F.6 (The dominant half)**.** *Under the hypotheses of Proposition F.5, suppose $q \geq 1/2$. Then $P_n(n-k) \leq P_n(k)$ for $2k \geq n$, and the right half has no strict valley:*

$$P_n(j) \geq \min\{P_n(i), P_n(k)\} \qquad \text{if } i < j < k \text{ and } 2i + 1 \geq n.$$

Equivalently, its first differences contain no negative value followed by a positive one. Reflection gives the corresponding left-half result when $q \leq 1/2$.

Proof. Temporarily display the bias in $P_{n,q}$ and put

$$A_n(k) = P_{n,q}(k) - P_{n,1-q}(k) = P_{n,q}(k) - P_{n,q}(n-k).$$

The forward equation gives $A_{n+1}(k) = (1 - r_n(k))A_n(k) + r_n(k-1)A_n(k-1)$. Induction yields $A_n(k) \geq 0$ for $2k \geq n$: both parents lie in the previous upper half unless k is the new centre, where $A_{n+1}(k) = 0$ by reflection; terms outside the support vanish. In particular, the central first difference of an odd row is nonnegative.

Restrict the augmented row Z_t in the preceding proof to spatial indices $2k + 1 \geq t$. At $t = 1$ its pattern is contained in $(+, -)$: the central entry is $2q - 1 \geq 0$, and the subsequent entries are the future right boundary values $r_s^*(q) = -\ell_s(1-q)$. Since $1 - q \leq 1/2$, their pattern is $(+, -)$ by (10). The positive two-parent recurrence preserves absence of a negative entry followed by a positive one. The sole possible parent outside the previous half occurs at the new central difference of an odd row. That entry is nonnegative, so it cannot begin the forbidden pair. More explicitly, if a negative entry at index i of row $t + 1$ satisfies $2i + 1 \geq t + 1$, then $2i > t$; both its parents satisfy $2k + 1 \geq t$. Parents of a later positive entry follow them strictly, as in the preceding proof. Induction therefore excludes the pair in every restricted augmented row. A strict valley would contain such two adjacent differences by telescoping, which proves the claim. □

G The certificate for the threshold

We prove Theorem 4.4. This appendix states the steps, and Section S6 gives their full proofs. Fix $a \in (\frac{1}{2}, 1)$ and write $L = \mathcal{L}_a$ and $\phi(t) = \mathbb{E}e^{itL}$. Put $\theta_j = 1$ for even j and $\theta_j = a$ for odd j, let $m_0 = m_1 = 1$ and $m_n = (na - \theta_n)^{-1} \sum_{j=1}^{n-1} \theta_j m_j m_{n-j}$ for $n \geq 2$. Then $\mathbb{E}L^n = n!\, c_n$ with $c_n = m_n/\Gamma(1 + an)$ [11, Theorem 1.4]. As $\mathbb{E}e^{t|L|} < \infty$ for real t [11, Corollary 2.11], $\phi(t) = \sum_n c_n (it)^n$ for all real t. By [11, Theorem 1.1 and (2.10)], L has the law of $V^a L' + \xi(1-V)^a L''$, with V uniform on $[0, 1]$, $\xi = \pm 1$, L' and L'' distributed as L, and all four independent. Since $|\phi(-s)| = |\phi(s)|$, this gives

$$|\phi(t)| \leq \int_0^1 |\phi(tv^a)|\, |\phi(t(1-v)^a)|\, dv \qquad (t \in \mathbb{R}). \tag{11}$$

The density g_a of $\mathcal{L}_a^\circ$ has the characteristic function $\operatorname{Re}\phi$.

Step 1, one value of a. If $m_j \leq AB^j$ for $1 \leq j \leq N$, with $A = ((N+1)a - 1)/N$, then $m_n \leq AB^n$ for all $n \geq 1$, by induction, since $(na-1)/(n-1)$ increases in n. This bounds the tail

of the moment series and of its derivatives by geometric series. Taylor's formula then bounds $|\phi|$ on cells covering $[0, T_3]$ by s_j, from finitely many coefficients. The next step turns these bounds into a bound for all t. Let $T_3 \geq 2^a T_2$, with T_2 an end point of a cell, and $\beta > 3$ with $a\beta > 1$. Let C be the largest $s_j r_j^\beta$ over the cells $[l_j, r_j]$ that meet $[T_2, T_3]$, and

$$R = T_2^{-1/a} \sum_{r_j \leq T_2} s_j \Big(1 - \tfrac{1}{2}(r_j/T_2)^{1/a}\Big)^{-a\beta} \big(r_j^{1/a} - l_j^{1/a}\big) + \frac{2^{a\beta} C T_2^{-\beta}}{a\beta - 1}.$$

If $R < 1$, then $|\phi(t)| \leq Ct^{-\beta}$ for all $t \geq T_3$ (Lemma S6.4). The proof puts this bound into the right side of (11), gets back a smaller bound, and closes with a continuity argument. It is an explicit form of Step 4 in the proof of [11, Theorem 2.5]. By Fourier inversion g_a then has two continuous derivatives, and $g_a''(0) = -\frac{1}{\pi} \int_0^\infty t^2 \operatorname{Re} \phi(t)\, dt$. We integrate the moment series term by term on $[0, T_3]$, and the two error terms are explicit.

Step 2, intervals of a. With a as an interval, ball arithmetic fails, as the series cancels terms of size about 10^{60}. On $I = [a_0 - w, a_0 + w]$ we put $\rho = w/q$ with $0 < q < 1$ and expand in $a - a_0$. The coefficients are computed at the exact point a_0 by power series arithmetic. If h is analytic with $|h| \leq M$ on the disc $|z - a_0| \leq \rho$, then Cauchy's estimate bounds the remainder after degree K by $Mq^{K+1}/(1-q)$ on I. We bound M by enclosing $c_n(z)$ on a square that contains the disc, where nothing cancels, and beyond degree N by the complex form of Step 1 with the product formula for $|\Gamma(x)/\Gamma(x+iy)|^2$ [45, 5.8.3]. This gives a polynomial G and a number η with $|g_a''(0) - G(a - a_0)| \leq \eta$ for every $a \in I$. It also shows that $a \mapsto g_a''(0)$ is continuous on I (Proposition S6.7).

Step 3, a near $\frac{1}{2}$. Put $y = 2a - 1$ and $\mu_n = m_n y^{\lfloor n/2 \rfloor}$. Then $\mu_2 = a$, and for $n \geq 3$ the recursion for μ_n has the factor y in its terms with n even and j odd, and no other change. So every μ_n is analytic for $\operatorname{Re} a > \frac{1}{4}$ (Lemma S6.8). The variable $X = L/\sigma$ with $\sigma^2 = \mathbb{E}L^2 = 1/(y\Gamma(2a))$ has $\mathbb{E}X^n/n! = \mu_n \Gamma(2a)^{\lfloor n/2 \rfloor}/\Gamma(1 + an)$ for even n, and the same times $\sqrt{y\Gamma(2a)}$ for odd n. It satisfies the same fixed-point equation. Steps 1 and 2 in this variable reach $a = \frac{1}{2}$ by continuation of the scaled moment bounds. They enclose $\sigma^3 g_a''(0)$, the second derivative at zero of the density of the fair-start scaled variable, whose sign agrees with $g_a''(0)$ for $a > \frac{1}{2}$. The unscaled density g_a is not defined at $a = \frac{1}{2}$.

Step 4, the Hankel test. This step uses no Fourier analysis.

Lemma G.1 (a Hankel test)**.** *Let $s_j = (2j+1)\mathbb{E}L^{2j}$. Suppose that for some $k \geq 1$ and $x \in \mathbb{R}^{k+1}$ we have $\sum_{i,j \leq k} x_i x_j s_{i+j} < 0$ or $\sum_{i,j \leq k} x_i x_j s_{i+j+1} < 0$. Then case (a) of Theorem 4.2 fails, so case (b) holds.*

Proof. In case (a), $\mathcal{L}_a^\circ$ is unimodal about 0. By Khinchine's theorem [41] it has the law of UZ, with U uniform on $(0,1)$ and independent of Z. So $s_j = (2j+1)\mathbb{E}(\mathcal{L}_a^\circ)^{2j} = \mathbb{E}W^j$ with $W = Z^2 \geq 0$, and the two sums are $\mathbb{E}(\sum_i x_i W^i)^2 \geq 0$ and $\mathbb{E}W(\sum_i x_i W^i)^2 \geq 0$. □

The script `certify_threshold.py` checks every hypothesis in ball arithmetic [40]. It evaluates R with a as a ball over I and evaluates G on subintervals of I, bisecting until $G \pm \eta$ has one sign on each. For the Hankel test it takes k and a rational x from a floating-point eigenvector at the centre of a box of a, and checks the sum on the whole box with a Taylor model as in Step 2. Choices made in floating point, such as N, the precision and x, affect only whether a run succeeds. The cells have half-width $\frac{1}{32}$, the Taylor order in t is 12, $\beta = 4$, and the degree K is chosen so that $Mq^{K+1} \leq e^{-60}$. Table 2 lists the runs.

range of a	method	pieces	parameters	time
$[\frac{1}{2}, 0.668]$	Steps 1 to 3	40	$q = \frac{1}{8}$, $T_2 \in [3, 10]$	2.1 h
$[0.668, 0.6712]$	Steps 1 and 2	8	$q = \frac{1}{32}$, $\rho \le \frac{1}{100}$, $T_2 \in [4, 10]$	13 min
$[0.6705, 1]$	Step 4	115	$k \le 56$	25 min

Table 2 The runs of the certificate. The times are summed over the processes of each run.

Step 5, the unique transition. Put $I = [0.66881285, 0.66881287]$, which contains $[z_-, z_+]$. Differentiating the fixed-point equation, rather than its error bound, gives inhomogeneous integral inequalities for $|\partial_t \phi_a(t)|$ and $|\partial_a \phi_a(t)|$. The same continuity argument as in Step 1 propagates the bounds $2t^{-8}$ and $4t^{-7}$, respectively, from $[6, 10]$ to $[6, \infty)$. The latter justifies differentiation under the Fourier integral, with tail error at most $1/(\pi 10^4)$ beyond $t = 10$. The script `certify_transition.py` certifies, uniformly for $a \in \operatorname{int} I$,

$$0.68 < \partial_a g_a''(0) < 0.70, \qquad -0.09 < g_a^{(4)}(0) < -0.07.$$

The finite-frequency bounds use parameter Taylor models with their differentiated Cauchy remainders. The full argument and numerical constants are in Section S6.D.

Proof of Theorem 4.4. The 40 intervals of the first run cover $[\frac{1}{2}, 0.668]$. On each, the scaled enclosure from Step 3 is negative, so $g_a''(0) < 0$ for $a \in (\frac{1}{2}, 0.668]$. The 8 intervals of the second run cover $[0.668, 0.6712]$. On the 2 intervals that cover $[0.668, 0.6688]$ the enclosure is negative, and on the 5 that cover $[0.6692, 0.6712]$ it is positive. On $[0.6688, 0.6692]$ we have $\eta < 1.1 \cdot 10^{-15}$. There the enclosure is negative on closed pieces that end at z_- and positive on closed pieces that start at z_+, and only (z_-, z_+) is undecided. The script prints these signs and checks this arrangement. So $g_a''(0) < 0$ on $(\frac{1}{2}, z_-]$ and $g_a''(0) > 0$ on $[z_+, 0.6712]$. The 115 boxes of the third run cover $[0.6705, 1]$, and on each the first sum of Lemma G.1 is negative.

By Step 1, g_a is even with two continuous derivatives, so $g_a'(0) = 0$. If $g_a''(0) < 0$, then $g_a' < 0$ on some $(0, \delta)$, which rules out case (b), where g_a is nondecreasing on $[0, c]$. If $g_a''(0) > 0$, then $g_a' > 0$ on some $(0, \delta)$, which rules out case (a). Lemma G.1 gives case (b) on $[0.6705, 1)$. By Step 2, $a \mapsto g_a''(0)$ is continuous on $[0.6688, 0.6692]$, so it has a zero in (z_-, z_+). Step 5 makes it strictly increasing throughout that bracket, so the zero a_* is unique. The signs already proved exclude other zeros in $(\frac{1}{2}, 0.6712]$ and give the stated shape on either side of a_*. At a_*, evenness and Step 5 give $g_{a_*}'(x) = g_{a_*}^{(4)}(0) x^3/6 + o(x^3) < 0$ for small positive x. This excludes case (b), so the critical density is unimodal too. □

Further results

This part is not in the journal version. It gives the center and the crossing to second order, the tail, further results for odd N, a likelihood bound for the reversible fluctuation assay and a convexity window between the humps. The main text does not use them. Section S6 gives the full proofs for the threshold certificate of Appendix G, which the proof of Theorem 4.4 uses. Numbers prefixed S refer to this part.

S1 The center and the crossing to second order

Theorem 2.6 gives $\log C_N$ up to an error $O(\varepsilon \log N)$. A lower bound that matches part (b) of that theorem brings the error down to $O(\varepsilon)$.

Proposition S1.1 (the center to second order)**.** *Let $N \geq 64$ be even. For every $0 < \varepsilon < \frac{1}{2}$,*

$$-11\varepsilon - \varepsilon^2(84 \log N + 1.05 \cdot 10^5) \leq \log C_N - \log \frac{4\varepsilon}{N+2} - 2\varepsilon \log N.$$

If moreover $\varepsilon(1+\log N) \leq \frac{1}{100}$, then the middle term is at most $5200\,\varepsilon$, and so $\left|\log C_N - \log \frac{4\varepsilon}{N+2} - 2\varepsilon \log N\right| \leq 5200\,\varepsilon$.

So C_N is $\frac{4\varepsilon}{N+2} N^{2\varepsilon}$ times a factor $1 + O(\varepsilon)$.

Idea of the proof. The proof is in Section S1.A. The upper bound is Theorem 2.6(b). For the lower bound we compare M after the first flip with Pólya's urn conditioned to end at h, which we call the bridge. Then $V_{t+1}(1)/\pi_t$ is the mean over the bridge of a product of factors, one for each step (Proposition S1.4). Each factor is 1 plus ε times a ratio that need not be bounded uniformly in N, and we control the factors through their first two moments under the bridge. By Jensen's inequality the mean of the product is at least the exponential of the mean of the sum of the logarithms. The step at time u adds about $2\varepsilon/u$ to this mean, and the sum over $u \leq N$ is about $2\varepsilon \log N$.

With this bound the proof of Theorem 2.7 gives one more term of the crossing.

Proposition S1.2 (the crossing to second order)**.** *As $N \to \infty$ over even N,*

$$x_N = g_N - \frac{2g_N \log N + g_N^2/2}{N(1 + 1/g_N)} + O\Big(\frac{\log N}{N}\Big).$$

More precisely, for every even N with $\log N \geq 30$ the error term is at most $20822 \log N/N$ in absolute value. So $x_N < g_N$ for every even N with $\log N \geq 3500$.

So the root g_N overshoots the crossing by an amount of order $\log^2 N/N$. The amount is $6 \log^2 N/N$ in the limit, but only about 60 percent of that at $N = 51200$ (Table 1).

Proof. We repeat the proof of Theorem 2.7(a) with one more term on each side. The center gains the term $2\varepsilon \log N$ and the end gains the term $-x^2/(2N)$, and both push F up.

Step 1. Both sides. Let F be as in the proof of Theorem 2.7, and let $1 \leq x \leq 2\Lambda_N$ and $\varepsilon = x/N$. For large N Proposition S1.1 applies, and its error $5200\,\varepsilon$ is $O(x/N)$, as is the term

$\log(1+2/N)$ in $\log\frac{4\varepsilon}{N+2}$. For the end we use $\log(1-y) = -y - \frac{y^2}{2} + O(y^3)$ for $0 \le y \le \frac{1}{2}$ with $y = x/N$. So

$$\log C_N = \log x + \log 4 - 2L + \frac{2xL}{N} + O\Big(\frac{x}{N}\Big), \qquad \log E_N = -\log 2 - x - \frac{x^2}{2N} + O\Big(\frac{x}{N}\Big).$$

In the second formula we multiplied the expansion by $N-1$, and the terms left over are $O(x/N)$ because x^2/N is bounded for $x \le 4L$. Subtracting gives

$$F(x) = G(x) - \Lambda_N + \frac{2xL + x^2/2}{N} + O\Big(\frac{x}{N}\Big) \qquad (1 \le x \le 2\Lambda_N).$$

Step 2. Solve for x_N. By Theorem 2.7, x_N lies in this range for large N. So $F(x_N) = 0$ gives $G(x_N) = \Lambda_N - \eta_N$ with $\eta_N = (2x_N L + x_N^2/2)/N + O(L/N)$. Since $G(g_N) = \Lambda_N$, the mean value theorem gives $x_N - g_N = -\eta_N/G'(\xi)$ for some ξ between x_N and g_N. Here $G'(\xi) = 1 + 1/\xi = (1 + 1/g_N)(1 + O(1/N))$, since $|\xi - g_N| = O(L^2/N)$ and $g_N \ge L$ for large N. As $\eta_N = O(L^2/N)$, we get $x_N - g_N = -\eta_N/(1 + 1/g_N) + O(L^2/N^2)$. Replacing x_N by g_N in the main term of η_N changes it by $O(L^3/N^2)$, again because $|x_N - g_N| = O(L^2/N)$. All these errors are $O(L/N)$, which gives the formula. Its main term is of order L^2/N and is larger than the error for large N, so $x_N < g_N$.

Step 3. The explicit bound. Let $L \ge 30$. By the proof of Theorem 2.7, $24L^4 + 20800L^3 \le N$, and $d = x_N - g_N$ has $|d| \le \rho_N \le 1/L^2$. Let $1 \le x \le 4L$ and $\varepsilon = x/N$. Then $\varepsilon(1 + \log N) \le \frac{1}{100}$, so Proposition S1.1 bounds $\log C_N - \log\frac{4\varepsilon}{N+2} - 2\varepsilon L$ by $5200\,\varepsilon$ in absolute value. Here $\log\frac{4\varepsilon}{N+2} = \log x + \log 4 - 2L - \log(1 + \frac{2}{N})$ with $0 \le \log(1+\frac{2}{N}) \le \frac{2x}{N}$. For $0 \le y \le \frac{1}{2}$ the series of the logarithm gives $|\log(1-y) + y + \frac{y^2}{2}| \le \frac{y^3}{3(1-y)} \le 2y^3$. So $(N-1)\log(1-\varepsilon) = -x + \varepsilon - \frac{x^2}{2N} + \frac{\varepsilon^2}{2} + R$ with $|R| \le 2N\varepsilon^3 = 2x^3/N^2 \le x/N$, because $2x^2 \le 32L^2 \le N$. Collecting terms, and using $\frac{\varepsilon^2}{2} \le \varepsilon$,

$$\Big|F(x) - G(x) + \Lambda_N - \frac{2xL + x^2/2}{N}\Big| \le 5200\varepsilon + \frac{2x}{N} + 2\varepsilon + \frac{x}{N} = \frac{5205\,x}{N}. \qquad (**)$$

Now $x_N \in [1, 4L]$ and $F(x_N) = 0$. Put $\eta = (2g_N L + g_N^2/2)/N$. Since $\Lambda_N/2 \le \Lambda_N - \log\Lambda_N \le g_N \le \Lambda_N$, we have $L/2 \le g_N \le 2L$, so $|d/g_N| \le \frac{1}{2}$. Also $G(x_N) - \Lambda_N = d + \log(1 + d/g_N)$, and $d/g_N - (d/g_N)^2 \le \log(1 + d/g_N) \le d/g_N$. The main term of $(**)$ at x_N differs from η by $d(2L + \frac{x_N + g_N}{2})/N$, which is at most $5L\rho_N/N \le L/N$ in absolute value. So $(**)$ at x_N gives

$$\Big|d\Big(1 + \frac{1}{g_N}\Big) + \eta\Big| \le \frac{5205\,x_N}{N} + \frac{L}{N} + \Big(\frac{d}{g_N}\Big)^2 \le \frac{20822\,L}{N},$$

since $5205\,x_N \le 20820L$ and $(d/g_N)^2 \le 4\rho_N^2/L^2 \le L/N$. The last inequality holds because $\rho_N \le 718L^2/N$ and $4 \cdot 718^2 L \le 20800L^3 \le N$. Dividing by $1 + 1/g_N \ge 1$ gives $|x_N - g_N + \eta/(1 + 1/g_N)| \le 20822L/N$, which is the explicit bound.

Finally let $L \ge 3500$. The tangent of the logarithm at 8192 gives $\log\Lambda_N \le \Lambda_N/8192 + \log 8192 - 1$, so $g_N \ge \Lambda_N - \log\Lambda_N \ge 1.996L$. Then $2g_N L + g_N^2/2 \ge 5.984L^2$ and $1 + 1/g_N \le 1 + 1/(1.996L)$. As $20822L + 20822/1.996 < 5.984L^2$ for $L \ge 3500$, we get $\eta/(1 + 1/g_N) > 20822L/N$, and the explicit bound gives $x_N < g_N$. □

Remark S1.3 (numerical values and open points). The formula of Proposition S1.2 without its error term is within 0.07 of x_N already at $N = 400$ (Table S1). Proposition S1.1 does not identify the term of order ε. At fixed N the center is a polynomial $c_1\varepsilon + c_2\varepsilon^2 + \cdots$ in ε, by Proposition 2.5(b). From its Taylor coefficients, computed in double precision (`verify_elephant_crossing.py`), we

get $c_2/c_1 - 2\log N = -1.585878$, -1.546572 and -1.541005 at $N = 1600$, 12800 and 51200. These values approach $2\gamma - 2 - \log 2 = -1.5387\ldots$, but we do not prove this limit here. We do not know whether, for some fixed k, the bound $\log C_N \le \log(c_1\varepsilon) + \varepsilon c_2/c_1 + O(\varepsilon^2(\log N)^k)$ holds uniformly in N in the range of Proposition S1.1. Table 1 has $x_N < g_N$ already at $N = 50$, while Proposition S1.2 proves it only for $\log N \ge 3500$.

Table S1 The crossings x_N of Table 1, their distance to $x_{2\mathrm{nd}}$, the formula of Proposition S1.2 without its error term, and the difference Υ_N of Theorem S1.7. These values are numerical and are not used in any proof.

N	x_N	$x_N - x_{2\mathrm{nd}}$	$\Upsilon_N \log N$
50	3.940487	+0.3452	−0.083
400	7.629217	+0.0621	−0.199
1600	10.240565	+0.0179	+0.215
3200	11.546834	+0.0097	+0.386
12800	14.158845	+0.0028	+0.607
51200	16.778468	+0.0008	+0.707

S1.A The bridge and the lower bound

Fix $1 \le t \le h$, and start the Pólya(c) chain at state 1 at time $t+1$. By Lemma A.1(b), given that it is at h at time N it moves up at time u from b with probability $\bar r_u(b) = \frac{h-b}{N-u}$, whatever c is. So its $h-1$ up steps are placed uniformly at random among the $N-t-1$ steps after time $t+1$. We call this law of the path $(M_u)_{t+1\le u\le N}$ the bridge, and we write $\mathbb{P}_t^{\mathrm{br}}$ and $\mathbb{E}_t^{\mathrm{br}}$ for it. For $1 \le b \le u-1$ let

$$\varpi_u^+(b) = 1 + \varepsilon\frac{u-2b}{b}, \qquad \varpi_u^-(b) = 1 - \varepsilon\frac{u-2b}{u-b},$$

$$\theta_u(b) = \frac{N(u-2b)^2}{2(N-u)\,b(u-b)}, \qquad Q_u(b) = (u-2b)^2\Big[\frac{\bar r_u(b)}{b^2} + \frac{1-\bar r_u(b)}{(u-b)^2}\Big],$$

and let $\varpi_u = \varpi_u^+(M_u)$ if $M_{u+1} = M_u + 1$ and $\varpi_u = \varpi_u^-(M_u)$ otherwise. Put

$$\mathcal{A}_t = \sum_{u=t+1}^{N-1} \mathbb{E}_t^{\mathrm{br}}\,\theta_u(M_u), \qquad \mathcal{B}_t = \sum_{u=t+1}^{N-1} \mathbb{E}_t^{\mathrm{br}}\,Q_u(M_u).$$

Both are nonnegative, and neither depends on ε.

Proposition S1.4 (bridge formula)**.** *Let $0 \le \varepsilon \le 1$ and $1 \le t \le h$. Then*

$$V_{t+1}(1) = \pi_t\,\mathbb{E}_t^{\mathrm{br}}\Big[\prod_{u=t+1}^{N-1}\varpi_u\Big].$$

If $0 < \varepsilon \le \frac12$, then $\log V_{t+1}(1) \ge \log\pi_t + \varepsilon\mathcal{A}_t - \varepsilon^2\mathcal{B}_t$.

Proof. We divide V by the hitting probability of Pólya's urn. The quotient satisfies a recursion whose weights are the step probabilities of the bridge, and each step leaves one factor ϖ_u. For $u \ge t+1$ and $b \in \mathcal{S}_u$ we have $\Phi_u^0(b) > 0$, and we put $R_u(b) = V_u(b)/\Phi_u^0(b)$. By Lemma A.1(a)

and (b) with $c = 0$, $\frac{b}{u}\Phi^0_{u+1}(b+1) = \bar r_u(b)\Phi^0_u(b)$ and $(1-\frac{b}{u})\Phi^0_{u+1}(b) = (1-\bar r_u(b))\Phi^0_u(b)$. Dividing $V_u = \mathcal{L}_u V_{u+1}$ by $\Phi^0_u(b)$ gives

$$R_u(b) = (1-\bar r_u(b))\,\frac{1-r_u(b)}{1-b/u}\,R_{u+1}(b) + \bar r_u(b)\,\frac{r_u(b)}{b/u}\,R_{u+1}(b+1),$$

where a term with an infeasible state is absent, and its bridge probability is 0 (if $b = h$ then $\bar r_u(b) = 0$, and if $u - b = h$ then $\bar r_u(b) = 1$). Also $\frac{r_u(b)}{b/u} = \varpi^+_u(b)$ and $\frac{1-r_u(b)}{1-b/u} = \varpi^-_u(b)$. So $R_u(b) = \mathbb{E}^{\mathrm{br}}_t[\varpi_u R_{u+1}(M_{u+1}) \mid M_u = b]$, and $R_N(h) = 1$. Iterating from $u = t+1$ and using $\Phi^0_{t+1}(1) = \pi_t$ gives the formula.

Now let $0 < \varepsilon \le \frac12$. We take the logarithm of the formula and bound it from below in three steps. First, every factor is at least $\frac12$. Indeed if $b \le u/2$ then $\varpi^+_u(b) \ge 1$ and $\varpi^-_u(b) - 1 \in [-\varepsilon, 0]$, and if $b \ge u/2$ the same holds with $+$ and $-$ exchanged. Second, by Jensen's inequality for the logarithm,

$$\log \mathbb{E}^{\mathrm{br}}_t\Big[\prod_u \varpi_u\Big] \ge \sum_u \mathbb{E}^{\mathrm{br}}_t \log \varpi_u \ge \sum_u \mathbb{E}^{\mathrm{br}}_t\big[(\varpi_u - 1) - (\varpi_u - 1)^2\big].$$

The second inequality is $\log(1+y) \ge y - y^2$ for $y \ge -\frac12$, which holds since the difference vanishes at 0 and its derivative $y(1+2y)/(1+y)$ has the sign of y. Third, we compute the two means given $M_u = b$. Write $\bar r = \bar r_u(b)$. The bridge moves up with probability $\bar r$, and then $\varpi_u - 1 = \varepsilon\frac{u-2b}{b}$. Otherwise $\varpi_u - 1 = -\varepsilon\frac{u-2b}{u-b}$. Since $u\bar r - b = \frac{N(u-2b)}{2(N-u)}$, this gives

$$\mathbb{E}^{\mathrm{br}}_t[\varpi_u - 1 \mid M_u = b] = \varepsilon(u-2b)\frac{u\bar r - b}{b(u-b)} = \varepsilon\,\theta_u(b), \qquad \mathbb{E}^{\mathrm{br}}_t[(\varpi_u - 1)^2 \mid M_u = b] = \varepsilon^2 Q_u(b).$$

So the sum above is $\varepsilon\mathcal{A}_t - \varepsilon^2\mathcal{B}_t$, which is the bound. □

Lemma S1.5 (the weights w_t)**.** *For each fixed t, $w_t \to t2^{-t-1}$ as $N \to \infty$. For every even $N \ge 2$, $\sum_t w_t(t+1)^3 \le 124$.*

Proof. Each fraction in $\pi_t = \frac{t}{N-t}\prod_{i=1}^{t-1}\frac{h-i}{N-i}$ tends to $\frac12$, so $N\pi_t \to t2^{1-t}$, and $Nc_1 \to 4$. This is the first claim. For the second claim we write w_t and $(t+1)^3$ through binomial coefficients in t. By the proof of Lemma 2.4 (with $v = t+1$ and $d = h$), π_t is a constant times $\binom{t}{1}\binom{N-1-t}{h-1}$, and the Chu–Vandermonde identity gives $\sum_t \binom{t}{j}\binom{N-1-t}{h-1} = \binom{N}{h+j}$. With $j = 1$ this gives the constant, namely $w_t = \binom{t}{1}\binom{N-1-t}{h-1}/\binom{N}{h+1}$. Now $t(t+1)^3 = 8\binom{t}{1} + 38\binom{t}{2} + 54\binom{t}{3} + 24\binom{t}{4}$, because both sides are polynomials of degree four that agree at $t = 0,1,2,3,4$. So $\sum_t w_t(t+1)^3$ is the sum of 8, 38, 54 and 24 times $\binom{N}{h+j}/\binom{N}{h+1}$ for $j = 1,2,3,4$. Since $\binom{N}{h+j} \le \binom{N}{h+1}$ for $j \ge 1$, it is at most $8 + 38 + 54 + 24 = 124$. □

Under $\mathbb{P}^{\mathrm{br}}_t$ let $K = M_u - 1$ and $K' = m - K$ be the numbers of up steps and of other steps among the $m = u - t - 1$ steps between times $t+1$ and u. So K is hypergeometric, with m draws from $N - t - 1$ steps of which $h - 1$ are up steps. Let $\Omega_u = \{u/8 \le M_u \le 7u/8\}$.

Lemma S1.6. *Let $N \ge 16$, $1 \le t \le h$ and $t+1 \le u \le N-1$.*

(a) The second moment is

$$\mathbb{E}^{\mathrm{br}}_t(u - 2M_u)^2 = \frac{(t-1)^2(N-u)^2}{(N-t-1)^2} + \frac{4(u-t-1)(h-1)(h-t)(N-u)}{(N-t-1)^2(N-t-2)} \le (t-1)^2 + u.$$

(b) If $t \le \frac{N+2}{4}$, then $\mathcal{A}_t \ge 2(H_{N-1} - H_t - 1 - \frac{1}{t}) \ge 2(\log N - \log t - 3)$.

(c) If $t \le \frac{N}{8}$ and $u \ge 8(t+1)$, then $\mathbb{P}_t^{\mathrm{br}}(\Omega_u^c) \le 2e^{-0.0943u}$.

(d) If $t \le \frac{N}{8}$, then $\mathcal{B}_t \le 84(t+1+\log N) + 683(t+1)^3 + 19100$.

Proof. (a) Here $u - 2M_u = u - 2 - 2K$ and $u - 2 = m + t - 1$. Since $N - 2 = 2(h-1)$, the mean $\mathbb{E}K = \frac{m(h-1)}{N-t-1}$ gives

$$\mathbb{E}(u - 2M_u) = \frac{(m+t-1)(N-t-1) - m(N-2)}{N-t-1} = \frac{(t-1)(N-t-1-m)}{N-t-1} = \frac{(t-1)(N-u)}{N-t-1}.$$

The variance of a hypergeometric variable with m draws from $N-t-1$ steps, of which $h-1$ are up steps and $h-t$ are not, is $\operatorname{Var} K = \frac{m(h-1)(h-t)(N-t-1-m)}{(N-t-1)^2(N-t-2)}$, and $N-t-1-m = N-u$. So the second term is $4\operatorname{Var} K$, and the identity is $\mathbb{E}(u-2M_u)^2 = (\mathbb{E}(u-2M_u))^2 + 4\operatorname{Var} K$. For the bound, the first term is at most $(t-1)^2$ because $N-u \le N-t-1$. In the second term $4(h-1)(h-t) = (N-2)(N-2t) = (N-t-1)^2 - (t-1)^2 \le (N-t-1)^2$, so it is at most $\frac{m(N-u)}{N-t-2}$. This is 0 for $u = t+1$, and for $u \ge t+2$ it is at most u, because $m \le u$ and $N - u \le N - t - 2$.

(b) We bound θ_u from below by a multiple of $(u-2b)^2$ and use only the variance term of (a). Since $b(u-b) \le u^2/4$, $\theta_u(b) \ge \frac{2N(u-2b)^2}{(N-u)u^2}$. By (a), $\mathbb{E}_t^{\mathrm{br}}(u-2M_u)^2 \ge 4\operatorname{Var} K$, and in $4\operatorname{Var} K$ the factor $N-u$ cancels. With $4(h-1)(h-t) = (N-2)(N-2t)$ this gives $\mathcal{A}_t \ge \nu_t \sum_{u=t+1}^{N-1} \frac{u-t-1}{u^2}$ with $\nu_t = \frac{2N(N-2)(N-2t)}{(N-t-1)^2(N-t-2)}$. We show that $\nu_t \ge 2$. Here $\nu_1 = \frac{2N}{N-3} > 2$, and for real $t \in [1, \frac{N+2}{4}]$, $\frac{d}{dt}\log\nu_t = -\frac{2}{N-2t} + \frac{2}{N-t-1} + \frac{1}{N-t-2} \ge \frac{3}{N-t-1} - \frac{2}{N-2t} \ge 0$, since the last inequality is equivalent to $4t \le N+2$. Next, the sum is $H_{N-1} - H_t - (t+1)\sum_{u=t+1}^{N-1} u^{-2}$, which is at least $H_{N-1} - H_t - \frac{t+1}{t}$. The sum is also nonnegative. So multiplying by $\nu_t \ge 2$ gives the first inequality, whether this lower bound is positive or not. For the second inequality use $H_{N-1} - H_t \ge \log\frac{N}{t+1} \ge \log N - \log t - \frac{1}{t}$.

(c) Here $m \ge \frac{7}{8}u$, so $\frac{u}{8} \le \frac{m}{7}$. Since $t \ge 1$, $\mathbb{E}K \ge \frac{m}{2}$, and since $t \le \frac{N}{8}$, $\mathbb{E}K' = \frac{m(h-t)}{N-t-1} \ge \frac{3m}{8}$. Hoeffding's inequality also holds for sampling without replacement [46, Section 6], so $\mathbb{P}(K - \mathbb{E}K \le -\sigma m) \le e^{-2\sigma^2 m}$ for $\sigma > 0$, and the same holds for K'. Now $\{M_u < \frac{u}{8}\} \subset \{K - \mathbb{E}K < -\frac{5m}{14}\}$, and since $u - M_u = t + K'$, also $\{u - M_u < \frac{u}{8}\} \subset \{K' - \mathbb{E}K' < -\frac{13m}{56}\}$. So $\mathbb{P}(\Omega_u^c) \le e^{-2(5/14)^2 m} + e^{-2(13/56)^2 m} \le 2e^{-0.10778m} \le 2e^{-0.0943u}$.

(d) We split each term $\mathbb{E}_t^{\mathrm{br}} Q_u(M_u)$ of $\mathcal{B}_t$ into the part on Ω_u and the part off Ω_u, and the three terms of the bound come from the parts on Ω_u, the parts off Ω_u with $u < 8(t+1)$, and the parts off Ω_u with $u \ge 8(t+1)$. In all of them we use that $\bar{r}(u-b)^2 + (1-\bar{r})b^2 \le u^2$, so that $Q_u(b) \le \frac{(u-2b)^2 u^2}{b^2(u-b)^2}$. On Ω_u we have $b(u-b) \ge \frac{7u^2}{64}$, so $Q_u \le \frac{4096}{49}\frac{(u-2b)^2}{u^2} \le 83.6\frac{(u-2b)^2}{u^2}$. By (a) these parts add up to at most $83.6\sum_{u>t}(\frac{(t-1)^2}{u^2} + \frac{1}{u}) \le 83.6(t + H_{N-1}) \le 84(t+1+\log N)$. Off Ω_u we only have $b(u-b) \ge u-1 \ge \frac{u}{2}$, which gives $Q_u \le 4(u-2b)^2 \le 4u^2$. For $u < 8(t+1)$ we bound the probability of Ω_u^c by 1, and these parts add up to at most $\sum_{u<8(t+1)} 4u^2 \le \frac{4}{3}(8(t+1))^3 \le 683(t+1)^3$. For $u \ge 8(t+1)$, (c) bounds the part by $8u^2 e^{-0.0943u}$, and with $q = e^{-0.0943}$ we have $8\sum_{u\ge 1} u^2 q^u = 8q(1+q)/(1-q)^3 < 19100$. □

Proof of Proposition S1.1. The upper bound is Theorem 2.6(b). For the lower bound let $0 < \varepsilon < \frac{1}{2}$ and $t_0 = \lfloor N/8 \rfloor \ge 8$. Every step below uses only this range of ε. By Proposition 2.5(a), $C_N = \varepsilon c_1 \sum_t w_t (1-\varepsilon)^{t-1} V_{t+1}(1)/\pi_t$. This is εc_1 times a mean over the probability vector (w_t). By Jensen's inequality the logarithm of the mean is at least the mean of the logarithms. For $t \le t_0$ we bound $\log(V_{t+1}(1)/\pi_t)$ from below by Proposition S1.4. For $t > t_0$ we use only that it is ≥ 0,

which is Lemma A.1(d). This gives

$$\log C_N \geq \log(\varepsilon c_1) + \sum_t w_t(t-1)\log(1-\varepsilon) + \sum_{t\leq t_0} w_t\big(\varepsilon\mathcal{A}_t - \varepsilon^2\mathcal{B}_t\big).$$

The first sum. By $\log(1-\varepsilon) \geq -\varepsilon-\varepsilon^2$ and Proposition 2.5(c), it is at least $-(\varepsilon+\varepsilon^2)\sum_t w_t(t-1) \geq -2\varepsilon - 2\varepsilon^2$.

The terms with $\mathcal{A}_t$. Since $t_0 \leq \frac{N+2}{4}$, Lemma S1.6(b) gives $\mathcal{A}_t \geq 2(\log N - \log t - 3)$ for $t \leq t_0$. Let $w_{\leq} = \sum_{t\leq t_0} w_t$. By Jensen's inequality $\sum_t w_t \log t \leq \log \sum_t w_t t \leq \log 3$, so

$$\sum_{t\leq t_0} w_t\,\varepsilon\mathcal{A}_t \geq 2\varepsilon(\log N - 3)w_{\leq} - 2\varepsilon\log 3 \geq 2\varepsilon\log N - 2\varepsilon(1-w_{\leq})\log N - 6\varepsilon - 2\varepsilon\log 3.$$

By Proposition 2.5(c), $1 - w_{\leq} \leq 2\sum_{t>t_0} t2^{-t} = 2(t_0+2)2^{-t_0}$, and $N < 8(t_0+1)$. As $(t_0+2)2^{-t_0}\log(8t_0+8)$ decreases for $t_0 \geq 8$, we get $2\varepsilon(1-w_{\leq})\log N \leq 4\cdot 10\cdot 2^{-8}\log 72\,\varepsilon < 0.67\,\varepsilon$.

The terms with $\mathcal{B}_t$. Here $t \leq t_0 \leq \frac{N}{8}$, so Lemma S1.6(d) applies. With $\sum_t w_t(t+1) \leq 4$ (Proposition 2.5(c)) and Lemma S1.5 it gives $\sum_{t\leq t_0} w_t\mathcal{B}_t \leq 84(3+1+\log N) + 683\cdot 124 + 19100 = 84\log N + 104128$.

Collecting terms, and using $c_1 = \frac{4}{N+2}$,

$$\log C_N - \log\frac{4\varepsilon}{N+2} - 2\varepsilon\log N \geq -(2+0.67+6+2\log 3)\,\varepsilon - \varepsilon^2(84\log N + 104130),$$

which gives the lower bound since $2+0.67+6+2\log 3 < 11$. For the last claim, the hypothesis on ε gives $\varepsilon^2(84\log N + 1.05\cdot 10^5) \leq \varepsilon\,\frac{84\log N + 1.05\cdot 10^5}{100(1+\log N)}$. The fraction decreases in N, and its value at $N = 64$ is below 205. Since $11 + 205 < 5200$, the two-sided bound follows. □

S1.B Comparison with the persistent walk

We compare with the persistent walk of the introduction of the paper, in which each step repeats the previous one with probability p and switches with probability $q = 1-p$. Here p and q refer to that walk. Put $T = Nq$. Its end is $p^{N-1}/2$, its crossing is p_N, and $z_N = N(1-p_N)$. So $x = N\varepsilon$ corresponds to T, and the crossing x_N corresponds to z_N. We compare the two crossings through the crossing equation of [16, Proposition 3.8], which says that for every $N \geq 2$

$$z_N + \tfrac{1}{2}\log z_N = L + c_0 + \frac{1}{8z_N} + O(L^{-2}), \qquad c_0 = \tfrac{1}{2}\log\tfrac{\pi}{8}, \tag{S1}$$

where the error term is at most $\frac{1}{4L^2} + \frac{L^2+L}{N}$ in absolute value, and that $L/2 \leq z_N \leq L$. This is the only result of [16] that we use. Doubling (S1) without its last two terms and adding $\log 2$ gives $2z_N + \log(2z_N) = \Lambda_N + \log(2\pi)$, so $2z_N - \log(2\pi)$ nearly solves $G(x) = \Lambda_N$. What is left is of order $1/L$.

Theorem S1.7. *Let z_N be the crossing value of T for the persistent walk with N steps, as above. As $N \to \infty$ over even N,*

$$\Upsilon_N := x_N - \big(2z_N - \log(2\pi)\big) = \frac{c_*}{\log N} + O\Big(\frac{\log\log N}{(\log N)^2}\Big), \qquad c_* = \tfrac{1}{2}\log(2\pi) - \tfrac{1}{4} = 0.668939\ldots$$

Proof. Write $\ell = \log(2\pi)$ and $\bar{z} = 2z_N$. We compute $G(\bar{z} - \ell)$ in three steps and then compare it with $G(x_N)$.

Step 1. The size of z_N. By the bounds after (S1), the error term there is $O(L^{-2})$ and $L/2 \le z_N \le L$. So $\log z_N = \log L + O(1)$ and $\frac{1}{8z_N} = O(1)$, and (S1) gives $z_N = L - \frac{1}{2}\log L + O(1)$. Hence $\bar{z} = 2L - \log L + O(1)$, and

$$\frac{1}{8z_N} = \frac{1}{8L} + \frac{L - z_N}{8Lz_N} = \frac{1}{8L} + O\Big(\frac{\log L}{L^2}\Big), \qquad \frac{1}{\bar{z}} = \frac{1}{2L} + O\Big(\frac{\log L}{L^2}\Big).$$

Step 2. The equation for $\bar{z}$. We double (S1) and add $\log 2$. The left side becomes $2z_N + \log z_N + \log 2 = \bar{z} + \log \bar{z}$. On the right, $2c_0 + \log 2 = \log \frac{\pi}{4}$, and $\frac{1}{4z_N} = \frac{1}{4L} + O(\log L/L^2)$ by Step 1. So $\bar{z} + \log \bar{z} = 2L + \log \frac{\pi}{4} + \frac{1}{4L} + O(\log L/L^2)$.

Step 3. The value $G(\bar{z} - \ell)$. We have $\log(\bar{z} - \ell) = \log \bar{z} + \log(1 - \ell/\bar{z})$, and $\log(1 - \ell/\bar{z}) = -\ell/(2L) + O(\log L/L^2)$ by Step 1. With Step 2 and $\log \frac{\pi}{4} - \ell = -\log 8$ this gives

$$G(\bar{z} - \ell) = \bar{z} + \log \bar{z} - \ell + \log\Big(1 - \frac{\ell}{\bar{z}}\Big) = \Lambda_N + \frac{1}{4L} - \frac{\ell}{2L} + O\Big(\frac{\log L}{L^2}\Big) = \Lambda_N - \frac{c_*}{L} + O\Big(\frac{\log L}{L^2}\Big).$$

Step 4. Compare with x_N. By Theorem 2.7(a), $G(x_N) = \Lambda_N + O(L^2/N)$. So $G(x_N) - G(\bar{z} - \ell) = c_*/L + O(\log L/L^2)$. Both x_N and $\bar{z} - \ell = 2z_N - \log(2\pi)$ are $2L + O(\log L)$, so $G'(x) = 1 + 1/x = 1 + O(1/L)$ between them. By the mean value theorem we may divide by G', and we get $x_N - (\bar{z} - \ell) = c_*/L + O(\log L/L^2)$. □

We computed z_N from the exact law of the persistent walk [16], with a tolerance of about 10^{-15}. The difference Υ_N changes sign between $N = 400$ and $N = 1600$, and $\Upsilon_N \log N$ is 0.707 at $N = 51200$ against the limit $c_* = 0.669$ (Table S1). So the asymptotic regime is reached slowly.

S2 The tail

We keep the conventions of Section 3 and Appendix B of the paper, and we use the lemmas of that appendix.

This section shows that the law falls like $x/(2m^2)$ beyond the humps.

Theorem S2.1 (the tail)**.** *Let N be even, and let $x = N\varepsilon \ge 1$ depend on N with $x \log^2 N/N \to 0$. Let $\omega_N \to \infty$ and $\delta'_N \to 0$. Then*

$$f_N(m) = \frac{x}{m(m+1)}\big(1 + o(1)\big) \qquad \text{and} \qquad P_N(N - m) = \frac{x}{2m^2}\big(1 + o(1)\big),$$

uniformly for $\omega_N x \log^2(x+1) \le m \le \delta'_N N$. We do not compute the absolute constants behind the $o(1)$ terms.

For m well beyond $x \log^2 x$ and well below N, the chance of landing in the value at distance m from an end is about half the expected number of flipped families with m steps, namely $x/(2m^2)$. The half is there because the first step must point to that end.

Idea of the proof. The proof is below. For the lower bound we use the first flipped family, which has size m with probability about $x/(m(m+1))$ (Lemma B.3), while the other flips change the count only by a relative amount of order $x \log m/m$. For the upper bound we show $\mathbb{P}(M_N \ge m) \le \frac{x}{m}(1 + O(x \log^2 m/m))$, and the unimodality of f_N turns this into a local bound. The mirror term $f_N(N - m)$ is at most $C_N = O(x/N^2)$, which gives the constant $\frac{1}{2}$.

Lemma S2.2 (late start)**.** *Let $j \ge 2$, $t \ge j-1$ and $z \ge 0$. Then* $\mathbb{P}(M_t^{[j-1]} \ge z) \le \exp\{-\frac{j-1}{2}(\frac{z}{t} - \varepsilon \log \frac{2t}{j-1})\}$.

Proof. This is a Chernoff bound with an exponent that changes with time. A chain that starts at 0 at the late time $j-1$ is like an urn that already holds $j-1$ balls of one color, so the fraction $M_k^{[j-1]}/k$ moves slowly. For this reason we take an exponent θ_k of size about $(j-1)/(2k)$.

Write $G_k = M_k^{[j-1]}$ and $\nu = (j-1)/2$. For $k \ge j-1$ let $\theta_k = \log \frac{k}{k-\nu}$, so that $e^{\theta_k} - 1 = \frac{\nu}{k-\nu}$. We claim that for $k \ge j-1$

$$\mathbb{E}e^{\theta_{k+1} G_{k+1}} \le e^{\varepsilon\nu/(k+1-\nu)}\, \mathbb{E}e^{\theta_k G_k}.$$

To see this, condition on $G_k = c$. The chain moves up with probability $r_k(c) \le \varepsilon + c/k$, and $1 + y \le e^y$. So

$$\mathbb{E}\big[e^{\theta_{k+1}G_{k+1}} \;\big|\; G_k = c\big] = e^{\theta_{k+1}c}\big(1 + r_k(c)(e^{\theta_{k+1}} - 1)\big) \le \exp\Big\{c\Big(\theta_{k+1} + \frac{\nu}{k(k+1-\nu)}\Big) + \frac{\varepsilon\nu}{k+1-\nu}\Big\}.$$

It remains to check that $\theta_{k+1} + \frac{\nu}{k(k+1-\nu)} \le \theta_k$. Now $\theta_k - \theta_{k+1} = \log(1+w)$ with $w = \frac{\nu}{(k-\nu)(k+1)}$, and $\log(1+w) \ge \frac{w}{1+w} = \frac{\nu}{k(k+1-\nu)}$. This proves the claim.

We start from $G_{j-1} = 0$ and use the claim for $k = j-1, \dots, t-1$. Since $\sum_{k=j}^{t} \frac{1}{k-\nu} \le \log \frac{t-\nu}{\nu} \le \log \frac{2t}{j-1}$, we get $\mathbb{E}e^{\theta_t G_t} \le \exp\{\varepsilon\nu \log \frac{2t}{j-1}\}$. Finally $\theta_t = -\log(1 - \frac{\nu}{t}) \ge \frac{\nu}{t}$, so Markov's inequality gives $\mathbb{P}(G_t \ge z) \le e^{-\nu z/t}\, \mathbb{E}e^{\theta_t G_t}$, which is the claim. □

Proposition S2.3 (cumulative upper bound)**.** *Assume* (H)*. For $6 \le m \le N/2$ with $m \ge 16x \log m$ we have* $\mathbb{P}(M_N \ge m) \le \frac{x}{m}\big(1 + 800\,\frac{x \log^2 m}{m}\big)$.

Idea of the proof. Given the first flip at time j, the count M_N is at most $|\mathcal{T}_V| + Z_j$, the size of the first flipped family plus the count from the rest of the tree. The event $|\mathcal{T}_V| \ge m$ gives the main term x/m. The other terms come from a shift of $|\mathcal{T}_V|$ by Z_j, from a large Z_j, and from a late first flip, and each is smaller by a factor of order $x \log^2 m/m$.

Proof. Step 1. Split by the first flip. Let $J = \lfloor N/4 \rfloor$ and $Z_j = Z_{N-1}^{(j)}$. First let $2 \le j \le J$. By Lemma 3.3, given $V = j$ the count M_N has the law of $|\mathcal{T}_V| - Y_{|\mathcal{T}_V|} + Z^{(j)}_{N-|\mathcal{T}_V|}$, where, given $V = j$, $|\mathcal{T}_V|$ has the law β_j and is independent of Y and $Z^{(j)}$. This is at most $|\mathcal{T}_V| + Z_j$, because $Y \ge 0$ and $Z^{(j)}$ is nondecreasing in time. So $M_N \ge m$ needs $Z_j > m/8$, or $Z_j \le m/8$ and $|\mathcal{T}_V| \ge m - Z_j$. Hence

$$\mathbb{P}(M_N \ge m \mid V = j) \le \mathbb{P}\Big(Z_j > \frac{m}{8}\Big) + \mathbb{E}\Big[\bar\beta_j(m - Z_j);\, Z_j \le \frac{m}{8}\Big].$$

We compare $\bar\beta_j(m-z)$ with $\bar\beta_j(m)$ for an integer $0 \le z \le m/8$. By (6), each step down from s to $s-1$ with $s \le m$ multiplies $\bar\beta_j$ by at most $\varrho_j := 1 + \frac{j-1}{N-m-j+2}$, and $\varrho_j \le 1 + \frac{4(j-1)}{N}$ because $N - m - j \ge N/4$. Since $\varrho^z - 1 \le z(\varrho - 1)\varrho^z$ for $\varrho \ge 1$, we get $\bar\beta_j(m-z) \le \bar\beta_j(m)\varrho_j^z \le \bar\beta_j(m)\big(1 + z(\varrho_j - 1)\varrho_j^{m/8}\big)$. So the mean above is at most $\bar\beta_j(m)\big(1 + (\varrho_j - 1)\varrho_j^{m/8}\, \mathbb{E}Z_j\big)$.

On $\{V > J\}$ there is no mark on $2, \dots, J$, so by Lemma 3.1 M_N is a copy of $M_N^{[J]}$ independent of $\{V > J\}$. Altogether

$$\mathbb{P}(M_N \ge m) \le \sum_{j=2}^{J} \mathbb{P}(V = j)\Big[\bar\beta_j(m)\big(1 + (\varrho_j - 1)\varrho_j^{m/8}\, \mathbb{E}Z_j\big) + \mathbb{P}\Big(Z_j > \frac{m}{8}\Big)\Big] + \mathbb{P}(V > J)\, \mathbb{P}\big(M_N^{[J]} \ge m\big).$$

This gives four terms, which we bound in turn. The main term is $\sum_j \mathbb{P}(V=j)\bar\beta_j(m) \le \mathbb{P}(|\mathcal{T}_V| \ge m) \le \frac{x}{m}$ by Lemma B.3.

Step 2. The shift term. This is the sum over $2 \le j \le J$ of $\mathbb{P}(V=j)\,\bar\beta_j(m)\,(\varrho_j - 1)\,\varrho_j^{m/8}\,\mathbb{E}Z_j$. We bound the five factors by

$$\mathbb{P}(V=j) \le \varepsilon, \quad \bar\beta_j(m) \le e^{-\frac{(j-1)(m-1)}{N-1}}, \quad \varrho_j - 1 \le \frac{4(j-1)}{N}, \quad \varrho_j^{m/8} \le e^{\frac{(j-1)m}{2N}}, \quad \mathbb{E}Z_j \le x \log \frac{N}{j-1}.$$

The second is (5), the fourth follows from the third and $1+y \le e^y$, and the last is Lemma B.2. The two exponentials combine. Since $m \ge 6$ we have $\frac{m-1}{N-1} - \frac{m}{2N} \ge \frac{m-2}{2N} \ge \frac{m}{3N}$. So with $a = \frac{m}{3N}$ and $i = j-1$ the shift term is at most

$$\frac{4\varepsilon x}{N} \sum_{1 \le i < N/4} i\, e^{-ai} \log \frac{N}{i}.$$

Here $i \le J - 1 < N/4$, so every term is nonnegative. In this sum only the first flips up to a time of order $1/a = 3N/m$ matter, because a later first flip rarely has a family of size m. To bound the sum we write $\log \frac{N}{i} = \log \frac{m}{3} + \log \frac{1}{ai}$. The first part is nonnegative, and $\sum_{i \ge 1} i e^{-ai} \le a^{-2}$. The second part is positive only for $i < 1/a$. There $i \log \frac{1}{ai} \le \frac{1}{ea}$, so these at most $1/a$ terms add up to at most $\frac{1}{ea^2}$. Hence the shift term is at most

$$\frac{4\varepsilon x}{N} \cdot \frac{\log m + 1}{a^2} = \frac{36x^2(\log m + 1)}{m^2}.$$

Step 3. The big-Z term. This is $\sum_{j=2}^{J} \mathbb{P}(V=j)\mathbb{P}(Z_j > m/8)$, and again $\mathbb{P}(V=j) \le \varepsilon$. Recall that Z_j counts the disagreeing steps of a walk whose first $j-1$ steps agree. For small j we only have the general tail bound of Lemma B.2, which is of order $x \log m/m$. For large j the late start makes $Z_j \ge m/8$ exponentially unlikely (Lemma S2.2). We split at $j_{\mathcal{K}} = \lfloor \mathcal{K}N/m \rfloor + 1$ with $\mathcal{K} = 32 \log(m/x)$. Note that $m \ge 16x \log 6 > 28x$, so $\mathcal{K} \ge 2$.

Let $2 \le j \le j_{\mathcal{K}}$. Lemma B.2 at the integer $\lfloor m/8 \rfloor + 1$ gives $\mathbb{P}(Z_j > m/8) \le 8xH_m/m$. There are at most $\mathcal{K}N/m$ such j, and $\mathcal{K} \le 32 \log m$. So they contribute at most

$$\varepsilon \cdot \frac{\mathcal{K}N}{m} \cdot \frac{8xH_m}{m} = \frac{8\mathcal{K}x^2 H_m}{m^2}.$$

Let $j > j_{\mathcal{K}}$, so that $j - 1 > \mathcal{K}N/m$. We use Lemma S2.2 with $t = N-1$ and the level $m/8$. Its exponent contains the term $\varepsilon \log \frac{2(N-1)}{j-1}$. Here $\frac{2(N-1)}{j-1} \le \frac{2m}{\mathcal{K}} \le m$, so this term is at most $\frac{x \log m}{N} \le \frac{m}{16N}$, by the hypothesis $m \ge 16x \log m$. The other term in the exponent is $\frac{m}{8(N-1)} \ge \frac{m}{8N}$. So the lemma gives $\mathbb{P}(Z_j \ge m/8) \le e^{-(j-1)m/(32N)}$, which is a geometric sequence in j. Summing over $j - 1 \ge j_{\mathcal{K}} > \mathcal{K}N/m$, these j contribute at most

$$\varepsilon \frac{e^{-\mathcal{K}/32}}{1 - e^{-m/(32N)}} \le \varepsilon \cdot \frac{x}{m}\Big(\frac{32N}{m} + 1\Big) \le \frac{x}{m} \cdot \frac{33x}{m}.$$

Here $e^{-\mathcal{K}/32} = x/m$ by the choice of $\mathcal{K}$, and we used $\frac{1}{1-e^{-y}} \le \frac{1}{y} + 1$ for $y > 0$ and $\varepsilon \le x/m$.

Step 4. No early flip. If there is no flip up to time J, the walk agrees with X_1 for about $N/4$ steps, and Lemma S2.2 makes $M_N \ge m$ exponentially unlikely in m. Namely $\mathbb{P}(V > J) = (1-\varepsilon)^{J-1} \le e^{\varepsilon - 3x/16}$, since $J \ge 3N/16$, and Lemma S2.2 with $j - 1 = J \in [\frac{3N}{16}, \frac{N}{4}]$ and $t = N$

gives $\mathbb{P}(M_N^{[J]} \ge m) \le \exp\{-\frac{3m}{32} + \frac{x}{8}\log\frac{32}{3}\}$. As $x \le m/28$ and $\varepsilon \le \frac{1}{4}$, the product of the two is at most $e^{1/4-m/13}$.

Step 5. Collect. Put $\ell = \log m \ge \log 6 > 1.79$, so that $\ell^2 > 3.2$. We compare each of the four error terms with $x^2\ell^2/m^2$. First $36(\ell+1) \le 40\ell^2$. Next $\mathcal{K} \le 32\ell$ because $x \ge 1$, and $H_m \le 1 + \ell \le 1.6\,\ell$, so $8\mathcal{K}H_m \le 410\ell^2$. Also $33 \le 12\ell^2$. Finally $y^2e^{-y} \le 4$ for $y \ge 0$, which with $y = m/13$ gives $e^{1/4-m/13} \le 676e^{1/4}/m^2 \le 304\ell^2x^2/m^2$, since $x \ge 1$ and $676e^{1/4} < 870 < 304 \cdot 3.2$. As $40 + 410 + 12 + 304 \le 800$, the four error terms add up to at most $800x^2\ell^2/m^2$, which is the claim. □

Proposition S2.4 (local upper bound)**.** *Let $N \ge 64$ and $x \ge 1$, and put $\bar{c} = \frac{1}{40}$ and $A_1 = 34000$. If $\varepsilon \log N \le \bar{c}$ and $A_1 x \log^2(x+1) \le m \le N/8$, then*

$$f_N(m) \le \frac{x}{m(m+1)}\bigl(1 + 20000\,\Delta(m)^{1/2}\bigr), \qquad \Delta(m) = \frac{x\log^2 m}{m} + \varepsilon\log N + \frac{m}{N}.$$

Idea of the proof. To the right of its mode f_N is nonincreasing, so $(k+1)f_N(m)$ is at most the probability of $\{m - k \le M_N \le m\}$, which is a difference of two tails. We bound the first tail from above by Proposition S2.3 and the second from below by summing Proposition B.4. Then we choose k to balance the errors.

Proof. Step 0. Three facts. Put $L_1 = \log(x+1) \ge \log 2$, so that $xL_1^2 \ge 0.48$, $m \ge 16200$ and $\log m \ge 9$. Also $\log N \ge 4$, so $\varepsilon \le \frac{1}{160}$. For $0 < A \le 59000$ we have $\log(AxL_1^2) \le 11 + 3L_1$, since $\log A \le 11$, $\log x \le L_1$ and $\log L_1 \le L_1$. First, $\log(x+2) \le 2L_1 \le 2.9L_1^2$, so $804x\log(x+2) \le 8A_1xL_1^2 \le 8m \le N$, and Lemma B.5 gives $m_h \le 15400xL_1^2 \le m/2$. Second, let $m' \ge m/2 \ge s_0 := 17000xL_1^2$. As $\log s/s$ decreases for $s \ge e$, we get $16x\log m'/m' \le 16x(11+3L_1)/s_0 \le 1$, because $16(11+3L_1) \le 17000L_1^2$. So $m' \ge 16x\log m'$, and also $m' \ge 6$. Third, $\log^2 s/s$ decreases for $s \ge e^2$. At $s_1 = A_1xL_1^2$ we have $\log^2 s_1 \le (11+3L_1)^2 \le 242 + 18L_1^2$, and $x(242+18L_1^2) \le s_1$. So $x\log^2 m/m \le 1$, and $\Delta(m) \le 1 + \frac{1}{40} + \frac{1}{8} \le 1.15$.

Step 1. A difference of tails. Let $0 \le k \le m/2$. Then $m - k \ge m/2 \ge m_h$, and Lemma B.1 with $a = m - k$ and $b = m$ gives $(k+1)f_N(m) \le \mathbb{P}(M_N \ge m-k) - \mathbb{P}(M_N \ge m+1)$.

Step 2. Bound each tail. Write $\Delta = \Delta(m)$. Here (H) holds. Proposition S2.3 at $m - k$, which applies by Step 0, bounds the first term by

$$\frac{x}{m-k} + \frac{800x^2\log^2(m-k)}{(m-k)^2} \le \frac{x}{m-k} + \frac{3200x^2\log^2 m}{m^2} \le \frac{x}{m-k} + \frac{3200x\Delta}{m}.$$

For the second we sum Proposition B.4 over $m < \ell \le \lfloor N/4 \rfloor$. The main terms give $x\bigl(\frac{1}{m+1} - \frac{1}{\lfloor N/4\rfloor+1}\bigr) \ge \frac{x}{m+1}\bigl(1 - \frac{8m}{N}\bigr)$. As $\frac{\log(\ell+1)}{\ell}$ decreases for $\ell \ge 3$ and $\sum_{\ell>m}\frac{1}{\ell(\ell+1)} \le \frac{1}{m+1}$, the terms with $\frac{x\log(\ell+1)}{\ell}$ cost at most $\frac{x}{m+1} \cdot 20x\frac{\log(m+2)}{m+1} \le \frac{x}{m+1} \cdot \frac{40x\log^2 m}{m}$. The terms with $\varepsilon\log N$ cost at most $\frac{x}{m+1} \cdot 20\varepsilon\log N$. The last terms are $\frac{20x}{\ell(\ell+1)}\ell e^{-\ell/9} \le \frac{20x}{m+1}e^{-\ell/9}$, and they cost at most $\frac{20x}{m+1} \cdot \frac{e^{-m/9}}{e^{1/9}-1} \le \frac{x}{m+1} \cdot 200e^{-m/9}$. Since $e^{-y} \le 2/y^2$ for $y > 0$ and $m \ge 16200$, we have $200e^{-m/9} \le 32400/m^2 \le 2/m \le 2x\log^2 m/m$. So $\mathbb{P}(M_N \ge m+1) \ge \frac{x}{m+1}(1 - 42\Delta)$. As $\frac{x}{m-k} - \frac{x}{m+1} = \frac{x(k+1)}{(m-k)(m+1)}$ and $\frac{42x\Delta}{m+1} \le \frac{42x\Delta}{m}$, this gives

$$f_N(m) \le \frac{x}{(m-k)(m+1)} + \frac{3242\,x\Delta}{(k+1)m}.$$

Step 3. Choose k. Put $\delta = \Delta^{1/2}$ and $X = \frac{x}{m(m+1)}$. Note that $\frac{1}{m} \le \frac{x\log^2 m}{m} \le \Delta$, and $\frac{1}{m^2} \le \frac{2}{m(m+1)}$. If $\Delta \le \frac{1}{16}$, take $k = \lceil m\delta \rceil$, so that $k \le \frac{m}{4}+1 \le \frac{m}{2}$. Then $\frac{m}{m-k} \le 1 + \frac{2k}{m} \le 1 + 2\delta + \frac{2}{m} \le 1 + 4\delta$,

because $\frac{1}{m} \le \Delta \le \delta$. Also $\frac{\Delta}{k+1} \le \frac{\delta}{m}$. So $f_N(m) \le X(1+4\delta) + \frac{3242x\delta}{m^2} \le X(1+4\delta+6484\delta)$, which gives the claim. If $\Delta > \frac{1}{16}$, take $k = \lfloor m/2 \rfloor$, so that $m-k \ge \frac{m}{2}$ and $k+1 \ge \frac{m}{2}$. Then $f_N(m) \le 2X + \frac{6484x\Delta}{m^2} \le X(2+12968\Delta)$. Here $\frac{1}{4} < \delta \le 1.073$, so $1 < 4\delta$ and $\Delta \le 1.073\,\delta$. Hence $2+12968\Delta \le 1+4\delta+13915\delta$, which again gives the claim. □

Proof of Theorem S2.1. We check that Propositions B.4 and S2.4 apply on the whole range of m and that their error terms tend to 0 uniformly there. Then we show that the mirror term $f_N(N-m)$ is negligible.

Step 1. The hypotheses hold. Put $\ell_x = \log(x+1) \ge \log 2$ and let $\mathcal{R}_N$ be the range $\omega_N x\ell_x^2 \le m \le \delta'_N N$. Since $x\log^2 N/N \to 0$, for large N we have $\varepsilon(1+\log N) \le \min(\bar{c}, \frac{1}{100})$, $\omega_N \ge \max(A_1, 16)$, $\delta'_N \le \frac{1}{8}$ and $804x\log(x+2) \le N$. So (H) holds as well. If $\mathcal{R}_N$ is empty there is nothing to prove.

Step 2. The errors are small on $\mathcal{R}_N$. The function $\frac{x\log^2 m}{m}$ decreases for $m \ge e^2$. At $m = \omega_N x\ell_x^2$ we have $m \ge e^2$ and $\log m \le \log\omega_N + 3\ell_x$, as $\log x$ and $\log\ell_x$ are at most ℓ_x. So on $\mathcal{R}_N$, uniformly in $x \ge 1$, $\frac{x\log^2 m}{m} \le \frac{(\log\omega_N+3\ell_x)^2}{\omega_N\ell_x^2} \le \frac{1}{\omega_N}(\frac{\log\omega_N}{\log 2}+3)^2 \to 0$. Also $m \ge \omega_N \log^2 2 \to \infty$ on $\mathcal{R}_N$. Since $\log(m+1) \le 2\log^2 m$ for $m \ge 3$, Proposition B.4 has $e_1(m) \le C(\frac{x\log^2 m}{m} + \varepsilon\log N + me^{-m/9}) \to 0$ uniformly on $\mathcal{R}_N$. Also $\Delta(m) \le \frac{x\log^2 m}{m} + \varepsilon\log N + \delta'_N \to 0$ there, and Proposition S2.4 applies. So $f_N(m) = \frac{x}{m(m+1)}(1+o(1))$ uniformly on $\mathcal{R}_N$.

Step 3. The mirror term. Lemma B.5 gives $m_h \le A_0 x\ell_x^2 \le h$ for large N, since $x\ell_x^2 \le x\log^2(N+1) = o(N)$. As $N-m \ge h$, Lemma 3.4 gives $0 \le f_N(N-m) \le f_N(h) = C_N$. By Theorem 2.6(b) and $\varepsilon\log N \to 0$, $C_N \le \frac{4x}{N(N+2)}N^{2\varepsilon}e^{5200\varepsilon} = \frac{4x}{N^2}(1+o(1))$. On $\mathcal{R}_N$ this is at most $\frac{x}{m(m+1)}\cdot 8\delta'^2_N(1+o(1))$, since $m(m+1) \le 2\delta'^2_N N^2$. Hence $P_N(N-m) = \frac{1}{2}[f_N(m)+f_N(N-m)] = \frac{x}{2m(m+1)}(1+o(1))$ uniformly on $\mathcal{R}_N$, and $m(m+1) = m^2(1+o(1))$ because $m \to \infty$. □

S3 Odd N

We show that the crossing for odd N behaves as for even N. As in Proposition 3.21, $N \ge 3$ is odd, $c = 1-2\varepsilon$, $\widetilde{C}_N$ is the probability of each of the two central values, and $x_N = N\varepsilon_N$ is the crossing.

Proposition S3.1 (odd N, the crossing)**.** *(a) As $N \to \infty$ over odd N, $x_N + \log x_N = \Lambda_N + O(L^2/N)$. So both parts of Theorem 2.7 hold for odd N as well. More precisely, if $L \ge 30$, then $|x_N - g_N| \le \tilde{\rho}_N$ and $|x_N + \log x_N - \Lambda_N| \le \tilde{\rho}_N$, where $\tilde{\rho}_N = (24L^2+20812L)/N \le 1/L^2$. In particular the last claim of Corollary 3.11(b) holds at the crossing for every large odd N.*

(b) For every odd $N \ge 1689$ we have $x_N > 2$, and at the crossing $P_N(0) < P_N(1)$. So Corollary S5.3 applies to these N with $x = x_N$.

(c) There is N'_1 such that the law at the crossing is convex between the humps for every odd $N \ge N'_1$. We do not compute the threshold N'_1.

Proof. (a) We follow the proof of Theorem 2.7(a), with $\widetilde{F}(x) = \log\widetilde{C}_N - \log E_N$ at $\varepsilon = x/N$ in place of F. Let $1 \le x \le 2\Lambda_N$. For large N, Theorem 2.6 applies to the even number $N+1$ at this ε and gives $\log C_{N+1} = \log\frac{4\varepsilon}{N+3} + O(\varepsilon L)$. By Proposition 3.21(a), $\log\widetilde{C}_N = \log C_{N+1} + O(1/N)$. Here $\log\frac{4\varepsilon}{N+3} = \log x + \log 4 - 2L + O(1/N)$, and $1/N \le xL/N$. So

$$\log\widetilde{C}_N = \log x + \log 4 - 2L + O\Big(\frac{xL}{N}\Big),$$

which is the estimate for $\log C_N$ in Step 1 of that proof. The formula $E_N = \frac{1}{2}(1-\varepsilon)^{N-1}$ holds for every N (Lemma 2.2). So $(*)$ holds for $\widetilde{F}$, and Step 2 of that proof gives $|x_N - g_N| = O(L^2/N)$ and the claim. Part (b) of Theorem 2.7 follows from part (a) as before.

For the explicit bound let $L \geq 30$. As in the proof of Theorem 2.7, $e^L \geq L^{12}/12!$ gives $26L^4 + 20802L^3 \leq N$. Let $1 \leq x \leq 4L$ and $\varepsilon = x/N$. Since $\log(N+1) \leq L+1$, we get $\varepsilon(1+\log(N+1)) \leq 4L(L+2)/N \leq \frac{1}{100}$, so Theorem 2.6 applies to $N+1$. By Proposition 3.21(a), $\log \widetilde{C}_N = \log C_{N+1} - \log(1-u)$ with $u = c/N \leq \frac{1}{N}$, and $0 \leq -\log(1-u) \leq u + u^2 \leq \frac{2}{N}$. Next $\log \frac{4\varepsilon}{N+3} = \log x + \log 4 - 2L - \log(1+\frac{3}{N})$ with $0 \leq \log(1+\frac{3}{N}) \leq \frac{3}{N}$. Theorem 2.6 gives $\log(1-2\varepsilon) \leq \log C_{N+1} - \log \frac{4\varepsilon}{N+3} \leq 2\varepsilon \log(N+1) + 5200\varepsilon$. Here $\log(1-2\varepsilon) \geq -2\varepsilon - 4\varepsilon^2$, $4\varepsilon^2 \leq \frac{1}{N}$, $2\varepsilon\log(N+1) \leq \frac{8L^2+8L}{N}$ and $5200\varepsilon \leq \frac{20800L}{N}$. For the end, $\log E_N = (N-1)\log(1-\varepsilon) - \log 2$ lies between $-\log 2 - x - \frac{16L^2}{N}$ and $-\log 2 - x + \varepsilon$, by $-y - y^2 \leq \log(1-y) \leq -y$. Adding up, and using $\varepsilon \leq \frac{4L}{N}$, we find that $\widetilde{F}(x) - G(x) + \Lambda_N$ lies between $-\frac{12L+4}{N}$ and $\frac{24L^2+20808L+2}{N}$. So $(*)$ holds for $\widetilde{F}$ with $\tilde{\rho}_N = (24L^2 + 20812L)/N$. Then $\tilde{\rho}_N \leq 1/L^2$, both ranges in Step 2 of that proof are nonempty, and Step 2 gives $|x_N - g_N| \leq \tilde{\rho}_N$ and $|G(x_N) - \Lambda_N| \leq \tilde{\rho}_N$.

For the last claim, $x_N = 2\log N - \log\log N + O(1)$ for odd N, so $\log N \leq x_N \leq 2\log N$ for every large odd N, and Corollary 3.11(b) applies with $x = x_N$.

(b) We argue as in the proof of Proposition 3.7. Let $N \geq 1689$ be odd and take $\varepsilon = 2/N$. Then $\varepsilon(1+\log(N+1)) \leq \frac{1}{100}$, so Theorem 2.6(b) for $N+1$ and Proposition 3.21(a) give

$$\widetilde{C}_N \leq \frac{N}{N-1} C_{N+1} \leq \frac{N}{N-1} \cdot \frac{8}{N(N+3)} (N+1)^{4/N} e^{10400/N}.$$

The right side decreases in N and is below $1.4 \cdot 10^{-3}$ at $N = 1689$. Also $E_N = \frac{1}{2}(1-\frac{2}{N})^{N-1} > 0.06$, as in that proof. So $\widetilde{C}_N < E_N$ at $\varepsilon = 2/N$, and hence $\varepsilon_N > 2/N$ and $x_N > 2 \geq 2(1-\varepsilon_N)$. Theorem 3.5 then gives $P_N(N-1) > E_N$ at the crossing, which is $P_N(0) < P_N(1)$ by symmetry. This is the hypothesis of the second case of Corollary S5.3.

(c) Proposition S5.4 holds for every large N, even or odd, and every ε with $\log N \leq x \leq 2\log N$. By (a), $\log N \leq x_N \leq 2\log N$ for every large odd N. So Proposition S5.4 applies at the crossing. □

S4 A likelihood for reversible fluctuation assays

This section proves likelihood bounds for the reversible fluctuation assay of Bokes, Hlubinová and Singh [27]. It uses the notation of Section 1.2 of the paper and the bound δ_Σ of Theorem 3.8.

With one sensitive founder, uniform parent choice, and independent daughter-type switching with probability ε at each birth in either direction, the resistant count at population t increases from k with probability $\varepsilon + (1-2\varepsilon)k/t$. Its law at size N is therefore f_N. Write $P_{N,x}$ for this law at $\varepsilon = x/N$, and $Q_{N,x} = \mathrm{CP}_N$ with masses $q_{N,x}(k)$. The CP law truncates jump sizes, not the total count. Its likelihood uses the full law on $\mathbb{N}_0$.

Theorem S4.1 (constrained compound Poisson likelihood)**.** *Let $N \geq 2, m \geq 1$ be integers, $a_N = 1 - 1/N$, and fix in advance $\Theta = [\ell, u]$ with $0 < \ell < u < N$ and $a_N u > 1$. Let $K_1, \ldots, K_m$ be independent observations with law P_{N,x_0}, where $x_0 \in \Theta$. Define $\widehat{x}$ as the smallest maximizer of $L_x = \prod_{i=1}^m q_{N,x}(K_i)$ on Θ, and put*

$$W = u - \ell, \qquad \omega = (m a_N)^{-1}, \qquad J = \lceil W/\omega \rceil + 1,$$
$$v_x = \min\{1, \delta_\Sigma(x)\}, \qquad \eta_x = 1 - (1-v_x)^m \leq \min\{1, m v_x\},$$

where $\delta_\Sigma(x)$ is evaluated at $\varepsilon = x/N$. Probabilities and expectations below are under $P_{N,x_0}^{\otimes m}$. For every $r > \omega$,

$$\mathbb{P}\{|\widehat{x} - x_0| \geq r\} \leq \min\left\{1, \eta_{x_0} + e^{1/2} J \exp\left(-\frac{m(r-\omega)^2}{2e^2u^2}\right)\right\}. \tag{S2}$$

For $0 < \alpha < 1$, set $r_\alpha = \omega + eu\sqrt{2\log(e^{1/2}J/\alpha)/m}$. Then

$$|\mathbb{E}\widehat{x} - x_0| \leq \mathbb{E}|\widehat{x} - x_0| \leq \min\{W, r_\alpha + W(\alpha + \eta_{x_0})\}. \tag{S3}$$

In particular, for fixed $0 < a < b$, $\Theta = [a\log N, b\log N]$ and $m = \lceil (\log N)^3 \rceil$, uniformly in $x_0 \in \Theta$ as $N \to \infty$,

$$\mathbb{E}|\widehat{x} - x_0| = O_{a,b}\left(\sqrt{\frac{\log\log N}{\log N}} + \frac{1}{\log N} + \frac{\log^8 N}{N}\right) = o(1).$$

Thus $\widehat{x}$ is absolutely consistent in probability and its mean bias tends to zero uniformly. For $\widehat{\varepsilon} = \widehat{x}/N$, divide (S3) by N.

The estimator is constrained and uses a sensitive founder. Its mean-bias bound is absolute, without a signed bias expansion.

Proof. For each integer $k \geq 0$, $q_{N,x}(k)$ is $e^{-a_N x}$ times a finite polynomial in x, and is positive, since the size-one jumps alone contribute $e^{-a_N x}(x/2)^k/k!$. Thus the likelihood attains its maximum on Θ. Its smallest maximizer is measurable on the countable sample space $\mathbb{N}_0^m$.

Put $G_N(s) = \sum_{d=1}^{N-1}(1 - s^d)/[d(d+1)]$. This function decreases continuously and strictly from a_N to 0 on $[0,1]$. Since $a_N u > 1$, choose $s \in (0,1)$ with $G_N(s) = 1/u$. The statistic $s^K \in [0,1]$ has mean $e^{-x/u}$ under $Q_{N,x}$, hence $\mathrm{TV}(Q_{N,x}, Q_{N,y}) \geq e^{-1}|x - y|/u$ for $x, y \in \Theta$. For the affinity $\rho(x,y) = \sum_k \sqrt{q_{N,x}(k)q_{N,y}(k)}$, Cauchy–Schwarz gives $\mathrm{TV}^2 \leq 1 - \rho^2$, and therefore

$$\rho(x,y) \leq \exp\{-|x-y|^2/(2e^2u^2)\}.$$

The CP semigroup gives $q_{N,y}(k) \geq e^{-a_N(y-x)}q_{N,x}(k)$ for $y \geq x$. Choose J equally spaced grid points including ℓ, u, with spacing at most ω. If $|\widehat{x} - x_0| \geq r$, the grid point y at or immediately above $\widehat{x}$ has $|y - x_0| \geq r - \omega$ and $L_y \geq e^{-ma_N\omega}L_{\widehat{x}} \geq e^{-1}L_{x_0}$. For each fixed grid point y, under $Q_{N,x_0}^{\otimes m}$, Markov's inequality gives

$$\mathbb{P}\{L_y/L_{x_0} \geq e^{-1}\} \leq e^{1/2}\mathbb{E}\sqrt{L_y/L_{x_0}} = e^{1/2}\rho(x_0,y)^m.$$

The union bound over the grid proves (S2) without η_{x_0} for CP observations. Independent couplings from Theorem 3.8 give $\mathrm{TV}(P_{N,x_0}^{\otimes m}, Q_{N,x_0}^{\otimes m}) \leq 1 - (1 - v_{x_0})^m = \eta_{x_0}$, proving the stated bound. Substitute $r = r_\alpha$ and use $|\widehat{x} - x_0| \leq W$ to split its expectation at r_α, which proves (S3).

Finally, with $L = \log N$, the stated interval and sample size give $\sup_{x\in\Theta}\delta_\Sigma(x) = O_{a,b}(L^4/N)$, $\omega = O(L^{-3})$, $J = O_{a,b}(L^4)$ and $\sup_{x\in\Theta}\eta_x = O_{a,b}(L^7/N)$. Taking $\alpha = L^{-2}$ in (S3) proves the uniform mean-error bound. Markov's inequality gives absolute consistency. □

S5 Convexity between the humps

This section concerns a stronger, prescribed convexity window between the two humps of the law at the crossing. We prove it for every large N, with a threshold N_1 that we do not compute. A certified computation gives it at the crossing for $5 \leq N \leq 2000$, and it is open in between (Conjecture S5.6). Proposition S5.1 collects these statements. The shape of Theorem 1.4(b) does not need this window. For a mode $a^* \geq N/2$ of P_N we put $m^* = N - a^*$, as in Section 3.

Proposition S5.1 (an eventual convexity window)**.** *There is N_1 such that the following holds for every $N \geq N_1$, even or odd, and every ε with $\log N \leq x \leq 2\log N$. For every mode $a^* \geq N/2$ of P_N we have $D_N(a) > 0$ for $2m^* + 1 \leq a \leq N - 2m^* - 1$ (Proposition S5.4). In particular the window holds at the crossing for every large even N. We do not compute N_1. At the crossing, a certified computation gives the window for every $5 \leq N \leq 2000$ (Proposition S5.5). For $N > 2000$ it is a conjecture (Conjecture S5.6).*

Definition S5.2 (convex between the humps)**.** Let $N \geq 5$. The law at the crossing is *convex between the humps* if, for $x = x_N$ and every mode $a^* \geq N/2$ of P_N with $m^* = N - a^*$, the second difference $P_N(a-1) - 2P_N(a) + P_N(a+1)$ is positive for $2m^* + 1 \leq a \leq N - 2m^* - 1$.

With the symmetry of the law, this makes P_N strictly convex on $[2m^*, N - 2m^*]$, so h is its unique minimizer there (the two central values for odd N).

Corollary S5.3 (one inequality is enough)**.** *Let $N \geq 8$ be even and $x = x_N$, or let $N \geq 5$ and $0 < \varepsilon < \frac{1}{2}$ be such that $P_N(0) < P_N(1)$. Let a^* be any mode of P_N and $m^* = N - a^*$. If $D_N(a_1) > 0$ for some integer a_1 with $m^* < a_1 \leq 2m^* + 1$ and $a_1 \leq N/2$, then $D_N(a) > 0$ for every $2m^* + 1 \leq a \leq N - 2m^* - 1$. In particular P_N is strictly convex on $[2m^*, N - 2m^*]$ as soon as $D_N(2m^* + 1) > 0$.*

Proof. At the crossing with $N \geq 8$ even, $P_N(0) = E_N < P_N(N-1) = P_N(1)$ by Proposition 3.7. Let a_0 be the smallest maximizer of P_N. Then $1 \leq a_0 \leq m^*$, because $P_N(0) < P_N(1)$ and $P_N(m^*) = P_N(a^*)$ is the maximum. Also $P_N(a_0 - 1) < P_N(a_0) \geq P_N(a_0 + 1)$, so $D_N(a_0) < 0$. Now Theorem 3.17 applies to $a_0 < a_1$. □

The condition $a_1 \leq N/2$ costs nothing at $a_1 = 2m^* + 1$, because $2m^* + 1 \leq N/2$ whenever the range $2m^* + 1 \leq a \leq N - 2m^* - 1$ is not empty. If $a^* < N/2$, then $m^* > N/2$, and no a_1 meets the conditions.

Proposition S5.4 (convexity between the humps for large N)**.** *There is N_1 such that the following holds for every $N \geq N_1$, even or odd, and every ε with $\log N \leq x \leq 2\log N$. For every mode $a^* \geq N/2$ of P_N with $m^* = N - a^*$ we have $4m^* + 4 \leq N$ and*

$$P_N(a-1) - 2P_N(a) + P_N(a+1) > 0 \qquad \text{for } 2m^* + 1 \leq a \leq N - 2m^* - 1.$$

In particular, after enlarging N_1, the law at the crossing is convex between the humps for every even $N \geq N_1$. We do not compute the threshold N_1 (Remark S5.7).

Idea of the proof. If there were no point of strict convexity between m^* and $2m^* + 1$, concavity would force more than half the total mass into an interval strictly to the left of the center. Symmetry rules this out, and Corollary S5.3 then gives the prescribed window.

Proof. Let $\log N \leq x \leq 2\log N$ and let $a^* \geq N/2$ be any mode. Write $m = m^* = N - a^*$ and $M = P_N(m)$. All estimates below are uniform in x and in the choice of mode, for either parity of N. By Corollary 3.11(b),

$$m = x\log x + \lambda_{\mathrm{L}} x + O(\sqrt{x\log x}), \qquad \lambda_m \longrightarrow \lambda_{\mathrm{L}}.$$

The error in Theorem 3.9(b) tends uniformly to zero, since $(1 + \log x)/x = o(1)$ and $x\delta_B = O(\log^5 N/N)$. Consequently

$$2xM \geq x f_N(m) = f_L(\lambda_m) + o(1) = f_L(\lambda_{\mathrm{L}}) + o(1).$$

It follows that $M(m+3) > 1$, $m \geq 1$ and $2m+2 \leq N/2$ for all sufficiently large N.

Suppose that $D_N(a) \leq 0$ for every integer $m < a \leq 2m+1$. Then P_N is concave on $[m, 2m+2]$, and $P_N(2m+2) \geq 0$ gives

$$\sum_{a=m}^{2m+1} P_N(a) \geq \frac{M}{m+2} \sum_{j=1}^{m+2} j = \frac{M(m+3)}{2} > \frac{1}{2}.$$

But this interval and its reflection are disjoint and have equal mass, a contradiction. Thus $D_N(a_1) > 0$ for some $m < a_1 \leq 2m+1 \leq N/2$. Theorem 3.5 gives $P_N(0) < P_N(1)$ because $x \geq \log N > 2$, and $\varepsilon < 1/2$ for large N. Corollary S5.3 now proves the required positivity. The final assertion follows from Theorem 2.7(b). □

Proposition S5.5 (small N, computer-assisted)**.** *For every $5 \leq N \leq N_c = 2000$, even or odd, the law at the crossing is convex between the humps.*

Proof. For $10 \leq N \leq 2000$, the certified computation described in Section 5 brackets ε_N and encloses P_N throughout that bracket. It lists every possible mode and checks $D_N(a) > 0$ on each required range. When the range is nonempty, it also checks $D_N(a_0) < 0$ at some $1 \leq a_0 \leq 2m^*$ and $D_N(2m^*+1) > 0$. Theorem 3.17 then gives a second check of the conclusion.

For $5 \leq N \leq 9$, Lean brackets the crossing by exact rational arithmetic and Proposition 2.3. Over the whole bracket it checks $\varepsilon < \frac{1}{2}$ and $P_N(0) < P_N(1)$. For each possible mode it checks either that the prescribed window is empty, or that $2m^*+1 \leq N/2$ and $D_N(2m^*+1) > 0$. Corollary S5.3 then gives the claim. These are outward-rounded integer computations with kernel-checked soundness (Section 5). At $N = 5$ the only possible right-hand mode is $a^* = 4$, whose window is empty. □

In double precision the law at the crossing is also convex between the humps for $N = 3200$, 6400, 12800 and 25600. The definition leaves out the two endpoints $a = 2m^*$ and $a = N - 2m^*$, because there the second difference is not positive for some $N \leq 51$. Heuristically, between the humps $P_N(a)$ is close to $\frac{x}{2}\big[\frac{1}{a(a+1)} + \frac{1}{(N-a)(N-a+1)}\big]$, which is strictly convex in a. We also leave open the tail for m between about $x \log x$ and $x \log^2 x$, below the range of the tail theorem in Theorem S2.1.

The constants of the large-N result above are not computed, so the following is open.

Conjecture S5.6 (convexity between the humps)**.** *For every $N > 2000$ the law at the crossing is convex between the humps.*

Remark S5.7 (what is still open)*.* The remaining gap concerns the prescribed convexity window. The minimum and two-hump shape hold for every even $N \geq 8$, and Corollary 3.20 gives a nonempty convexity window there. The threshold N_1 in Proposition S5.4 is not computed, and its proof uses mode location and the local mass bound to obtain $P_N(m^*)(m^*+3) > 1$. At the crossing $m^* \approx x \log x$ and $P_N(m^*) \approx f_L(\lambda_L)/(2x)$, so this condition needs roughly $f_L(\lambda_L) \log x > 2$, that is x larger than about $6.4 \cdot 10^4$. Since $x_N \approx 2 \log N$, no computation can reach this. In double precision the product $P_N(m^*)(m^*+3)$ is about 0.235, 0.239 and 0.24 at $N = 1000$, 2000 and 4000, so it grows very slowly. These numbers are heuristic and are not used in any proof.

S6 Proofs for the threshold certificate

This section proves the lemmas that Appendix G summarizes, and so completes the proof of Theorem 4.4. We use the notation of that appendix, and in particular the inequality (11). We write $\Phi_N(t) = \sum_{n\le N} c_n (it)^n$ for the partial sums of the moment series.

S6.A One value of the memory

Lemma S6.1 (moment bounds)**.** *Let $N \ge 3$.*

(a) Put $A = ((N+1)a - 1)/N$. If $m_j \le AB^j$ for $1 \le j \le N$, then $m_n \le AB^n$ for all $n \ge 1$.

(b) Let $S \subset \mathbb{C}$ be compact, with $\frac12 < \alpha \le \operatorname{Re} z$, $|\operatorname{Im} z| \le \rho$ and $|z| \le 1$ for $z \in S$. Define $m_n(z)$ by the recursion with z in place of a, and $c_n(z) = m_n(z)/\Gamma(1+nz)$. Then every m_n is analytic near S. Put $A = ((N+1)\alpha - 1)/N$. If $|m_j| \le AB^j$ on S for $1 \le j \le N$, then for all $n \ge 1$ and $z \in S$

$$|m_n(z)| \le AB^n, \qquad |c_n(z)| \le \frac{\tilde A \tilde B^n}{\Gamma(1+n\alpha)}, \qquad \tilde A = Ae^{\rho^2/(2\alpha^2)}, \quad \tilde B = Be^{\rho^2/(2\alpha)}.$$

Proof. For (a) we use induction on $n > N$. As $\theta_j \le 1$ and $na - \theta_n \ge na - 1 > 0$, the bound for smaller indices gives $m_n \le (n-1)A^2B^n/(na-1)$. The ratio $(na-1)/(n-1)$ increases in n, since its derivative is $(1-a)/(n-1)^2$, and it equals A at $n = N+1$. So $m_n \le AB^n$.

For (b), $\operatorname{Re}(nz - \theta_n) \ge n\alpha - 1 > 0$ on S, because $\alpha \le 1$. So m_n is analytic near S, and the induction of (a) goes through with $|\theta_j| \le 1$. For $x > 0$ and real y,

$$\Big|\frac{\Gamma(x)}{\Gamma(x+iy)}\Big|^2 = \prod_{k\ge0}\Big(1 + \frac{y^2}{(x+k)^2}\Big) \le \exp\Big(\frac{y^2}{x^2} + \frac{y^2}{x}\Big)$$

by [45, 5.8.3]. We take $x = 1 + n\operatorname{Re} z \ge 1 + n\alpha$ and $|y| \le n\rho$. Then $y^2/x^2 \le \rho^2/\alpha^2$ and $y^2/x \le n\rho^2/\alpha$. Finally $\Gamma(1 + n\operatorname{Re} z) \ge \Gamma(1+n\alpha)$, as Γ increases on $[1.47, \infty)$. □

Lemma S6.2 (truncation)**.** *Let $0 < \alpha \le 1$, and suppose that $|c_n| \le \tilde A \tilde B^n/\Gamma(1+n\alpha)$ for $n > N$. Let $t > 0$, $0 \le k \le N$ and $n_0 = N+1$, and put*

$$u = \frac{\tilde A \tilde B^{n_0} n_0^k t^{n_0-k}}{\Gamma(1+\alpha n_0)}, \qquad r = \tilde B t\Big(1 + \frac{1}{n_0}\Big)^k \frac{\Gamma(1+\alpha n_0)}{\Gamma(1+\alpha n_0+\alpha)}.$$

If $r < 1$, then $\big|\frac{d^k}{dt^k}\sum_{n>N} c_n(it)^n\big| \le \tau_k(t) := u/(1-r)$, and τ_k increases on $(0,t]$.

Proof. The kth derivative of $(it)^n$ has modulus at most $n^k t^{n-k}$. The ratio of consecutive terms of the majorant $\tilde A\tilde B^n n^k t^{n-k}/\Gamma(1+\alpha n)$ is $\tilde B t(1+1/n)^k\Gamma(1+\alpha n)/\Gamma(1+\alpha n+\alpha)$. It decreases in n, because $\log\Gamma$ is convex. So the majorant is at most a geometric series with first term u and ratio r. Both u and r increase in t. □

Lemma S6.3 (cells)**.** *Let ν be even, $t_c > 0$ and $|\delta| \le h$. Then $|\phi(t_c+\delta)| \le \sum_{k<\nu}|\phi^{(k)}(t_c)|h^k/k! + c_\nu h^\nu$.*

Proof. This is Taylor's formula, with $|\phi^{(\nu)}| \le \mathbb{E}|L|^\nu = \mathbb{E}L^\nu = \nu!\, c_\nu$. □

Lemma S6.4 (the tail)**.** *Let $0 < T_2 < T_3$ with $T_3 \ge 2^a T_2$, and let $\beta > 3$ with $a\beta > 1$. Cover $[0, T_3]$ by cells $[l_j, r_j]$ with $l_0 = 0$ and with T_2 among the end points, and let $s_j \ge \sup_{[l_j, r_j]} |\phi|$. Let C be the largest $s_j r_j^\beta$ over the cells that meet $[T_2, T_3]$, and put*

$$R = T_2^{-1/a} \sum_{r_j \le T_2} s_j \Big(1 - \frac{1}{2}\Big(\frac{r_j}{T_2}\Big)^{1/a}\Big)^{-a\beta} \big(r_j^{1/a} - l_j^{1/a}\big) + \frac{2^{a\beta} C T_2^{-\beta}}{a\beta - 1}.$$

If $R < 1$, then $|\phi(t)| \le C t^{-\beta}$ for every $t \ge T_3$.

Proof. Let $b = s_j$ on the cells in $[0, T_3)$ and $b(t) = Ct^{-\beta}$ for $t \ge T_3$. On a cell that meets $[T_2, T_3]$ we have $s_j \le C r_j^{-\beta} \le C t^{-\beta}$. So $b(t) \le Ct^{-\beta}$ for $t \ge T_2$, and $|\phi| \le b$ on $[0, T_3]$. Let $J(t) = \int_0^1 b(tv^a)\, b(t(1-v)^a)\, dv$.

We show that $J(t) \le R b(t)$ for $t \ge T_3$. By symmetry $J(t)$ is twice the integral over $v \le \frac12$. There $t(1-v)^a \ge 2^{-a} T_3 \ge T_2$, so $b(t(1-v)^a) \le b(t)(1-v)^{-a\beta}$. Hence $J(t)/b(t)$ is at most $2\int_0^{1/2} b(tv^a)(1-v)^{-a\beta}\, dv$. Put $v_0 = (T_2/t)^{1/a} \le \frac12$. On $v < v_0$ we substitute $u = tv^a$, and this part becomes

$$\frac{2}{a} t^{-1/a} \int_0^{T_2} s(u)\big(1 - (u/t)^{1/a}\big)^{-a\beta} u^{1/a - 1}\, du.$$

Here $t^{-1/a} \le \frac12 T_2^{-1/a}$ and $(u/t)^{1/a} \le \frac12 (u/T_2)^{1/a}$. The weight increases in u, so on each cell we take it at r_j. As $\int_l^r u^{1/a-1} du = a(r^{1/a} - l^{1/a})$, this part is at most the first term of R. On $v \ge v_0$ we use $b(tv^a) \le Ct^{-\beta} v^{-a\beta}$ and $(1-v)^{-a\beta} \le 2^{a\beta}$. This part is at most $2^{1+a\beta} T_2^{-\beta} (T_2/t)^{1/a} C/(a\beta - 1)$, which is at most the second term of R.

Let ς be the supremum of the $\tau \ge T_3$ with $|\phi| \le b$ on $[0, \tau]$, and suppose that $\varsigma < \infty$. By continuity $|\phi| \le b$ on $[0, \varsigma]$. The arguments in (11) at $t = \varsigma$ lie in $[0, \varsigma]$, so $|\phi(\varsigma)| \le J(\varsigma) \le R b(\varsigma) < b(\varsigma)$. As $|\phi|$ and b are continuous on $[T_3, \infty)$, we get $|\phi| < b$ near ς, against the choice of ς. □

This is an explicit form of Step 4 in the proof of [11, Theorem 2.5], with computed constants.

Lemma S6.5 (the second derivative)**.** *If $|\phi(t)| \le Ct^{-\beta}$ for $t \ge T_3$, with $\beta > 3$, then g_a has two continuous derivatives and*

$$g_a''(0) = -\frac{1}{\pi}\Big(\sum_{\substack{n \le N \\ n \text{ even}}} (-1)^{n/2} c_n \frac{T_3^{n+3}}{n+3} + E\Big), \qquad |E| \le \frac{T_3^3}{3}\tau_0(T_3) + \frac{C T_3^{3-\beta}}{\beta - 3}.$$

Proof. As $t^2 \operatorname{Re}\phi(t)$ is integrable, Fourier inversion gives $g_a(x) = \frac{1}{2\pi} \int e^{-itx} \operatorname{Re}\phi(t)\, dt$, and dominated convergence allows two derivatives under the integral [43, Section 3.3]. So $g_a''(0) = -\frac{1}{\pi}\int_0^\infty t^2 \operatorname{Re}\phi(t)\, dt$. On $[0, T_3]$ we replace ϕ by Φ_N, whose real part is $\sum_{n \text{ even}} (-1)^{n/2} c_n t^n$, and integrate term by term. By Lemma S6.2 the error is at most $\int_0^{T_3} t^2 \tau_0(t)\, dt \le \frac13 T_3^3 \tau_0(T_3)$. The integral over $t > T_3$ is at most $C T_3^{3-\beta}/(\beta - 3)$. □

S6.B Intervals of the memory

Ball arithmetic with a as an interval fails here. Near the threshold the series for ϕ cancels terms of size about 10^{60}, so the boxes in a would have to be narrower than about 10^{-65}. We expand in a around one exact point instead.

Lemma S6.6 (Cauchy remainder). *Let h be analytic near the closed disc $|z-a_0|\le\rho$, with $|h|\le M$ there and Taylor coefficients h_i at a_0. If $0<q<1$ and $|a-a_0|\le q\rho$, then $|h(a)-\sum_{i\le K}h_i(a-a_0)^i|\le Mq^{K+1}/(1-q)$.*

Proof. Cauchy's estimate gives $|h_i|\le M\rho^{-i}$. Sum the geometric tail. □

Let $I=[a_0-w,a_0+w]$, $0<q<1$ and $\rho=w/q$, and let S be the square $a_0+[-\rho,\rho]+i[-\rho,\rho]$, which contains the disc of Lemma S6.6. Suppose that $\alpha=a_0-\rho>\frac12$, that $|z|\le1$ on S, and that $\tilde A,\tilde B$ are as in Lemma S6.1(b). Let τ_k be the bound of Lemma S6.2 with these constants. It holds for every real $a\in I$. For a cell with centre t_c write $\partial_t^k\Phi_N(z,t_c)=E_k(z)+iO_k(z)$, where E_k is the sum over even n and iO_k the sum over odd n. Both E_k and O_k are real for real z. Let $E_{k,i}$ and $O_{k,i}$ be their Taylor coefficients at a_0, and put $M_k=\sum_{n\le N}\sup_S|c_n|\,n(n-1)\cdots(n-k+1)\,t_c^{n-k}$.

Proposition S6.7 (enclosure on an interval). *(a) For every $a\in I$ and every t in the cell with centre t_c and half-width h,*

$$|\phi_a(t)|\le\sum_{k<12}\Big(\sum_{i\le K}w^i\sqrt{E_{k,i}^2+O_{k,i}^2}+\frac{M_kq^{K+1}}{1-q}+\tau_k(t_c)\Big)\frac{h^k}{k!}+h^{12}\sup_S|c_{12}|.$$

(b) Let T_2, $T_3\ge2^{a_0+w}T_2$, β and the cells be as in Lemma S6.4, with s_j the bound of (a), and suppose that $R<1$ for every $a\in I$. Let $F(z)=\sum_{n\le N,\,n\ even}(-1)^{n/2}c_n(z)T_3^{n+3}/(n+3)$, with Taylor coefficients F_i at a_0, and let M_F be the same sum with $\sup_S|c_n|$ in place of $(-1)^{n/2}c_n(z)$. Then for every $a\in I$

$$\Big|g_a''(0)+\frac1\pi\sum_{i\le K}F_i(a-a_0)^i\Big|\le\eta=\frac1\pi\Big(\frac{M_Fq^{K+1}}{1-q}+\frac{T_3^3}{3}\tau_0(T_3)+\frac{CT_3^{3-\beta}}{\beta-3}\Big),$$

and $a\mapsto g_a''(0)$ is continuous on I.

Proof. For (a) we apply Lemma S6.3 with $\nu=12$, and Lemma S6.2 to replace $\phi_a^{(k)}$ by $\partial_t^k\Phi_N$. For real a we have $|\partial_t^k\Phi_N(a,t_c)|=|(E_k(a),O_k(a))|$. Lemma S6.6 for E_k and O_k, with $\sup|E_k|+\sup|O_k|\le M_k$ on S, and the triangle inequality in $\mathbb{R}^2$ give the bracket. Also $c_{12}(a)\le\sup_S|c_{12}|$.

For (b), $T_3\ge2^aT_2$ for every $a\in I$, so Lemma S6.4 applies at each a. Then Lemma S6.5 and Lemma S6.6 for F give the enclosure. By Lemma S6.1(b) the series for $\phi_a(t)$ converges uniformly for $a\in I$ and t in compact sets. So $\phi_a(t)$ is continuous in a. Moreover $t^2|\phi_a(t)|$ is at most t^2 on $[0,T_3]$ and $Ct^{2-\beta}$ beyond, uniformly in $a\in I$. Dominated convergence gives the continuity. □

Near $a=\frac12$ the bounds of Lemma S6.1 degenerate, since $m_2=a/(2a-1)$. We rescale.

Lemma S6.8 (the scaled variable). *Put $y=2a-1$ and $\mu_n=m_ny^{\lfloor n/2\rfloor}$.*

(a) We have $\mu_0=\mu_1=1$, $\mu_2=a$, and for $n\ge3$

$$(na-\theta_n)\mu_n=\sum_{j=1}^{n-1}\theta_jy^{e_{nj}}\mu_j\mu_{n-j},$$

where $e_{nj}=1$ if n is even and j is odd, and $e_{nj}=0$ otherwise. Here $na-\theta_n$ is $(n-1)a$ for odd n and $na-1$ for even n. So every μ_n is a rational function of a with no pole in $\operatorname{Re}a>\frac14$.

(b) Let $\sigma^2 = \mathbb{E}L^2$, $X = L/\sigma$ *and* $\hat{c}_n = \mu_n \Gamma(2a)^{\lfloor n/2 \rfloor}/\Gamma(1+an)$. *Then* $\sigma^2 = 1/(y\Gamma(2a))$,

$$\mathbb{E}e^{iuX} = \sum_{n \text{ even}} \hat{c}_n (iu)^n + \sqrt{y\Gamma(2a)} \sum_{n \text{ odd}} \hat{c}_n (iu)^n, \qquad g_a''(0) = -\frac{1}{\pi\sigma^3} \int_0^\infty u^2 \operatorname{Re} \mathbb{E}e^{iuX}\, du.$$

Proof. Multiply the recursion for m_n by $y^{\lfloor n/2 \rfloor}$. The exponent $\lfloor n/2 \rfloor - \lfloor j/2 \rfloor - \lfloor (n-j)/2 \rfloor$ is 1 when n is even and j is odd, and 0 otherwise. Also $m_2 = a/y$. This proves (a), as the zeros of the factors are $a = 0$ and $a = 1/n \le \frac{1}{4}$. For (b), $\sigma^2 = 2m_2/\Gamma(1+2a) = 1/(y\Gamma(2a))$ and $\mathbb{E}X^n/n! = c_n \sigma^{-n} = \mu_n \Gamma(2a)^{\lfloor n/2 \rfloor} (y\Gamma(2a))^{(n \bmod 2)/2}/\Gamma(1+an)$. The formula for $g_a''(0)$ follows from the substitution $t = u/\sigma$ in Lemma S6.5. □

The fixed-point equation is linear, so X satisfies it too, and (11) and Lemmas S6.3 to S6.5 hold for X. The proof of Lemma S6.1(b) carries over to μ_n for $n \ge 3$ and $\alpha > \frac{1}{4}$, with two changes. The constant A becomes $((N+1)\alpha - 1)/(N\kappa)$, where κ is the largest value of $\max(1, |z|) \max(1, |2z-1|)$ on S. And $\tilde{B}$ gains the factor $\max(1, \sup_S |\Gamma(2z)|)^{1/2}$. For real a the odd part carries the factor $\sqrt{y\Gamma(2a)}$. In the cell bounds we treat the even and odd parts as separate analytic functions. We multiply the odd part by $f = \sup_I \sqrt{y\Gamma(2a)}$ only for real a, and the remainders by $\max(1, f)$. The real part of $\mathbb{E}e^{iuX}$ involves only the even part. So the disc may cross $a = \frac{1}{2}$, and the cover reaches $a = \frac{1}{2}$. The sign of $g_a''(0)$ is the opposite of the sign of the integral, since $\sigma^3 > 0$ and the Fourier formula has a minus sign.

S6.C From the runs to the theorem

The script accepts an interval of a only when $G \pm \eta$ has one sign on each piece of a subdivision of it. In the 7 scaled runs every piece of every interval is negative, so $g_a''(0) < 0$ on $(\frac{1}{2}, 0.668]$. The zero run has 8 intervals that cover $[0.668, 0.6712]$. On the 2 intervals that cover $[0.668, 0.6688]$ every piece is negative. On the 5 intervals that cover $[0.6692, 0.6712]$ every piece is positive. On $[0.6688, 0.6692]$ the pieces up to z_- are negative, the pieces from z_+ on are positive, and only (z_-, z_+) is undecided. Proposition S6.7(b) turns these signs into $g_a''(0) < 0$ on $(\frac{1}{2}, z_-]$ and $g_a''(0) > 0$ on $[z_+, 0.6712]$. The order matters. A positive interval left of z_- would give another zero, and the sign classification of Theorem 4.4 would fail. So the script prints the sign of each accepted interval with its exact end points. Its mode `zero` fails unless the signs are $-$ before the undecided piece and $+$ after it. Its mode `check` requires the same of the stored records, together with the sign $-$ on every interval of the scaled runs.

S6.D Uniqueness of the transition

The derivative required to close the bracket cannot be obtained by differentiating the error bound in Lemma S6.5. We instead differentiate the smoothing equation and bound its new tail.

Lemma S6.9 (differentiated smoothing bound)**.** *Fix a compact memory interval* $I \subset (1/2, 1)$. *Suppose that the cell certificate of Lemma S6.4 gives, uniformly on* I, $f(t) := |\phi_a(t)| \le Ct^{-\beta}$ *for* $t \ge T_2$, *and*

$$H_\beta(t) := 2 \int_0^{1/2} f(tv^a)(1-v)^{-a\beta}\, dv \le R < 1 \quad (t \ge T), \qquad T \ge 2^a T_2.$$

Let $\beta > 4$, $\gamma = \beta - 1$, $a\gamma > 1$, and $u = |\partial_t \phi_a|$, $w = |\partial_a \phi_a|$. Suppose that $u \le M$ everywhere, $w \le M_a$ on $[0, T_2]$, and cells give $u(t) < D_1 t^{-\beta}$ and $w(t) < D_a t^{-\gamma}$ on $[T_2, T]$. Put

$$A_1 = \frac{MC2^{a(\beta-1)}}{a+1}, \qquad J = 2C2^{a\beta} T^{\gamma-\beta-1/a},$$
$$\kappa = R + \frac{JT_2^{1/a-\gamma}}{a\gamma - 1}, \qquad A_0 = JM_a T_2^{1/a},$$
$$A_S = MC2^{a(\beta-1)} \left\{ \frac{\log 2}{a+1} + \frac{1}{(a+1)^2} \right\} + D_1 (\log 2) R + CR/T.$$

If $A_1 + RD_1 < D_1$ and $A_0 + A_S + \kappa D_a < D_a$, with $\kappa < 1$, then the two derivative bounds hold for every $t \ge T_2$. Consequently, in the interior of I,

$$\frac{d}{da} g_a''(0) = -\frac{1}{\pi} \int_0^\infty t^2 \operatorname{Re} \partial_a \phi_a(t)\, dt, \qquad \left| \frac{1}{\pi} \int_T^\infty t^2 \operatorname{Re} \partial_a \phi_a(t)\, dt \right| \le \frac{D_a T^{3-\gamma}}{\pi(\gamma - 3)}. \tag{S4}$$

Proof. The complex majorant in Lemma S6.1(b) makes the moment series locally uniformly convergent in the frequency and memory. Cauchy's estimate on a smaller parameter disc gives the same conclusion for its memory derivative and every fixed frequency derivative. Thus these are actual derivatives of ϕ_a. The smoothing equation, with $p = (1 + a)/2$, is

$$\phi_a(t) = \int_0^1 \phi_a(tv^a) \{ p\phi_a(t(1-v)^a) + (1-p)\phi_a(-t(1-v)^a) \}\, dv.$$

On compact frequency sets, its differentiation under the integral is justified by local boundedness of these derivatives and boundedness of $v^{\inf I} |\log v|$. Conjugate symmetry gives even moduli. Writing $x = tv^a$ and $y = t(1-v)^a$, we obtain

$$u(t) \le \int_0^1 \{ v^a u(x) f(y) + (1-v)^a f(x) u(y) \}\, dv,$$
$$w(t) \le \int_0^1 \{ w(x) f(y) + f(x) w(y) \}\, dv + S(t),$$
$$S(t) \le \int_0^1 \{ x |\log v| u(x) f(y) + y |\log(1-v)| f(x) u(y) + f(x) f(y) \}\, dv.$$

The final term includes the derivative of p: its coefficient is one because $p' = 1/2$ and $|\phi_a(y) - \phi_a(-y)| \le 2f(y)$.

Split each integral at $1/2$ and reflect its second half. For the frequency derivative, the smaller-argument term is at most $A_1 t^{-\beta}$, by $u \le M$ and

$$2 \int_0^{1/2} v^a (1-v)^{-a\beta}\, dv \le \frac{2^{a(\beta-1)}}{a+1}.$$

The larger-argument term is at most $RD_1 t^{-\beta}$. For the memory derivative, split the smaller-argument term again at $v_0 = (T_2/t)^{1/a}$. Its bound is

$$2C2^{a\beta} t^{-\beta-1/a} \left\{ M_a T_2^{1/a} + \frac{D_a T_2^{1/a-\gamma}}{a\gamma - 1} \right\}.$$

The larger-argument term is at most $RD_a t^{-\gamma}$, since $H_\gamma \le H_\beta$. As $\gamma - \beta - 1/a < 0$ and $t \ge T$, these two terms are bounded by $(A_0 + \kappa D_a)t^{-\gamma}$. In S, the smaller logarithmic term uses

$$\int_0^{1/2} v^a(-\log v)\,dv = 2^{-a-1}\left\{\frac{\log 2}{a+1} + \frac{1}{(a+1)^2}\right\};$$

the larger uses $-\log(1-v) \le \log 2$ and $H_\beta \le R$. The mixing-weight term is at most $CRt^{-\beta}$. Thus $S(t) \le A_S t^{-\gamma}$.

For completeness, the propagation is an inhomogeneous compact-maximum argument. On any $[T_2, U]$, maximize the continuous weighted derivative. If its maximum lies in $[T_2, T]$, use the cell bound. Otherwise repeat the preceding inequalities with that maximum in place of the proposed constant, and rearrange the strict contraction inequality. This first proves the frequency bound, then the memory bound, independently of U. The integrable uniform bound $t^2 w(t) \le D_a t^{2-\gamma}$ now permits differentiation of the Fourier formula for $g_a''(0)$ on the interior of I and gives (S4). □

The computation uses

$$I = [0.66881285, 0.66881287], \quad T_2 = 6, \quad T = 10, \quad \beta = 8, \quad \gamma = 7,$$

and the conservative constants

$$C = 17/1000, \quad R = 83/1000, \quad M = 2, \quad D_1 = 2, \quad M_a = D_a = 4.$$

It checks $A_1 < 1$, $\kappa < 1/10$ and $A_0 + A_S < 2$, so both contractions have strict slack. The new tail is at most $1/(10000\pi) < 1/30000$. The old zero bracket lies strictly inside I.

For the finite-frequency integral, let w_0 be the halfwidth of I, $\rho > w_0$ the Cauchy radius and $q = w_0/\rho$. Differentiating the finite Taylor polynomial in memory leaves the coefficient remainder factor

$$\frac{1}{\rho}\sum_{i>K} i q^{i-1} = \frac{q^K\{(K+1) - Kq\}}{\rho(1-q)^2}.$$

The omitted moment series has derivative bound $\tau_k(t)/(\rho - w_0)$ by Cauchy's estimate. Frequency cells use a Taylor expansion of order 48; its remainder bounds the actual high derivative by the complex coefficient majorant, not by a probability moment at complex memory. The signed interval checks give

$$0.68 < \frac{d}{da} g_a''(0) < 0.70, \qquad -0.09 < g_a^{(4)}(0) < -0.07 \quad (a \in \operatorname{int} I). \tag{S5}$$

The second enclosure uses the same Fourier bound with weight t^4; its tail is $CT^{-3}/(3\pi)$. The file `certify_transition.py` performs these checks, including the initial derivative-cell bounds. It was rerun with 1100 moments and 1600-bit precision under `python -O`; every check uses `require`, so optimization cannot remove it.

Continuity and the old sign bracket give a zero a_* of $g_a''(0)$. The first inequality in (S5) makes it unique within I. At a_*, evenness and the second inequality give

$$g'_{a_*}(x) = \tfrac{1}{6} g_{a_*}^{(4)}(0) x^3 + o(x^3) < 0 \quad \text{for all sufficiently small } x > 0.$$

The strict shape dichotomy therefore puts a_* in the unimodal branch. Together with the signs on either side inside I, and the existing shape certificate outside the bracket, this proves the unique-transition assertion in Theorem 4.4. No claim that central curvature has no other zero in the already bimodal region is needed.

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